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(54) **METHODS FOR TREATING CANCER USING COMBINATIONS OF EPIGENETIC THERAPIES AND RADIOCONJUGATE TARGETING AGENTS**

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(57)

ABSTRACT

The invention provides methods for treating a cancer, such as acute myeloid leukemia, in a mammalian subject that include administering to the subject (i) an epigenetic drug such as one or both an HDAC inhibitor and an LSD1/KDM1A inhibitor, and (ii) a radioisotope-labeled agent that targets cancer cells in the subject, wherein the amounts of the epigenetic drug(s) and radiolabeled agent, when administered in conjunction with one another, are therapeutically effective.

Specification includes a Sequence Listing.

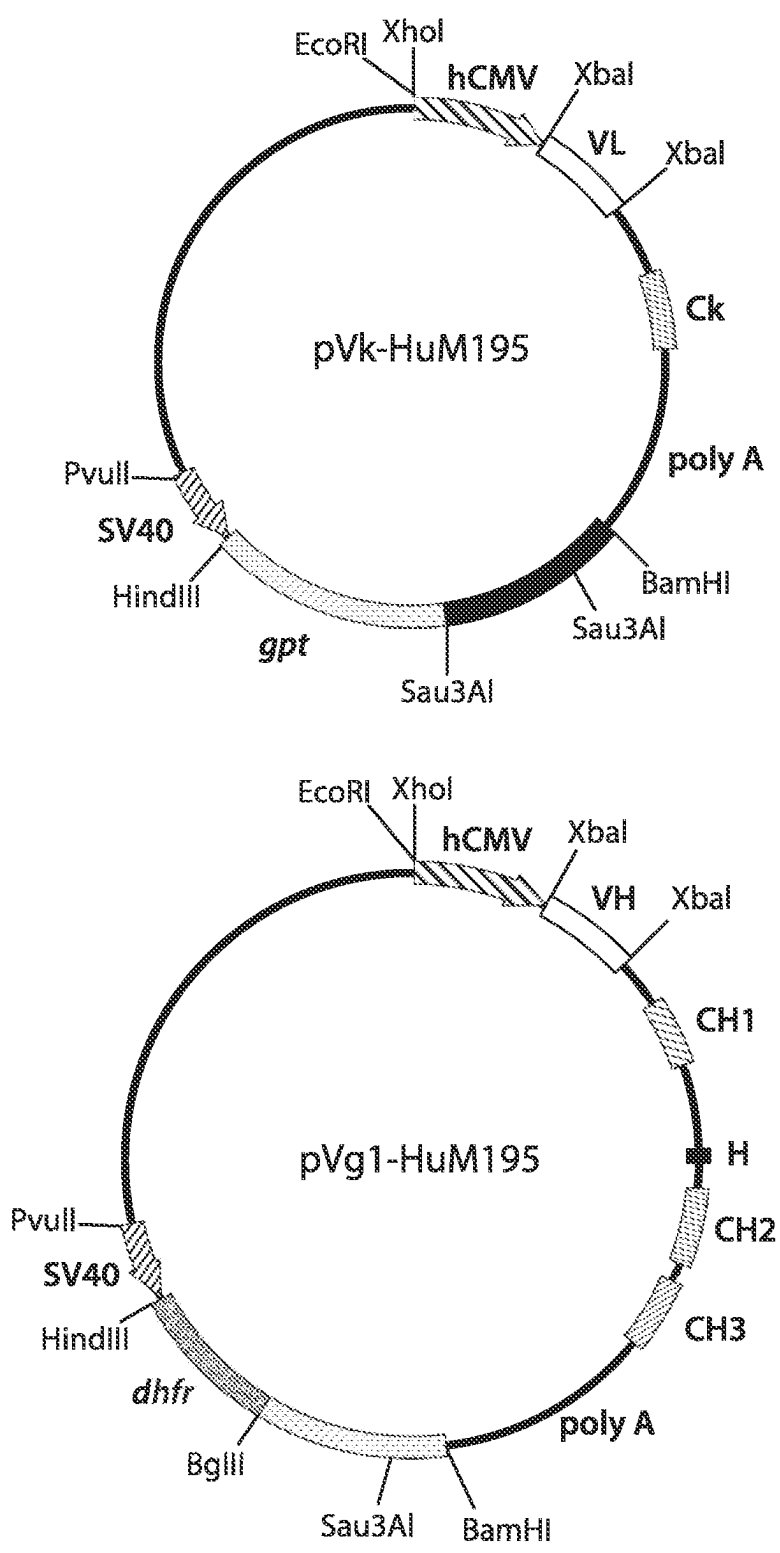


Figure 1

624 TCTAGACCACCATGGAGAAAGACACACTCCTGCTATGGGTCCTACTTCTCTGGGTTCAGGTTCACAGGTGACATTGAG
M E K D T L L L W V L L L W V P G S T G D I Q
704 ATGACCCAGTCTCCGAGCTCTCTGTCCGCATCAGTAGGAGACAGGGTCACCATCACATGCAGAGCCAGCGAAAGTGTCGA
M T Q S P S S L S A S V G D R V T I T C R A S E S V D
784 CAATTATGGCATTAGCTTTATGAAGTGGTTCCAACAGAAACCCGGAAGGCTCCTAAGCTTCTGATTACGCTGCATCCA
N Y G I S F M N W F Q Q K P G K A P K L L I Y A A S
864 ACCAAGGCTCCGGGGTACCCTCTCGCTTCTCAGGCAGTGGATCTGGGACAGACTTCACTCTCACCATTTCATCTCTGCAG
N Q G S G V P S R F S G S G S G T D F T L T I S S L Q
944 CCTGATGACTTCGCAACCTATTACTGTGAGCAAGTAAGGAGGTTCCGTGGACGTTCCGGTCAAGGGACCAAGGTGGAGAT
P D D F A T Y Y C Q Q S K E V P W T F G Q G T K V E I
1024 CAAACGTAAGTAGAATCCAAAGTCTAGAAATCTAAACTCTGAGGGGGTCGGATGACGTGGCCATTCTTTGCCCTAAAGCA
K R
1104 TTGAGTTTACTGCAAGGTGAGAAAGCATGCAAGCCCTCAGAATGGCTGCAAGAGCTCCAACAAAACAATTTAGAACT
1184 TTATTAAGGAATAGGGGAAGCTAGGAAGAACTCAAAACATCAAGATTTTAAATACGCTTCTTGGTCTCCTTGCTATAA
1264 TTATCTGGGATAAGCATGCTGTTTTCTGTCTGTCCCTAACATGCTCTGTGATTATCCGAAACAACACCCCAAGGGCAG
1344 AACTTTGTTACTTAAACACCATCCTGTTTGCTTCTTTCCCTCAGGAACTGTGGCTGCACCATCTGTCTTCATCTTCCCGCC
T V A A P S V F I F P P
1424 ATCTGATGAGCAGTTGAAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCCAGAGAGGCCAAAGTAC
S D E Q L K S G T A S V V C L L N N F Y P R E A K V
1504 AGTGGAAGGTGGATAACGCCCTCCAATCGGGTAACTCCAGGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACCTAC
Q W K V D N A L Q S G N S Q E S V T E Q D S K D S T Y
1584 AGCCTCAGCAGACCCCTGACGCTGAGCAAGCAGACTACGAGAAACACAAAGTCTACGCTGCGAAGTCACCCATCAGGG
S L S S T L T L S K A D Y E K H K V Y A C E V T H Q G
1664 CCTGAGCTCGCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTAGAGGGAGAAGTCCCCCACCTGCTCCTCAGTTCCA
L S S P V T K S F N R G E C .
1744 GCCTGACCCCTCCCATCCTTTGGCTCTGACCCCTTTTCCACAGGGGACCTACCCCTATTGCGGTCTCCAGCTCATCT
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1984 AAATATGTAGTCATCCTAAGCGGCATAACCATTATATAAAATCATCCTTCATTCTATTTTACCCTATCATCCTCTGCAAG
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2144 CCCCTCCTCAGCAAGCCCTCATAGTCCTTTTAAAGGGTGACAGGTCTTACAGTCATATATCCTTTGATTCAATTCCTGA
2224 GAATCAACCAAAGCAAATTTTCAAAGAAGAAACCTGCTATAAAGAGAATCATTCATTGCAACATGATATAAAATAACA
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2464 TTAATCCACACTATACTGTGAGATTAAAAACATTCATTAATGTTGCAAGGTTCTATAAAGCTGAGAGACAAATATAT
2544 TCTATAACTCAGCAATCCCACTTCTAGGATCC

Figure 2

624 TCTAGACCACCATGGGATGGAGCTGGATCTTTCTTCTCCTCTGTCCAGGAAGTCTGTCGGCGTGCCACTCTCAGGTTCACTGC
M G W S W I F L F L L S G T A G V H S Q V Q L

704 GTGCAGTCTGGAGCTGAGGTGAAGAAGCCTGGGAGCTCACTGAAGGTTTTCTGCAAAGCTTCTGGCTACACCTTCACTGA
V Q S G A E V K K P G S S V K V S C K A S G Y T F T D

784 CTACAACATGCACTGGGTGAGGCAGGCTCCTGGCCAAGGCTGGAATGGATTGGATATATTTATCCTTACAATGGTGGTA
Y N M H W V R Q A P G Q G L E W I G Y I Y P Y N G G

864 CCGGCTACAACCAGAAGTTCAAGAGCAAGGCCACAATTACAGCAGACGAGAGTACTAACACAGCCTACATGGAACTCTCC
T G Y N Q K F K S K A T I T A D E S T N T A Y M E L S

944 AGCCTGAGGCTCTGAGGACACTGCAGTCTATTACTGCGCAAGAGGGCGCCCCGCTATGGACTACTGGGGCCAAAGGACTCT
S L R S E D T A V Y Y C A R G R P A M D Y W G Q G T L

1024 GGTCACTGTCTCTTCAGGTAAGAATGGCTCTAGACCACCATGGGATGGAGCTTTCTGGGGCAGGCCAGGCTGACCTTG
V T V S S

1104 GCTTTGGGGCAGGGAGGGGGCTAAGGTGAGGCAGGTGGCGCCAGCCAGGTGCACACCCAATGCCCATGAGCCCAGACACT
G

1184 GGACGCTGAACCTCGCGGACAGTTAAGAACCAGGGGCTCTGCGCCCTGGGGCCAGCTCTGTCCCACACCGGGTCACA
G

1264 TGGCACCACTCTCTTGCAGCCTCCACCAAGGGCCCATCGGTCTTCCCCTGGCACCTCTCCAGAGCACCTCTGGGG
A S T K G P S V F P L A P S S K S T S G

1344 GCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGGAAGTCAAGGCGCCCTGACC
G T A A L G C L V K D Y F P E P V T V S W N S G A L T

1424 AGCGGCGTGCACACCTTCCCGCTGTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAG
S G V H T F P A V L Q S S G L Y S L S S V V T V P S S

1504 CAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCAGCAACACCAAGGTGGACAAGAAAGTTGGTGAGA
S L G T Q T Y I C N V N H K P S N T K V D K K V

1584 GGCCAGCACAGGGAGGGAGGGTCTCTGCTGGAAGCCAGGCTCAGCGCTCCTGCCTGGACGCATCCCGGCTATGCAGCCCC
G

1664 AGTCCAGGGCAGCAAGGCAGGCCCCGTCTGCCCTTTACCCGGAGGCTCTGCCCCCCCCACTCATGCTCAGGGAGAGGG
G

1744 TCTTCTGGCTTTTTTCCCAGGCTCTGGGCAGGCACAGGCTAGGTGCCCTTAACCCAGGCCCTGCACACAAAGGGCAGGT
G

1824 GCTGGGCTCAGACCTGCCAAGAGCCATATCCGGGAGGACCTGCCCCGACCTAAGCCCACCCCAAAGGCCAAACTCTCC
G

1904 ACTCCCTCAGCTCGGACACCTTCTCTCTCTCCAGATTCCAGTAAGTCCCAATCTTCTCTCTGCAGAGCCCAATCTTGTG
E P K S C

1984 ACAAACTCACACATGCCACCGTGCCAGGTAAGCCAGCCAGGCTCGCCCTCCAGCTCAAGGGGGACAGGTGCCCT
D K T H T C P P C P

2064 AGAGTAGCCTGCATCCAGGGACAGGCCCCAGCCGGGTGCTGACACGTCCACCTCCATCTTCTCTCAGCACCTGAACCTC
A P E L

2144 TGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCAAGGACACCTCATGATCTCCCGACCCCTGAGGTCACATGC
L G G P S V F L F P P K P K D T L M I S R T P E V T C

2224 GTGGTGGTGGACGTGAGCCACGAAGACCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAA
V V V D V S H E D P E V K F N W Y V D G V E V H N A K

2304 GACAAAGCCGCGGGAGGAGCAGTACAACAGCAGCTACCGTGTGGTCAGCGTCTCACCGTCTGCACCAGGACTGGCTGA
T K P R E E Q Y N S T Y R V V S V L T V L H Q D W L

2384 ATGGCAAGGAGTACAAGTGAAGGTCTCCAACAAAGCCCTCCAGCCCCATCGAGAAAACCATCTCCAAGGCCAAAGGT
N G K E Y K C K V S N K A L P A P I E K T I S K A K

2464 GGGACCCGTGGGGTGGAGGGGCCACATGGACAGAGGGCCGCTCGGCCCCACCTCTGGCCTGAGAGTGACCGCTGTACCAA
G

Figure 3

2544 CCTCTGTCCCTACAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGATGAGCTGACCAAGAACCAG
G Q P R E P Q V Y T L P P S R D E L T K N Q
2624 GTCAGCCTGACCTGGCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAA
V S L T C L V K G F Y P S D I A V E W E S N G Q P E N
2704 CAACTACAAGACCACGCCTCCCGTGTGGACTCCGACGGCTCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCA
N Y K T T P P V L D S D G S F F L Y S K L T V D K S
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R W Q Q G N V F S C S V M H E A L H N H Y T Q K S L S
2864 CTGTCTCCGGGTAAATGAGTGCACGGCCGGCAAGCCCCCGTCCCCGGGCTCTCGCGGTGCGACGAGGATGCTTGGCAC
L S P G K *
2944 GTACCCCTGTACATACTTCCCCGGCGCCCAGCATGGAATAAAGCACCAGCGCTGCCCTGGGCCCCCTGCGAGACTGTG
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3744 AGACAAACCCAGCCCTCCTCTCACAAGGGTGGCCCTGCAGCCGCCACACACACAGGGGATCACACACCACGTCACGTC
3824 CCTGGCCCTGGCCCACTTCCCAGTGGCCGCCCTTCCCTGCAGGATCC

Figure 3 Cont'd

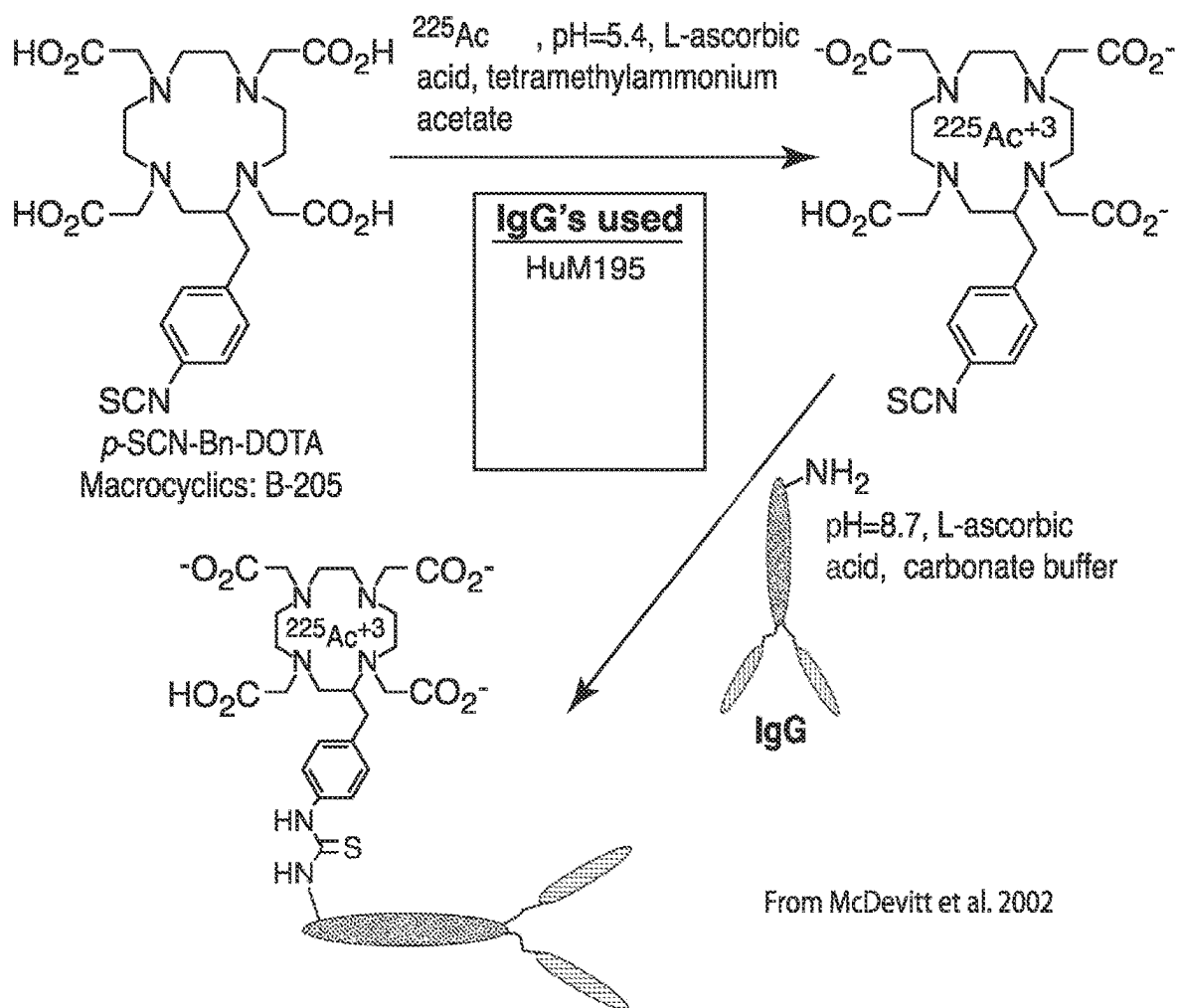


Figure 4

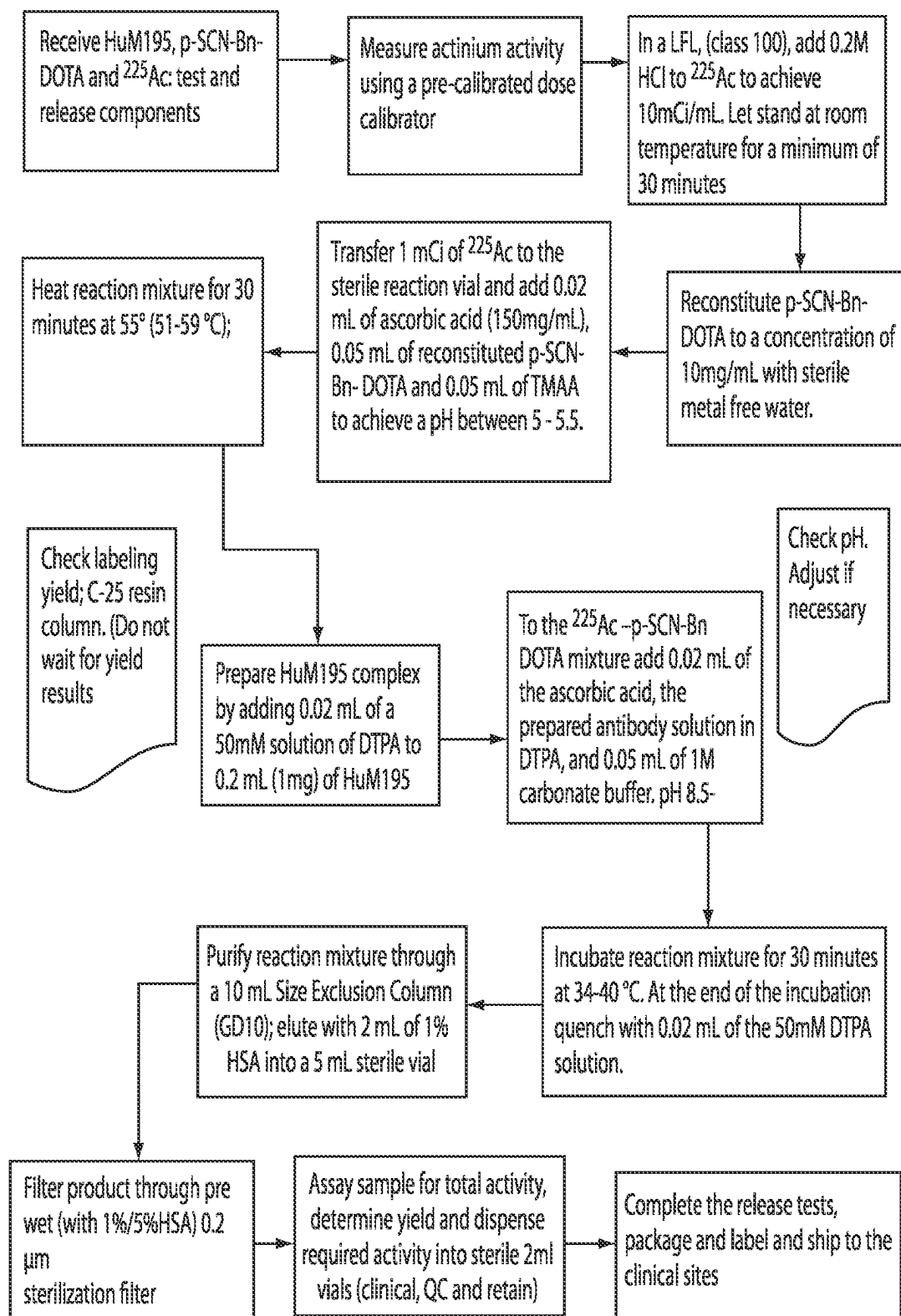
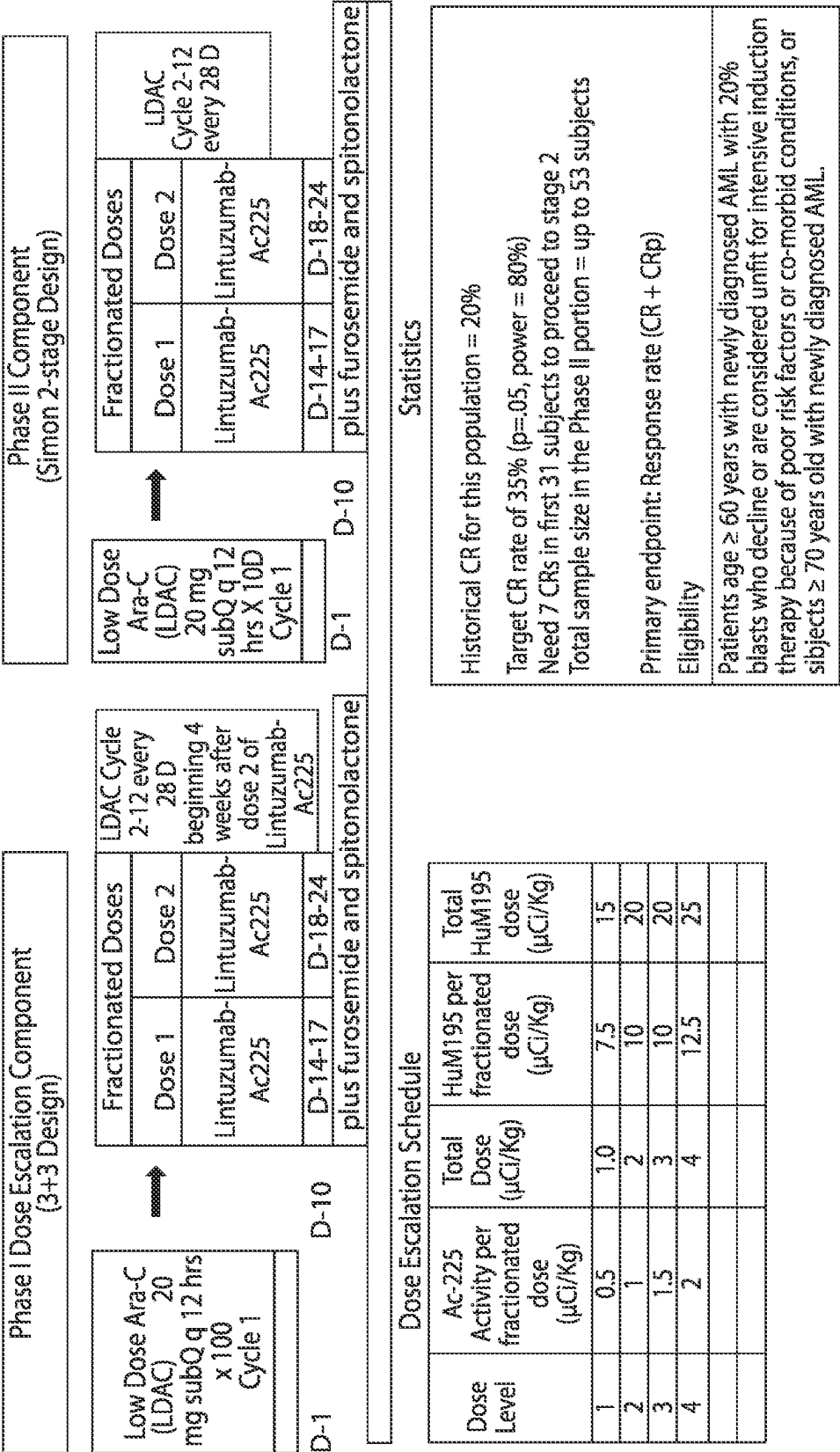


Figure 5

Protocol Schema



METHODS FOR TREATING CANCER USING COMBINATIONS OF EPIGENETIC THERAPIES AND RADIOCONJUGATE TARGETING AGENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. provisional application Ser. No. 63/118,181 filed Nov. 25, 2020 which is hereby incorporated by reference in its entirety.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Oct. 20, 2023, is named REPLACE-MENT_10202023.txt and is 184,125 bytes in size.

FIELD OF THE INVENTION

[0003] The present invention relates to the field of radiotherapeutics.

BACKGROUND OF THE INVENTION

HDAC Inhibitors

[0004] Histone deacetylases (“HDACs”) are a class of enzymes that remove acetyl groups from an epsilon N-acetyl lysine residue on a histone, thereby permitting the histone to wrap DNA more tightly. This action is opposite to the function of histone acetyltransferase. Tightly wrapping DNA is important because DNA expression is regulated by acetylation and de-acetylation.

[0005] Histone tails are normally positively charged due to the amine groups present on their lysine and arginine amino acids. These positively charged residues help the histone tails interact with, and bind to, the negatively charged DNA backbone.

[0006] Acetylation, which occurs normally, changes amines into amides. This change neutralizes the positive charges on the histone, thereby decreasing the ability of the affected histone to bind to DNA. This, in turn, permits chromatin expansion and, thus, genetic transcription to occur. HDACs remove acetyl groups, thus increasing the positive charge of the histone tails, thus increasing the binding affinity between histones and DNA, thus condensing DNA structure, and thus preventing transcription.

[0007] HDACs are involved in a number of biochemical pathways relating, for example, to cell growth and death, and to human diseases such as cancer. As such, HDAC inhibitors are of interest as anti-cancer therapeutics.

[0008] To date, the FDA has approved four HDAC inhibitor drugs (vorinostat, romidepsin, belinostat, and panobinostat) as anticancer agents. Vorinostat and romidepsin are approved for treating patients having cutaneous T-cell lymphoma. Belinostat is approved for treating patients having peripheral T-cell lymphoma. Panobinostat is approved for treating patients having multiple myeloma.

LSD1/KDM1A Inhibitors

[0009] Lysine-specific histone demethylase 1A (LSD1) also known as lysine (K)-specific demethylase 1A (KDM1A), is encoded by the human KDM1A gene and

upregulated in various solid tumor and hematological malignancies LSD1 is a flavin-dependent monoamine oxidase that can demethylate mono- and di-methylated lysines, specifically histone 3, lysines 4 and 9 (H3K4 and H3K9). LSD1 functions as a transcription co-repressor by demethylating H3K4me2/1 and shaping chromatin into a repressive conformation through complexes formed by LSD1 and other numerous proteins. LSD1 also functions as a demethylase of non-histone protein by minimizing the reaction of p53 and 53BP1, a tumor suppressor gene, by removing a methyl group from p53K370me2, thereby suppressing the role of p53. LSD1 upregulation stimulates tumorigenesis by regulating chromatin remodeling and aggregation. Further, LSD1 upregulation can affect the cell cycle of cancer cells resulting in the inhibition of the p53.

Radiopharmaceuticals

[0010] The term “radiopharmaceutical” (also called a “radioisotope-labeled agent”) refers to a source of ionizing radiation (e.g., alpha and beta particles, and gamma rays) linked to a targeting agent such as an antibody, an antigen-binding antibody fragment, peptide, or a non-antibody scaffold. A source of ionizing radiation linked to an antibody is referred to as an “antibody radioconjugate” (ARC). The antibody portion of an ARC may, for example, be an intact (full-length) antibody or an antigen-binding antibody fragment.

[0011] Actinium-225 (Ac-225) is an ideal source of radiation for such purpose. Ac-225 (Linear Energy Transfer=6.83 MeV; $T_{1/2}$ =10 days; path length 40-80 μ m) causes clustered DNA lesions including single-strand breaks (SSBs) and double-strand breaks (DSBs). Even one alpha emission may be lethal to a tumor cell. Importantly, the radiopharmaceutical (e.g., ARC), once bound to a tumor cell, does not need to enter the cell to kill it. So, the process for effecting cell killing is far simpler than that required for an antibody-drug (chemo) conjugate. In addition, because Ac-225 can emit four alpha particles as it decays over its 10-day half-life, a bound radiopharmaceutical (e.g., ARC) can kill not only the target cell but also adjacent unbound tumor cells, including those that may be target antigen-negative. Importantly, the short path length of an alpha emitter limits the field of damage to immediately adjacent cells (i.e., as few as 2-6 cell diameters). As a result, normal tissue is significantly spared.

SUMMARY OF THE INVENTION

[0012] In one aspect, the invention provides a method for treating a subject afflicted with cancer, including administering to the subject (i) an epigenetic therapy such as one or both of an HDAC inhibitor and an LSD1 inhibitor, in conjunction with (ii) a radioisotope-labeled agent that targets cancer cells in the subject, wherein the amounts of the HDAC inhibitor and labeled agent, when administered in conjunction with one another, are therapeutically effective.

[0013] In another aspect, the invention provides a method for treating a human subject afflicted with acute myeloid leukemia, including administering to the subject (i) an epigenetic therapy such as one or both of an HDAC inhibitor such as vorinostat, romidepsin, belinostat, and/or panobinostat in conjunction and an LSD1 inhibitor such as selicicicemstat, TCP (tranylcypromine), ORY-1001 (iadademstat), GSK2879552 (GSK), INCB059872, IMG-7289 (bomedemstat), ORY-2001 (vafidemstat), and/or CC-90011; with (ii) a

radiolabeled CD33 targeting agent such as ^{225}Ac -labeled HuM195, wherein the amounts of the HDAC inhibitor and the radiolabeled CD33 targeting agent, when administered in conjunction with one another, are therapeutically effective.

[0014] In a further aspect, the invention provides a method for inducing the death of a cancer cell, including contacting the cell with (i) one or both of an HDAC inhibitor and an LSD1 inhibitor in conjunction with (ii) a radioisotope-labeled agent that targets the cancer cell, wherein the amounts of HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled targeting agent, when contacted with the cell in conjunction with one another, are effective to induce the cell's death.

[0015] In still another aspect, the invention provides a method for inducing the death of an acute myeloid leukemia (AML) cell, including contacting the cell with (i) one or both of an HDAC inhibitor such as vorinostat, romidepsin, belinostat, or panobinostat and an LSD1 inhibitor such as seclidemstat, TCP (tranilcyproline), ORY-1001 (iadademstat), GSK2879552 (GSK), INCB059872, IMG-7289 (bomedemstat), ORY-2001 (vafidemstat), and/or CC-90011 in conjunction with (ii) a radiolabeled CD33 targeting agent such as ^{225}Ac -labeled HuM195, wherein the amounts of HDAC inhibitor and/or LSD1 inhibitor, and ^{225}Ac -labeled HuM195, when contacted with the cell in conjunction with one another, are effective to induce the cell's death.

[0016] In still further aspects, this invention provides two articles of manufacture. The first article of manufacture includes (i) one or both of an HDAC inhibitor (e.g., vorinostat, romidepsin, belinostat, or panobinostat) and an LSD1 inhibitor (e.g., seclidemstat, TCP (tranilcyproline), ORY-1001 (iadademstat), GSK2879552 (GSK), INCB059872, IMG-7289 (bomedemstat), ORY-2001 (vafidemstat), and/or CC-90011) and (ii) a label instructing the user (e.g., a healthcare provider) to treat a subject (e.g., a human) afflicted with a hematologic malignancy (e.g., acute myeloid leukemia or myelodysplastic syndrome) by administering the HDAC inhibitor and/or LSD1 inhibitor to the subject in conjunction with a radioisotope-labeled agent (e.g., a radiolabeled CD33 targeting agent such as ^{225}Ac -labeled HuM195) that targets cancer cells in the subject, wherein the amounts of the HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled targeting agent, when administered in conjunction with one another, are therapeutically effective.

[0017] The second article of manufacture provided includes (i) a radioisotope-labeled agent (e.g., ^{225}Ac -labeled HuM195) that targets cancer cells and (ii) a label instructing the user (e.g., a healthcare provider) to treat a subject (e.g., a human) afflicted with a hematologic malignancy (e.g., acute myeloid leukemia or myelodysplastic syndrome) by administering the labeled agent to the subject in conjunction with an HDAC inhibitor (e.g., vorinostat, romidepsin, belinostat, and/or panobinostat) and/or an LSD1 inhibitor (e.g., seclidemstat, TCP (tranilcyproline), ORY-1001 (iadademstat), GSK2879552 (GSK), INCB059872, IMG-7289 (bomedemstat), ORY-2001 (vafidemstat), and/or CC-90011), wherein the amounts of the HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled agent, when administered in conjunction with one another, are therapeutically effective. Additional features, advantages, and embodiments of the invention may be set forth or apparent from consideration of the following detailed description, drawings if

any, and claims. Moreover, it is to be understood that both the foregoing summary of the invention and the following detailed description are exemplary and intended to provide further explanation without limiting the scope of the invention as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 shows a schematic diagram of the expression plasmids for HuM195. The humanized VL and VH exons of HuM195 are flanked by XbaI sites. The VL exon was inserted into mammalian expression vector pVk, and the VH exon into pVg1 (Co, et al., J. Immunol. 148:1149-1154, 1992).

[0019] FIG. 2 shows the complete sequence of the HuM195 light chain gene cloned in pVk between the XbaI and BamHI sites. The nucleotide number indicates its position in the plasmid pVk-HuM195. The VL and CK exons are translated in single letter code; the dot indicates the translation termination codon. The mature light chain begins at the double-underlined aspartic acid (D). The intron sequence is in italics. The polyA signal is underlined. The full-length nucleotide sequence and the encoded amino acid sequence are disclosed as SEQ ID NO:114 and SEQ ID NO:115 respectively.

[0020] FIG. 3 shows the complete sequence of the HuM195 heavy chain gene cloned in pVg1 between the XbaI and BamHI sites. The nucleotide number indicates its position in the plasmid pVg1-HuM195. The VH, CH1, H, CH2 and CH3 exons are translated in single letter code; the dot indicates the translation termination codon. The mature heavy chain begins at the double-underlined glutamine (Q). The intron sequences are in italics. The polyA signal is underlined. The full-length nucleotide sequence and the encoded amino acid sequence are disclosed as SEQ ID NO:116 and SEQ ID NO:117 respectively.

[0021] FIG. 4 shows the structure of ^{225}Ac -Lintuzumab (^{225}Ac -HuM195).

[0022] FIG. 5 shows a flowchart for the production of ^{225}Ac -HuM195.

[0023] FIG. 6 shows a dosing protocol for ^{225}Ac -Lintuzumab (^{225}Ac -HuM195) treatment of AML, without HDACi.

DETAILED DESCRIPTION OF THE INVENTION

[0024] The invention provides combination therapies and related methods for treating cancer in a mammalian subject, such as a human patient, using one or both of an HDAC inhibitor and an LSD1 inhibitor, and a radioisotope-labeled cancer targeting-agent, such as but not limited to ^{225}Ac -labeled HuM195.

Definitions

[0025] In this disclosure, certain terms are used which shall have the meanings set forth as follows.

[0026] As used herein, "administer", with respect to an agent, means to deliver the agent to a subject's body via any known method. Specific modes of administration include, without limitation, intravenous, intramuscular, oral, sublingual, transdermal, subcutaneous, intraperitoneal, intrathecal and intra-tumoral administration. HDAC inhibitors and LSD1 inhibitors may, for example, be administered orally or

via injection, and radiolabeled targeting agents, such as antibody radioconjugates, may, for example, be administered intravenously.

[0027] In addition, in this invention, the various HDAC inhibitors, antibodies and other antigen-targeting agents used can be formulated using one or more routinely used pharmaceutically acceptable carriers/excipients. Such carriers are well known to those skilled in the art. For example, injectable drug delivery systems include solutions, suspensions, gels, microspheres and polymeric injectables, and can include excipients such as solubility-altering agents (e.g., ethanol, propylene glycol and sucrose) and polymers (e.g., polycaprylactones and PLGA's). Likewise, oral delivery systems include, for example, tablets and capsules. These can contain excipients such as binders (e.g., hydroxypropylmethyl-cellulose, polyvinyl pyrrolidone, other cellulosic materials and starch), diluents (e.g., lactose and other sugars, starch, dicalcium phosphate and cellulosic materials), disintegrating agents (e.g., starch polymers and cellulosic materials) and lubricating agents (e.g., stearates and talc).

[0028] As used herein, the term "agent", whether in reference to an HDAC inhibitor or a radioisotope-labeled agent, can be any type of compound or composition useful for such purpose. Types of agents include, without limitation, antibodies, other protein-based drugs, peptides, nucleic acids, carbohydrates and small molecules drugs.

[0029] As used herein, the term "antibody" includes, without limitation, (a) an immunoglobulin molecule including two heavy chains and two light chains and which recognizes an antigen; (b) polyclonal and monoclonal immunoglobulin molecules; (c) monovalent and divalent antigen binding fragments thereof such as Fab and Fab₂ fragments, and (d) bi-specific forms and multi-specific thereof. Immunoglobulin molecules may derive from any of the commonly known classes, including but not limited to IgA, secretory IgA, IgG and IgM. IgG subclasses are also well known to those in the art and include, but are not limited to, human IgG1, IgG2, IgG3 and IgG4. Antibodies can be both naturally occurring and non-naturally occurring. Furthermore, without limitation antibodies include chimeric antibodies, wholly synthetic antibodies, single chain antibodies, and fragments thereof, nanobodies, minibodies, affibodies, and scFv molecules. Antibodies may be human, humanized, or nonhuman.

[0030] As used herein, an "anti-CD33 antibody" is an antibody such as a monoclonal antibody, that binds to any available epitope of CD33 such as human CD33. In one embodiment, the anti-CD33 antibody binds to the epitope recognized by monoclonal antibody HuM195.

[0031] A "hematologic malignancy", also known as a blood cancer, is a cancer that originates in blood-forming tissue, such as the bone marrow or other cells of the immune system. Hematologic malignancies that may be treated according to the invention include, without limitation, leukemias, lymphomas, and related disorders such as such as acute myeloid leukemia (AML), acute promyelocytic leukemia, acute lymphoblastic leukemia (ALL), acute mixed lineage leukemia, chronic myeloid leukemia (CML), chronic lymphocytic leukemia (CLL), hairy cell leukemia and large granular lymphocytic leukemia, myelodysplastic syndrome (MDS), myeloproliferative disorders (polycythemia vera, essential thrombocytosis, primary myelofibrosis and chronic myeloid, leukemia, cutaneous T-cell lymphoma (CTCL), peripheral T-cell lymphoma

(CTCL), and multiple myeloma. Hematologic malignancies are characterized by hematologic malignancy-associated antigens. Such antigen may, for example, be a protein and/or carbohydrate marker found exclusively or predominantly on the surface of a cancer cell associated with that particular malignancy. Examples of hematologic malignancy-associated antigens that may be targeted include, without limitation, CD20, CD33, CD38, CD45, CD52, CD123 and CD319.

[0032] The antibody "HuM195" (also known as lintuzumab) is known, as are methods of making it. Likewise, methods of labeling HuM195 with ²²⁵Ac are known. These methods are exemplified, for example, in Scheinberg, et al., U.S. Pat. No. 6,683,162. This information is also exemplified in the examples and figures below.

[0033] As used herein, administering to a subject an HDAC inhibitor and/or an LSD1 inhibitor "in conjunction with" a radioisotope-labeled agent that targets cancer cells in the subject means administering the HDAC inhibitor before, during and/or after administration of the radiolabeled agent. This administration includes, without limitation, the following scenarios: (i) the epigenetic therapy(ies) is administered first, and the labeled agent is administered second; (ii) the H epigenetic therapy(ies) is administered concurrently with the labeled agent (e.g., the HDAC inhibitor is administered orally once per day for n days, and the labeled agent is administered intravenously in a single dose on one of days 2 through n-1 of the epigenetic therapy(ies) regimen); (iii) the epigenetic therapy(ies) is administered concurrently with the labeled agent (e.g., the epigenetic therapy(ies) is administered orally for a duration of greater than one month (e.g., orally once per day for 35 days, 42 days, 49 days, or a longer period during which the cancer being treated does not progress and during which the epigenetic therapy(ies) does not cause unacceptable toxicity), and the labeled agent is administered intravenously in a single dose on a day within the first month of the epigenetic therapy(ies) regimen); and (iv) the labeled agent is administered first (e.g., intravenously in a single dose or a plurality of doses over a period of weeks), and the epigenetic therapy(ies) is administered second (e.g., orally once per day for 21 days, 28 days, 35 days, 42 days, 49 days, or a longer period during which the cancer being treated does not progress and during which the epigenetic therapy(ies) does not cause unacceptable toxicity. Additional permutations are provided below in the Examples section. Those skilled in the art will recognize that when two or more therapeutic agents are administered in sufficient temporal proximity, the activity and/or the effect of each agent can temporally overlap with the other(s) in the treated subject.

[0034] As used herein, "HDAC" (i.e., histone deacetylase) includes, without limitation, all HDAC classes, such as class I, class IIA, class IIB, and class IV. Also included, without limitation, are all members of these HDAC classes, such as HDAC1, HDAC2, HDAC3, HDAC4, HDAC5, HDAC6, HDAC7, HDAC8, HDAC9, HDAC10, and HDAC11.

[0035] As used herein, "HDAC inhibitor", also referred to as HDACi, includes, without limitation, FDA-approved HDAC inhibitors and experimental HDAC inhibitors. Without limitation, HDAC inhibitors that may be used in the various embodiments of this invention include vorinostat, romidepsin, belinostat, panobinostat, tucidinostat, pracinostat, CG 200745, entinostat, tacedinaline, mocetinostat, abex-

inostat, givinostat, resminostat, quisinostat, mocetinostat, and rocilinostat, as well others disclosed herein.

[0036] As used herein, “inducing” the death of a cancer cell includes, without limitation, (i) directly causing the cell’s death, and (ii) indirectly causing the cell’s death (e.g., by triggering a cascade of biochemical events that ultimately leads to the cell’s death).

[0037] As used herein, an “LSD1 inhibitor” includes any agent that inhibits the lysine-specific histone demethylase activity of LSD1. Without limitation, LSD1 inhibitors that may be used in the various embodiments of the invention include seclidemstat, TCP (tranylcypromine), ORY-1001 (iadademstat), GSK2879552 (GSK), INCB059872 (Imago BioSciences), IMG-7289 (bomedemstat; Imago BioSciences), ORY-2001 (vafidemstat), and CC-90011 (Celgene), as well as others disclosed herein.

[0038] As used herein, “myelodysplastic syndrome”, also known as MDS, is a disorder that occurs when blood-forming cells in the bone marrow become abnormal (e.g., dysplastic), leading to low numbers of one or more types of blood cells. An example of a myelodysplastic syndrome cell is a mutated myeloid precursor cell lacking the ability to differentiate.

[0039] As used herein, a “radioisotope” can be an alpha-emitting isotope, a beta-emitting isotope, and/or a gamma-emitting isotope. Accordingly, as used herein, “radioisotope” is synonymous with “radionuclide.” Examples of radioisotopes that may be used in connection with the various embodiment of the invention include without limitation the following: ^{90}Y , ^{89}Sr , ^{153}Sm , ^{32}P , ^{225}Ac , ^{213}Bi , ^{213}Po , ^{211}At , ^{212}Bi , ^{213}Bi , ^{223}Ra , ^{227}Th , ^{149}Tb , ^{131}I , ^{137}Cs , ^{212}Pb , ^{103}Pd , ^{166}Ho , ^{186}Re , ^{188}Re , ^{67}Cu , ^{199}Au , ^{105}Rh , ^{211}As , and ^{177}Lu . Thus, the radiolabeled antibodies envisioned in this invention include, without limitation, ^{90}Y -HuM195, ^{89}Sr -HuM195, ^{153}Sm -HuM195, ^{32}P -HuM195, ^{225}Ac -HuM195, ^{213}Bi -HuM195, ^{213}Po -HuM195, ^{211}At -HuM195, ^{212}Bi -HuM195, ^{213}Bi -HuM195, ^{223}Ra -HuM195, ^{227}Th -HuM195, ^{149}Tb -HuM195, ^{131}I -HuM195, ^{137}Cs -HuM195, ^{212}Pb -HuM195, ^{103}Pd -HuM195, and ^{177}Lu -HuM195. Each of the antibody radioconjugates above is also envisioned, mutatis mutandis, and without limitation, for each of the following antibodies: alemtuzumab (Campath®), ibritumomab tiuxetan (Zevalin®), brentuximab vedotin (Adcetris®), trastuzumab (Herceptin®), trastuzumab emtansine (Kadcyla®), gemtuzumab ozogamicin (Mylotarg®), B1836858, BC8/apamistamab, daratumumab (Darzalex®), felzartamab/MOR202, isatuximab/SAR650984 (Sarcilar®), TAK-169, AV-203, CDX-3379, HMBD 001, patritumab, seribantumab, elgantumab/LJM716, lumretuzumab/RG7116/RO5479599, GSK2849330, mapatumumab, tigatuzumab/CS-1008, drozitumab, lexatumumab, conatumumab, MEDI10641, PF-06263507/A1mcMMAF, Anti-5T4 SYD1875, ASN004, AVA-020, Tb535, Hu3S193, B3, IGN311, BR96/BMS-182248-01, girentuximab, BAY 794620, BAY 2701439, MEDI-4276, XMT-1522, zolbetuximab, TST001, A315, labetuzumab, sacituzumab govitecan-hziy (Trodelvy™), SAR566658, DMOT4039, anetumab ravtansine, BAY 2287411, and amatuximab. Methods for affixing a radioiso-

tope to an antibody or other targeting agent (i.e., labeling the antibody or other targeting agent with a radioisotope) are well known in the art.

[0040] As used herein, the term “subject” includes, without limitation, a mammal such as a human, a non-human primate, a dog, a cat, a horse, a sheep, a goat, a cow, a rabbit, a pig, a rat and a mouse. Where the subject is human, the subject can be of any age. For example, the subject can be 60 years or older, 65 or older, 70 or older, 75 or older, 80 or older, 85 or older, or 90 or older. Alternatively, the subject can be 50 years or younger, 45 or younger, 40 or younger, 35 or younger, 30 or younger, 25 or younger, or 20 or younger. For a human subject afflicted with AML or MDS, the subject can be newly diagnosed, or relapsed and/or refractory, or in remission.

[0041] As used herein, “targeting agent” means a synthetic or biological molecule that preferentially and/or specifically binds a target molecule, such as a cancer-associated antigen or otherwise preferentially accumulates in cancer cells or and/or tumor tissue, in a subject and which may, for example, be radiolabeled for the delivery of therapeutic amounts of ionizing radiation to cancer cells within the subject. Without limitation, suitable types of targeting agents for use in the invention include antibodies such as IgG₁, IgG₂, IgG₃ and IgG₄, antigen-binding antibody fragments such as Fab and Fab₂, peptides such as GRPR-binding peptides such as RM-2, non-antibody antigen binding scaffolds/antibody mimetics, such as DARPins, and small molecules, such as PSMA-binding small molecules such as DUPA.

[0042] As used herein, an amount of HDAC inhibitor and/or LSD1 inhibitor, and an amount of radioisotope-labeled agent that targets cancer cells in the subject, when administered in conjunction with each other, are “therapeutically effective” if the subject is treated.

[0043] As used herein, “treating” a subject afflicted with a disorder shall include, without limitation, (i) slowing, stopping or reversing the disorder’s progression, (ii) slowing, stopping or reversing the progression of the disorder’s symptoms, (iii) reducing the likelihood of the disorder’s recurrence, and/or (iv) reducing the likelihood that the disorder’s symptoms will recur. In one embodiment, treating a subject afflicted with a disorder means (i) reversing the disorder’s progression, ideally to the point of eliminating the disorder, and/or (ii) reversing the progression of the disorder’s symptoms, ideally to the point of eliminating the symptoms, and/or (iii) reducing or eliminating the likelihood of relapse (i.e., consolidation, which is a common goal of post-remission therapy for AML and, ideally, results in the destruction of any remaining leukemia cells).

[0044] The treatment of hematologic malignancy, such as the treatment of AML, can be measured according to a number of clinical endpoints. These include, without limitation, survival time (such as weeks, months or years of improved survival time, e.g., one, two or more months’ of additional survival time), and response status (such as complete remission (CR), complete remission with incomplete platelet recovery (CRp), complete remission with incom-

plete peripheral blood recovery (CRi), morphologic leukemia-free state (MLFS) and partial remission (PR)).

[0045] In one embodiment, treatment of hematologic malignancy, such as the treatment of AML, can be measured in terms of remission. Included here are the following non-limiting examples. (1) Morphologic complete remission (“CR”): ANC $\geq 1,000/\text{mcl}$, platelet count $100,000/\text{mcl}$, $<5\%$ bone marrow blasts, no Auer rods, no evidence of extramedullary disease. (No requirements for marrow cellularity, hemoglobin concentration). (2) Morphologic complete remission with incomplete blood count recovery (“CRi”): Same as CR but ANC may be $<1,000/\text{mcl}$ and/or platelet count $<100,000/\text{mcl}$. (3) Partial remission (PR): ANC $1,000/\text{mcl}$, platelet count $>100,000/\text{mcl}$, and at least a 50% decrease in the percentage of marrow aspirate blasts to 5-25%, or marrow blasts $<5\%$ with persistent Auer rods. These criteria and others are known, and are described, for example, in SWOG Oncology Research Professional (ORP) Manual Volume I, Chapter 11A, Leukemia (2014).

Embodiments of the Invention

[0046] This invention combines the use of different agents to treat cancers and proliferative disorders such as pre-cancerous conditions. Here, one or both of an HDAC inhibitor and LSD1 inhibitor, are administered in conjunction with a radiolabeled targeting agent such as antibody radioconjugate to more effectively treat patients with solid tumors or hematologic malignancies.

[0047] Specifically, this invention provides a method for treating a mammalian subject afflicted with cancer or proliferative disorder, such as human patient, including administering to the subject (i) an HDAC inhibitor and/or an LSD1 inhibitor, in conjunction with (ii) a radioisotope-labeled agent that targets cancer cells in the subject, wherein the amounts of the HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled agent, when administered in conjunction with one another, are therapeutically effective.

[0048] The cancer treated may, for example, be a solid tumor, for example, breast cancer, HER2-positive breast cancer, HER2-negative breast cancer, HER3-positive breast cancer, tamoxifen-resistant breast cancer, triple negative

breast cancer (TNBC), ovarian cancer, prostate cancer, castration-resistant prostate cancer (CRPC) lung cancer, small cell lung cancer (SCLC), non-small cell lung cancer (NSCLC), squamous cell carcinoma of the head and neck, gastric cancer, colorectal cancer, pancreatic cancer, brain cancer (e.g., glioblastoma and neuroblastoma), liver cancer (e.g. hepatocellular carcinoma or cholangiocarcinoma), sarcoma or melanoma. The cancer may be metastatic or non-metastatic.

[0049] The cancer treated may, for example, be a hematologic malignancy, for example, a CD33-expressing hematological malignancy, a myeloid malignancy, multiple myeloma, acute myeloid leukemia, or myelodysplastic syndrome.

[0050] The radioisotope-labeled targeting agent may, for example, be a radioisotope labeled CD33 targeting agent such as a radiolabeled anti-CD33 monoclonal antibody, for example, radioisotope labeled lintuzumab (HuM195), gemtuzumab, or vadastuximab, or a CD33-binding fragment of one of the aforementioned antibodies.

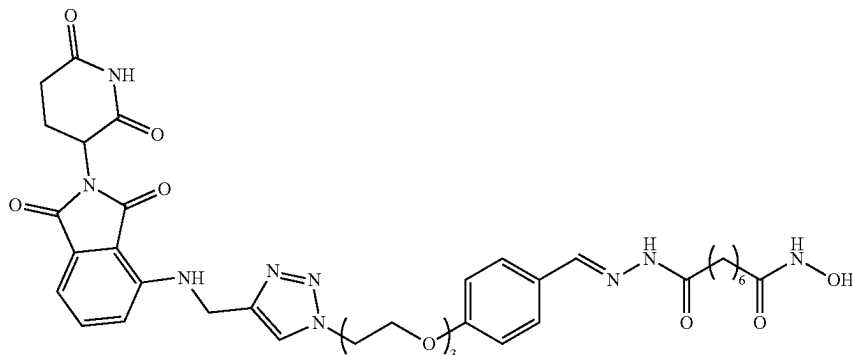
[0051] The radioisotope may, for example, be selected from the group consisting of ^{90}Y , ^{89}Sr , ^{153}Sm , ^{32}P , ^{225}Ac , ^{213}Bi , ^{213}Po , ^{211}At , ^{212}Bi , ^{213}Bi , ^{223}Ra , ^{227}Th , ^{149}Tb , ^{131}I , ^{137}Cs , ^{212}Pb , ^{103}Pd , ^{166}Ho , ^{186}Re , ^{188}Re , ^{67}Cu , ^{199}Au , ^{105}Rh , ^{211}As , and ^{177}Lu .

[0052] In a specific embodiment, the radioisotope-labeled agent is ^{225}Ac -labeled HuM195 (also referred to herein as ^{225}Ac -HuM195).

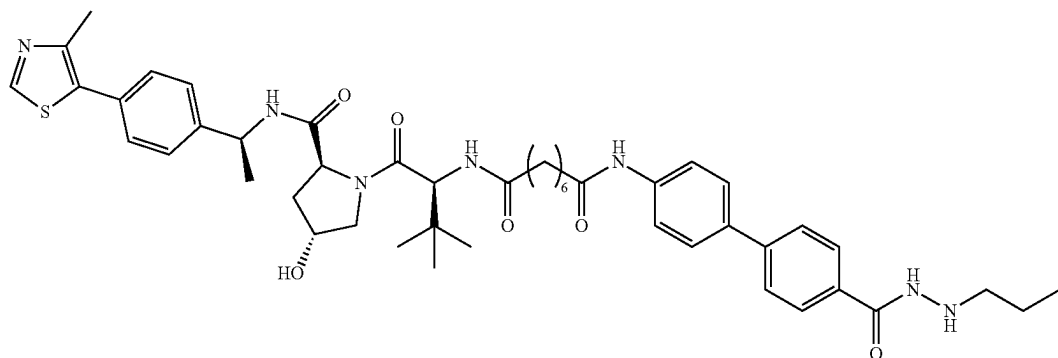
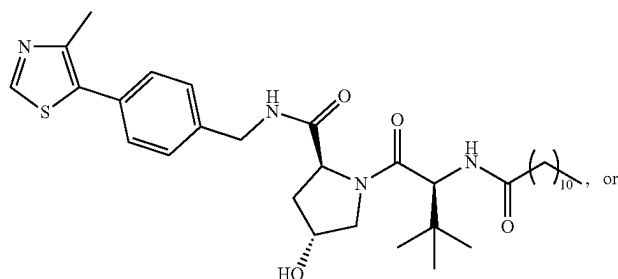
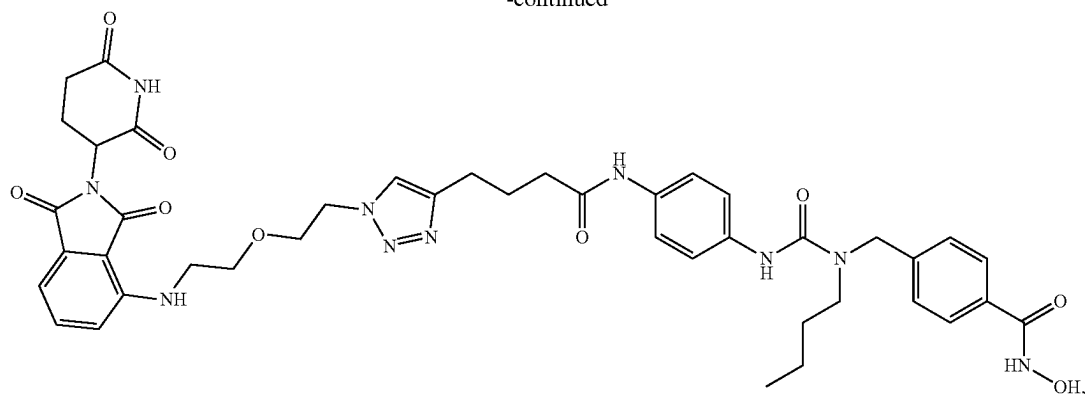
[0053] In another specific embodiment, the radioisotope-labeled agent is ^{177}Lu -labeled HuM195 (^{177}Lu -HuM195).

[0054] In the various embodiments of this invention involving an HDAC inhibitor, the HDAC inhibitor may be any agent performing that function. For example, the HDAC inhibitor may be vorinostat, romidepsin, belinostat, panobinostat, tucidinosat, pracinostat, CG 200745, entinostat, tacedinaline, mocetinostat, abexinostat, givinostat, resminostat, quisinostat, mocetinostat, rocilinostat, JB-802 (Jubilant Therapeutics Inc.), or JBI-295 (Jubilant Therapeutics Inc.).

[0055] An HDAC inhibitor that may be used may, for example, be a PROTAC (proteolysis-targeting chimera) agent such as any of the following compounds or a pharmaceutically acceptable salt thereof:



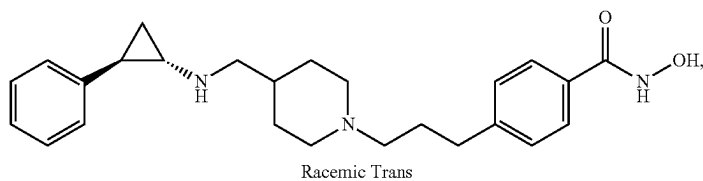
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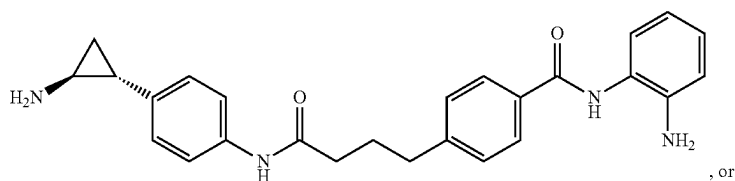
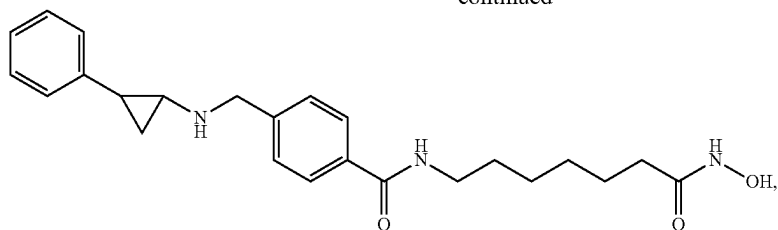
[0056] In the various embodiments of this invention involving an LSD1 inhibitor, the LSD1 inhibitor may be any agent performing that function, such as seclidemstat, TCP (tranylcypromine), ORY-1001 (iadademstat), GSK2879552 (GSK), INCB059872 (Imago BioSciences), IMG-7289 (Bo-medemstat; Imago BioSciences), ORY-2001 (Vafidemstat), and CC-90011 (Celgene).

[0057] An HDAC inhibitor that may be used may, for example, be an agent having one or more additional activities.

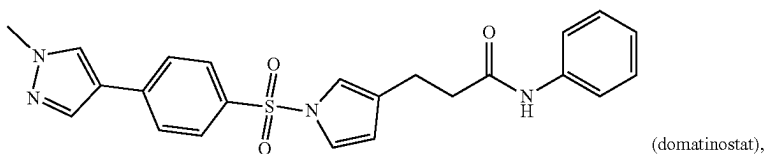
[0058] An HDAC inhibitor that may be used may also be a LSD1 inhibitor, such as any of the following dual inhibitors or a pharmaceutically acceptable salt thereof:



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, or



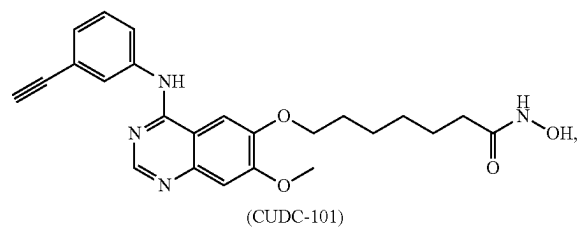
(domatinostat),

or

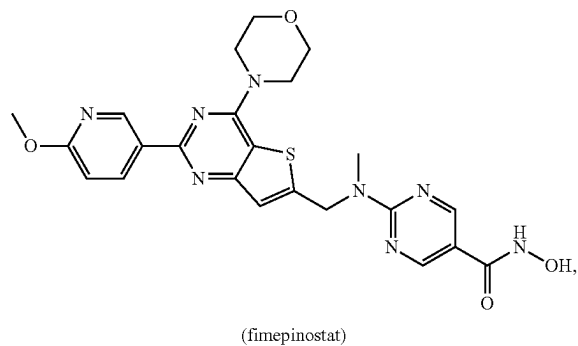
a dual HDAC/EGFR inhibitor, or

[0059] any of the dual LSD1/HDAC inhibitors disclosed in U.S. Pub. No. 20200308110.

[0060] An HDAC inhibitor that may be used may also be a kinase inhibitor, such as any of the following dual inhibitors or a pharmaceutically acceptable salt thereof:



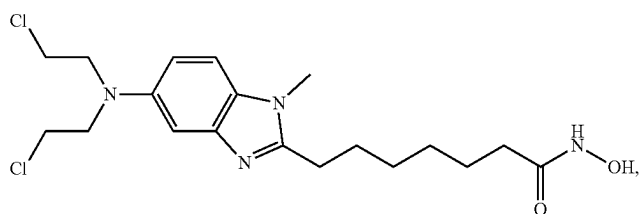
(CUDC-101)



(fimepinostat)

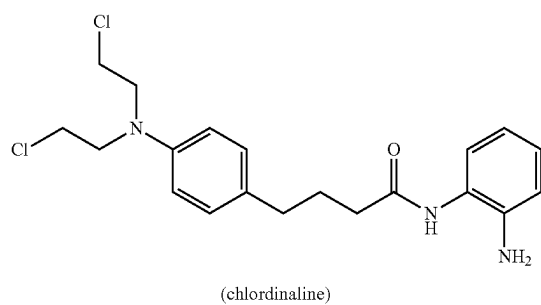
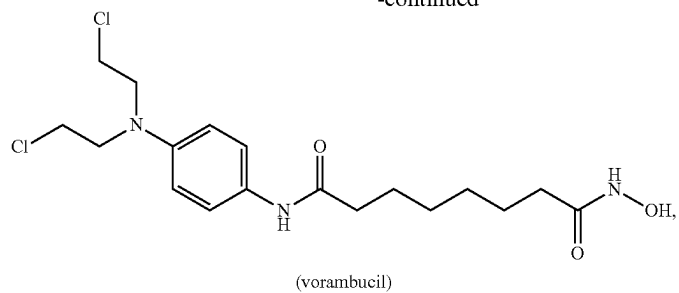
a dual PI3K/HDAC inhibitor.

[0061] An HDAC inhibitor that may be used may also have DNA-alkylating activity, such as any of the following dual activity agents or a pharmaceutically acceptable salt thereof:

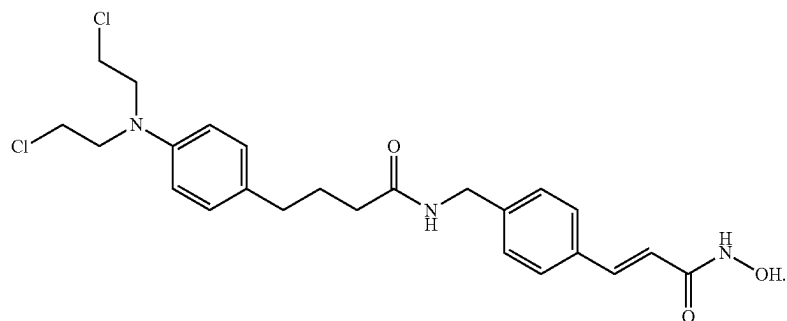


(tinostamustine)

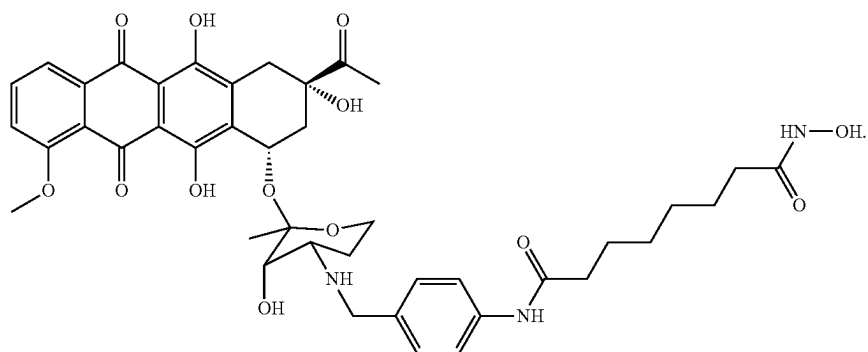
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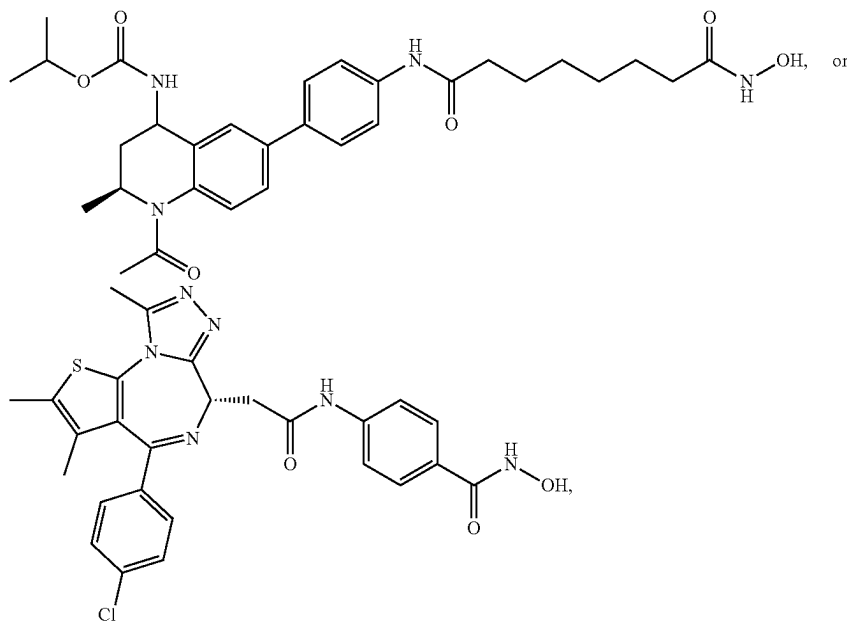


[0062] An HDAC inhibitor that may be used may also be a topoisomerase inhibitor, such as the following dual inhibitor or a pharmaceutically acceptable salt thereof:

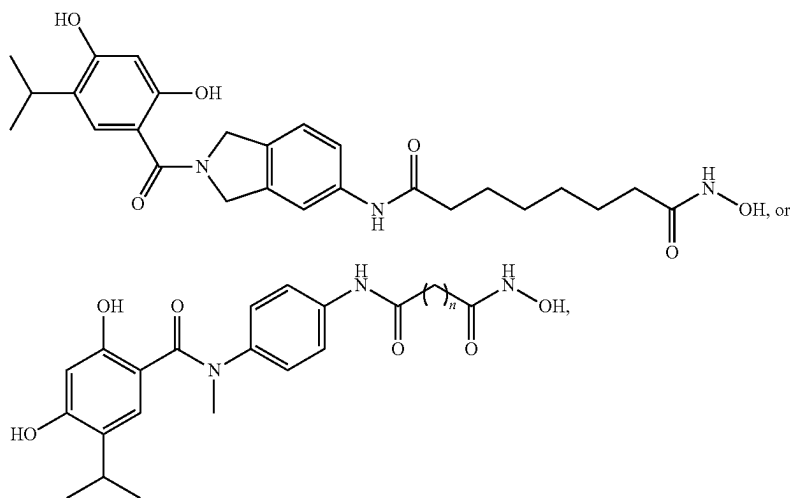


[0063] An HDAC inhibitor that may be used may also be a BET protein (BRD2, BRD3, BRD4, BRDT) inhibitor, such as any of the following dual inhibitors or a pharmaceutically acceptable salt thereof:

(DUAL946)



[0064] An HDAC inhibitor that may be used may also be a heat shock protein 90 (Hsp90) inhibitor, such as any of the following dual inhibitors or a pharmaceutically acceptable salt thereof:



for example, where n is 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10.

[0065] In the various embodiments of this invention, the radioisotope-labeled agent may be any agent, such as any known in the art, that targets cancer cells, for example, one approved by the FDA.

[0066] The radioisotope labeled targeting agent may, for example, preferentially and/or specifically bind to one or more of the following antigens, such as to the human forms of the following antigens: mesothelin, TSHR, CD19, CD123, CD22, CD30, CD45, CD171, CD138, CS-1, CLL-1, GD2, GD3, B-cell maturation antigen (BCMA), Tn Ag,

prostate specific membrane antigen (PSMA), GRPR, ROR1, FLT3, TROP2, T-cell receptor gamma (TCR γ) chain alternate reading frame protein (TRAP), fibroblast activation protein (FAP), calreticulin, phosphatidylserine, GRP78 (BiP), TAG72, CD38, CD44v6, CEA, EPCAM, B7H3, KIT, IL-13Ra2, interleukin-11 receptor α (IL-11Ra), PSCA, PRSS21, VEGFR2, LewisY, CD24, platelet-derived growth factor receptor-beta (PDGFR-beta), SSEA-4, CD20, Folate receptor alpha (FRA), ERBB2 (Her2/neu), Her3, MUC1, epidermal growth factor receptor (EGFR), EGFRvIII, NCAM, Prostase, PAP, ELF2M, Ephrin B2, IGF-I receptor, CAIX, LMP2, gp100, bcr-abl, tyrosinase, EphA2, Fucosyl GM1, sLe, GM3, DR5, 5T4, TGS5, HMWMAA, o-acetyl-GD2, Folate receptor beta, TEM1/CD248, TEM7R, CLDN6, GPRC5D, CXORF61, CD97, CD 179a, ALK, Polysialic acid, PLAC1, GloboH, NY-BR-1, UPK2, HAVCR1, ADRB3, PANX3, GPR20, LY6K, OR51E2, TARP, WT1, NY-ESO-1, LAGE-1a, MAGE-A1, legumain, HPV E6,E7, MAGE A1, MAGEA3, MAGEA3/A6, ETV6-AML, sperm protein 17, XAGE1, Tie 2, MAD-CT-1, MAD-CT-2, Fos-related antigen 1, prostein, survivin and telomerase, PCTA-1/Galectin 8, KRAS, MelanA/MART1, Ras mutant, hTERT, sarcoma translocation breakpoints, ML-IAP, ERG (TMPRSS2 ETS fusion gene), NA17, PAX3, Androgen receptor, Cyclin B 1, MYCN, RhoC, TRP-2, CYP1B 1, BORIS, SART3, PAX5, OY-TES 1, LCK, AKAP-4, SSX2, RAGE-1, human telomerase reverse transcriptase, RU1, RU2, intestinal carboxyl esterase, mut hsp70-2, CD79a, CD79b, CD72, LAIR1, FCAR, LILRA2, CD300LF, CLEC12A, BST2, EMR2, LY75, GPC3, FCRL5, GPA7, and IGLL1.

[0067] Exemplary DR5 (death receptor 5) targeting agents that may be radiolabeled for use in the invention include the monoclonal anti-DR5 antibodies mapatumumab, conatumumab, lexatumumab, tigatuzumab, drozitumab, and LB-135. Such DR5 targeting agents may, for example, be used in combination with an HDAC inhibitor and/or LSD1 inhibitor for the treatment of ovarian, breast, cervical prostate, gastric, bladder, lung, melanoma, colorectal and squamous cell carcinoma cancers and any of the cancers disclosed herein.

[0068] Exemplary 5T4 (Trophoblast glycoprotein (TBPG)) targeting agents that may be radiolabeled for use in the invention include the anti-5T4 monoclonal antibodies MED10641, ALG-APV-527, Tb535, H6-DM5, and ZV0508, as well as Naptumomab estafenatox or the Fab portion thereof. Such 5T4 targeting agents may, for example, be used in combination with an HDAC inhibitor and/or LSD1 inhibitor for the treatment of ovarian, head and neck, breast, prostate, gastric, bladder, lung, melanoma, colorectal and squamous cell carcinoma cancers and any of the cancers disclosed herein.

[0069] Exemplary HER2 (ERBB2) targeting agents that may be radiolabeled for use in the invention include the monoclonal antibodies trastuzumab and pertuzumab. Applicants have successfully conjugated Trastuzumab with p-SCN-DOTA and radiolabeled the composition with ^{225}Ac or ^{177}Lu . The anti-HER2 antibody may, for example, also be a multi-specific antibody, such as bispecific antibody, against any available epitope of HER3/HER2 such as MM-111 and MM-141/Istiratumab from Merrimack Pharmaceuticals, MCLA-128 from Merus NV, and MEHD7945A/Duligotumab from Genentech. Radiolabeled HER2 targeting agents may, for example, be used in com-

bination with an HDAC inhibitor and/or LSD1 inhibitor for treatment of HER2-expressing cancers such as ovarian, breast, metastatic breast, esophageal, lung, cervical, and endometrial cancers including but not limited to those that are both HER2- and HER3-positive.

[0070] In one embodiment, the radioisotope-labeled agent is an antibody that specifically binds to an antigen selected from the group consisting of CD33, CD45, CD38, Her3, DR5 (TRAIL-R2), 5T4, Lewis-Y, CAIX, Her2, Claudin 18.2, CEACAM5, MUC1, and mesothelin, such as the human forms of these antigens.

[0071] In another embodiment, the radioisotope-labeled targeting agent is an antibody selected from the group consisting of alemtuzumab, ibritumomab, ibritumomab tiuxetan, brentuximab, brentuximab vedotin, trastuzumab, trastuzumab emtansine, gemtuzumab, gemtuzumab ozogamicin, lintuzumab, B1836858, BC8/apamistamab, daratumumab, felzartamab/MOR202, isatuximab/SAR650984, TAK-169, AV-203, CDX-3379, HMBD 001, patritumab, seribantumab, elgentumab/LJM716, lumretuzumab/RG7116/RO5479599, GSK2849330, mapatumumab, tigatuzumab/CS-1008, drozitumab, lexatumumab, conatumumab, MED10641, PF-06263507/A1mcMMAF, Anti-5T4 SYD1875, ASN004, AVA-020, Tb535, Hu3S193, B3, IGN311, BR96/BMS-182248-01, girentuximab, BAY 794620, BAY 2701439, MEDI-4276, XMT-1522, zolbetuximab, TST001, A315, labetuzumab, labetuzumab govitecan, SAR566658, DMOT4039, anetumab, anetumab ravtansine, BAY 2287411, and amatuximab, or an antigen-binding fragment of one of said antibodies.

[0072] The radioisotope-labeled targeting agent may, for example, be an anti-CD33 antibody, such as an anti-CD33 monoclonal antibody, radiolabeled with an isotope selected from the group consisting of ^{90}Y , ^{89}Sr , ^{153}Sm , ^{32}P , ^{225}Ac , ^{213}Bi , ^{213}Po , ^{211}At , ^{212}Bi , ^{213}Bi , ^{223}Ra , ^{227}Th , ^{149}Tb , ^{131}I , ^{137}Cs , ^{212}Pb , ^{103}Pd , and ^{177}Lu . For example, the radiolabeled anti-CD33 antibody may be ^{225}Ac -labeled HuM195 or ^{177}Lu -labeled HuM195.

[0073] This invention further provides a method for treating a mammalian subject, such as a human subject, afflicted with a myeloid proliferative disorder such as acute myeloid leukemia or myelodysplastic syndrome, including administering to the subject (i) one or both of an HDAC inhibitor such as vorinostat, romidepsin, belinostat, panobinostat or any disclosed herein, and an LSD1 inhibitor, such as seclidemstat, TCP (translucypromine), ORY-1001 (iadademstat), GSK2879552 (GSK), INCB059872, IMG-7289 (bomedemstat), ORY-2001 (vafidemstat), CC-90011) or any disclosed herein, in conjunction with (ii) ^{225}Ac -labeled HuM195, wherein the amounts of the HDAC inhibitor and/or LSD1 inhibitor, and ^{225}Ac -labeled HuM195, when administered in conjunction with one another, are therapeutically effective. Specific exemplifications envisioned in this method are the following combinations: (i) ^{225}Ac -labeled HuM195 and vorinostat; (ii) ^{225}Ac -labeled HuM195 and romidepsin; (iii) ^{225}Ac -labeled HuM195 and belinostat; and (iv) ^{225}Ac -labeled HuM195 and panobinostat.

[0074] This invention further provides a method for treating a human subject afflicted with myelodysplastic syndrome, including administering to the subject (i) an HDAC inhibitor selected from the group consisting of vorinostat, romidepsin, belinostat, and panobinostat in conjunction with (ii) ^{225}Ac -labeled HuM195, wherein the amounts of the HDAC inhibitor and ^{225}Ac -labeled HuM195, when admin-

istered in conjunction with one another, are therapeutically effective. Specifically envisioned in this method are the following combinations: (i) ^{225}Ac -labeled HuM195 and vorinostat; (ii) ^{225}Ac -labeled HuM195 and romidepsin; (iii) ^{225}Ac -labeled HuM195 and belinostat; and (iv) ^{225}Ac -labeled HuM195 and panobinostat.

[0075] This invention provides a method for inducing the death of a mammalian cancer cell, including contacting the cell with (i) an HDAC inhibitor in conjunction with (ii) a radioisotope-labeled agent that targets the cancer cell, wherein the amounts of HDAC inhibitor and radiolabeled agent, when contacted with the cell in conjunction with one another, are effective to induce the cell's death. The cancer cell may, for example, be a human cancer cell.

[0076] In one embodiment of this method, the cancer cell is a tumor cell selected from the group consisting of a breast cancer cell, an ovarian cancer cell, a prostate cancer cell, a lung cancer cell, a squamous cell carcinoma of the head and neck cell, a gastric cancer cell, a pancreatic cancer cell, a brain cancer cell, a liver cancer cell, a sarcoma cell, and a melanoma cell.

[0077] In another embodiment of this method, the cancer cell is a hematologic cancer cell, such as a multiple myeloma cell, an acute myeloid leukemia cell or a myelodysplastic syndrome cell (e.g., a mutated myeloid precursor cell lacking the ability to differentiate).

[0078] The radioisotope-labeled targeting agent may, for example, specifically binds to a cancer-associated antigen selected from the group consisting of CD33, CD45, CD38, Her3, DR5 (TRAIL-R2), 5T4, Lewis-Y, CAIX, Her2, Her3, Claudin 18.2, CEACAM5, MUC1, and mesothelin. Such targeting agent may, for example, be an antibody, an antigen-binding antibody fragment, an antibody mimetic, a peptide, a ligand, or a small molecule.

[0079] In another embodiment, the radioisotope-labeled agent includes or is the antibody which is, or the antibody portion of, one of the following: alemtuzumab, ibritumomab, ibritumomab tiuxetan, brentuximab, brentuximab vedotin, trastuzumab, trastuzumab emtansine, gemtuzumab, gemtuzumab ozogamicin, lintuzumab, B1836858, BC8/apamistamab, daratumumab, felzartamab/MOR202, isatuximab/SAR650984, TAK-169, AV-203, CDX-3379, HMBD 001, patritumab, seribantumab, elgemtumab/LJM716, lumretuzumab/RG7116/RO5479599, GSK2849330, mapatumumab, tigatuzumab/CS-1008, drozitumab, lexatumumab, conatumumab, MED10641, PF-06263507/A1mcMMAF, Anti-5T4 SYD1875, ASN004, AVA-020, Tb535, Hu3S193, B3, IGN311, BR96/BMS-182248-01, girentuximab, BAY 794620, BAY 2701439, MEDI-4276, XMT-1522, zolbetuximab, TST001, A315, labetuzumab, labetuzumab govitecan, SAR566658, DMOT4039, anetumab, anetumab ravtansine, BAY 2287411, and amatuximab.

[0080] The amino acid sequences of the heavy chain and the light chain of Trastuzumab reported by DrugBank Online are: heavy chain (SEQ ID NO:102) and light chain (SEQ ID NO:103) and an anti-HER2 binding antibody including one or both of said chains may be embodied in or used in the various embodiments of the invention.

[0081] The amino acid sequences of the heavy chain and the light chain of Pertuzumab reported by DrugBank Online are: heavy chain (SEQ ID NO:104) and light chain (SEQ ID NO:105) and an anti-HER2 binding antibody including one or both of said chains may be embodied in or used in the various embodiments of the invention.

[0082] An exemplary HER3 antibody that may be radio-labeled and embodied in and/or used in the presently disclosed invention may, for example, include a murine monoclonal antibody against HER3 including a heavy chain having the amino acid sequence as set forth in SEQ ID NO:9 or 11 and/or a light chain having the amino acid sequence as set forth in SEQ ID NO:10 or 12, or an antibody such as a humanized antibody derived from one or more of said sequences. An exemplary HER3 antibody that may be radiolabeled and embodied in and/or used in the presently disclosed invention may include or a heavy chain with an N-terminal region having the sequence set forth in SEQ ID NO:13 and/or a light chain with an N-terminal region having the sequence as set forth in SEQ ID NO:14. A HER3 antibody that may be similarly embodied or used in various aspect of the invention may, for example, include the heavy chain variable region having the amino acid sequence as set forth in SEQ ID NO:7, and/or a light chain variable region having an amino acid sequence as set forth in SEQ ID NO:8, and/or a heavy chain including one or more of CDR1, CDR2 and CDR3 having the amino acid sequences respectively set forth in SEQ ID NOS:1-3, and/or a light chain with one or more of the CDR1, CD2 and CDR3 having the amino acid sequences respectively set forth in SEQ ID NOS:4-6. A HER3 antibody embodied in and/or used in any of the aspects of the invention may, for example, include any combination of the aforementioned light chain sequences and/or heavy chain sequences.

[0083] An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including a CDR-H1 including SEQ ID NO:15, a CDR-H2 including SEQ ID NO:16, and a CDR-H3 including SEQ ID NO:17, and/or an immunoglobulin light chain variable region including a CDR-L1 including SEQ ID NO:18, a CDR-L2 including SEQ ID NO:19, and a CDR-L3 including SEQ ID NO:20. An exemplary An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including SEQ ID NO:21 and/or an immunoglobulin light chain variable region including SEQ ID NO:22. An exemplary HER3 antibody includes an immunoglobulin heavy chain amino acid sequence of SEQ ID NO:23 and/or an immunoglobulin light chain amino acid sequence of SEQ ID NO:24.

[0084] An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including a CDR-H1 including SEQ ID NO:25, a CDR-H2 including SEQ ID NO:26, and a CDR-H3 including SEQ ID NO:27; and/or an immunoglobulin light chain variable region including a CDR-L1 including SEQ ID NO:28, a CDR-L2 including SEQ ID NO:29, and a CDR-L3 including SEQ ID NO:30. An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including SEQ ID NO:31 and/or an immunoglobulin light chain variable region including SEQ ID NO:32. An exemplary HER3 antibody includes an immunoglobulin heavy chain amino acid sequence of SEQ ID NO:33 and/or an immunoglobulin light chain amino acid sequence of SEQ ID NO:34.

[0085] An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including a CDR-H1 including SEQ ID NO:35, a CDR-H2 including SEQ ID NO:36, and a CDR-H3 including SEQ ID NO:37; and/or an immunoglobulin light chain variable region including a CDR-L1 including SEQ ID NO:38, a CDR-L2 including SEQ ID NO:39, and a CDR-L3 including SEQ ID NO:40. An exemplary HER3 antibody includes an immunoglobulin

heavy chain variable region including SEQ ID NO:41, and/or an immunoglobulin light chain variable region SEQ ID NO:42. An exemplary HER3 antibody includes an immunoglobulin heavy chain amino acid sequence of SEQ ID NO:43 and an immunoglobulin light chain amino acid sequence of SEQ ID NO:44.

[0086] An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including a CDR-H1 including SEQ ID NO:45, a CDR-H2 including SEQ ID NO:46, and a CDR-H3 including SEQ ID NO:47; and/or an immunoglobulin light chain variable region including a CDR-L1 including SEQ ID NO:48, a CDR-L2 including SEQ ID NO:29, and a CDR-L3 including SEQ ID NO:49. An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including SEQ ID NO:50 and/or an immunoglobulin light chain variable region including SEQ ID NO:51. An exemplary HER3 antibody includes an immunoglobulin heavy chain amino acid sequence of SEQ ID NO:52 and/or an immunoglobulin light chain amino acid sequence of SEQ ID NO:53.

[0087] An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including a CDR-H1 including SEQ ID NO:54, a CDR-H2 including SEQ ID NO:55, and a CDR-H3 including SEQ ID NO:56; and/or an immunoglobulin light chain variable region including a CDR-L1 including SEQ ID NO:28, a CDR-L2 including SEQ ID NO:29, and a CDR-L3 including SEQ ID NO:30. An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including SEQ ID NO:57 and/or an immunoglobulin light chain variable region including SEQ ID NO:58. An exemplary HER3 antibody includes an immunoglobulin heavy chain amino acid sequence of SEQ ID NO:59 and/or an immunoglobulin light chain amino acid sequence of SEQ ID NO: 60.

[0088] An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including a CDR-H1 including SEQ ID NO:61, a CDR-H2 including SEQ ID NO:62, and a CDR-H3 including SEQ ID NO:63; and/or an immunoglobulin light chain variable region including a CDR-L1 including SEQ ID NO:64, a CDR-L2 including SEQ ID NO:65, and a CDR-L3 including SEQ ID NO:66. An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including SEQ ID NO:67, and/or an immunoglobulin light chain variable region including SEQ ID NO:68. An exemplary HER3 antibody includes an immunoglobulin heavy chain amino acid sequence of SEQ ID NO:69 and an immunoglobulin light chain amino acid sequence of SEQ ID NO:70.

[0089] An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including a CDR-H1 including SEQ ID NO:71, a CDR-H2 including SEQ ID NO:72, and a CDR-H3 including SEQ ID NO:66; and/or an immunoglobulin light chain variable region including a CDR-L1 including SEQ ID NO:28, a CDR-L2 including SEQ ID NO:29, and a CDR-L3 including SEQ ID NO:30. An exemplary HER3 antibody includes an immunoglobulin heavy chain variable region including SEQ ID NO:73, and/or an immunoglobulin light chain variable region including SEQ ID NO:74. An exemplary HER3 antibody includes an immunoglobulin heavy chain amino acid sequence of SEQ ID NO:75 and/or an immunoglobulin light chain amino acid sequence of SEQ ID NO:76.

[0090] An exemplary HER3 antibody includes an immunoglobulin heavy chain amino acid sequence of SEQ ID NO:77 and/or an immunoglobulin light chain amino acid sequence of SEQ ID NO:78.

[0091] An exemplary HER3 antibody includes an immunoglobulin light chain variable region including SEQ ID NOS:86, 87, 88, 89, 90 or 91 and/or a heavy chain variable region including SEQ ID NOS:79, 80, 81, 82, 83, 84 or 85.

[0092] An exemplary HER3 antibody includes an immunoglobulin heavy chain sequence including SEQ ID NO:92, 94, 95, 98 or 99 and/or an immunoglobulin light chain sequence including SEQ ID NO:93, 96, 97, 100 or 101.

[0093] Exemplary HER3 antibodies also include Baretamab (ISU104) from Isu Abxis Co and any of the HER3 antibodies disclosed in U.S. Pat. No. 10,413,607.

[0094] Exemplary HER3 antibodies also include HMBD-001 (10D1F) from Hummingbird Bioscience Pte. and any of the HER3 antibodies disclosed in International Pub. Nos. WO 2019185164 and WO2019185878, U.S. Pat. No. 10,662,241; and U.S. Pub. Nos. 20190300624, 20210024651, and 20200308275.

[0095] Exemplary HER3 antibodies also include the HER2/HER3 bispecific antibody MCLA-128 (i.e., Zenocutuzumab) from Merus N.V.; and any of the HER3 antibodies, whether monospecific or multi-specific, disclosed in U.S. Pub. Nos. 20210206875, 20210155698, 20200102393, 20170058035, and 20170037145.

[0096] Exemplary HER3 antibodies also include the HER3 antibody Patritumab (U3-1287), an antibody including heavy chain sequence SEQ ID NO:106 and/or light chain sequence SEQ ID NO:107 which are reported chains of Patritumab, and any of the HER3 antibodies disclosed in U.S. Pat. Nos. 9,249,230 and 7,705,130 and International Pub. No. WO2007077028.

[0097] Exemplary HER3 antibodies also include the HER3 antibody MM-121 and any of the HER3 antibodies disclosed in U.S. Pat. No. 7,846,440 and International Pub. No. WO2008100624. Exemplary HER3 antibodies also include the EGFR/HER3 bispecific antibody DL1 and any of the HER3 antibodies, whether monospecific or multi-specific, disclosed in U.S. Pat. Nos. 9,327,035 and 8,597,652, U.S. Pub. No. 20140193414, and International Pub. No. WO2010108127.

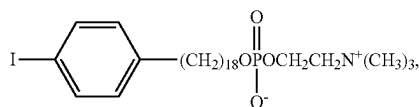
[0098] Exemplary HER3 antibodies also include the HER2/HER3 bispecific antibody MM-111 and any of the HER3 antibodies, whether monospecific or multi-specific, disclosed in U.S. Pub. Nos. 20130183311 and 20090246206 and International Pub. Nos. WO2006091209 and WO2005117973.

[0099] According to certain aspects, the HER3 targeting agent includes an anti-HER3 antibody that binds to an epitope of HER3 recognized by Patritumab from Daiichi Sankyo, Seribantumab (MM-121) from Merrimack Pharmaceuticals, Lumretuzumab from Roche, Elgemtumab from Novartis, GSK2849330 from GlaxoSmithKline, CDX-3379 of Celldex Therapeutics, EV20 and MP-RM-1 from MediPharma, Baretamab (ISU104) from Isu Abxis Co., HMBD-001 (10D1F) from Hummingbird Bioscience Pte., REGN1400 from Regeneron Pharmaceuticals, and/or AV-203 from AVEO Oncology. According to certain aspects, the anti-HER3 antibody is selected from one or more of Patritumab, Seribantumab or an antibody including heavy chain sequence SEQ ID NO:108 and/or light chain sequence SEQ ID NO:109 which are reported for Seribantumab,

Lumretuzumab or an antibody including heavy chain sequence SEQ ID NO:110 and/or light chain sequence SEQ ID NO:111 which are reported for Lumretuzumab, Elgemitumab or an antibody including heavy chain sequence SEQ ID NO:112 and/or light chain sequence SEQ ID NO:113 which are reported for Elgemitumab, AV-203, CDX-3379, GSK2849330, EV20, MP-RM-1, ISU104, HMBD-001 (10D1F), and REGN1400.

[0100] It should be understood that wherever in this disclosure specific antibodies, specific antibody heavy chains and specific antibody light chains are disclosed, against any target, also intended to be disclosed for embodiment in or use in the various aspects of the invention are antibodies, such as but not limited to immunoglobulins, such as but not limited to IgG, that (i) include the heavy chain variable region of the disclosed antibody or heavy chain, (ii) include 1, 2 or 3 of the heavy chain CDRs (e.g., by Kabat definition) of the disclosed antibody or heavy chain, (iii) include the light chain variable region of the disclosed antibody or light chain, and/or (iv) include 1, 2 or 3 of the light chain CDRs (e.g., by Kabat definition) of the disclosed antibody or light chain. It should also be understood that wherever in this disclosure an antibody heavy chain or an antibody light chain is disclosed that includes an N-terminal leader sequence, also intended to be disclosed for embodiment in and use in the various aspects of the invention are corresponding heavy chains and corresponding light chains that lack the leader sequence.

[0101] In still further embodiments of the invention, the radiolabeled targeting agent used in combination or conjunction with the epigenetic therapy(ies) for the treatment of a cancer or proliferative disorder such as any of those disclosed herein in a mammal, such as a human, includes a phospholipid-based cancer targeting agent. In certain embodiments, the phospholipid-based cancer targeting agent includes any of the radioactive phospholipid metal chelates disclosed in U.S. Pub. No. 20200291049, incorporated by reference herein, such as but not limited to

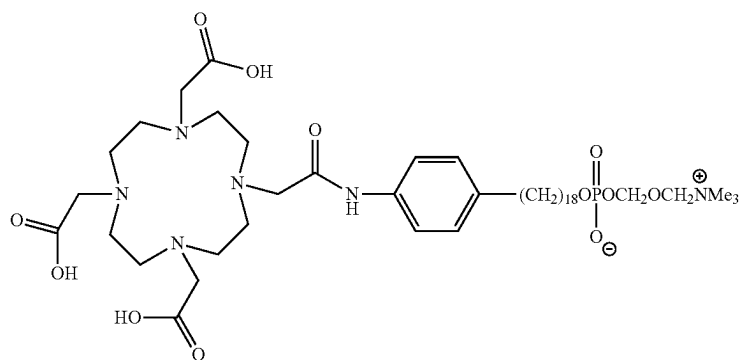


i.e., 18-(p-iodophenyl)octadecyl phosphocholine, wherein iodine is ^{131}I (a/k/a NM404 I-131, and CLR 131), or a pharmaceutically acceptable salt thereof.

[0103] In certain aspects, the phospholipid-based radiolabeled targeting agent used with the epigenetic therapy(ies) includes any of the phospholipid drug conjugate compounds disclosed in U.S. Pat. No. 9,480,754, incorporated by reference herein.

[0104] In one embodiment, the radioisotope-labeled targeting agent is an anti-CD33 antibody, such as a monoclonal antibody or an antigen-binding fragment thereof, or an antibody mimetic, labeled with an isotope selected from the group consisting of ^{90}Y , ^{89}Sr , ^{153}Sm , ^{32}P , ^{225}Ac , ^{213}Bi , ^{213}Po , ^{211}At , ^{212}Bi , ^{213}Bi , ^{223}Ra , ^{227}Th , ^{149}Tb , ^{131}I , ^{137}Cs , ^{212}Pb , ^{103}Pd , ^{166}Ho , ^{186}Re , ^{188}Re , ^{67}Cu , ^{199}Au , ^{105}Rh , ^{211}As , and ^{177}Lu . For example, the radiolabeled anti-CD33 antibody may be ^{225}Ac -labeled HuM195 or ^{177}Lu -labeled HuM195.

[0105] This invention further provides a method for inducing the death of a CD33-expressing hematological proliferative disorder cell such as an acute myeloid leukemia cell, a multiple myeloma cell, or a myelodysplastic syndrome cell, including contacting the cell with (i) one or both of an HDAC inhibitor such as any of those disclosed herein and an LSD1 inhibitor such as any of those disclosed herein, in conjunction with (ii) a radiolabeled CD33 targeting agent such as ^{225}Ac -labeled HuM195, wherein the amounts of HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled CD33 targeting agent, when contacted with the cell in conjunction with one another, are effective to induce the cell's death. Specific exemplifications envisioned in this



(a/k/a NM600) or a pharmaceutically acceptable salt thereof, chelated with a radionuclide, such as ^{225}Ac , ^{177}Lu , or ^{90}Y

[0102] In certain aspects, the lipid based radiolabeled targeting agent used with the epigenetic therapy includes any of the radiolabeled phospholipid compounds disclosed in U.S. Pub. No. 20140030187 or U.S. Pat. No. 6,417,384, each incorporated by reference herein, such as but not limited to

method are the following combinations: (i) ^{225}Ac -labeled HuM195 and vorinostat; (ii) ^{225}Ac -labeled HuM195 and romidepsin; (iii) ^{225}Ac -labeled HuM195 and belinostat; and (iv) ^{225}Ac -labeled HuM195 and panobinostat.

[0106] This invention further provides a method for inducing the death of an acute myeloid leukemia cell, including contacting the cell with (i) one or both of an HDAC inhibitor such as any of those disclosed herein and an LSD1 inhibitor such as any of those disclosed herein, in conjunction with

(ii) a radiolabeled CD33 targeting agent such as ^{225}Ac -labeled HuM195, wherein the amounts of HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled CD33 targeting agent, when contacted with the cell in conjunction with one another, are effective to induce the cell's death. Specific exemplifications envisioned in this method are the following combinations: (i) ^{225}Ac -labeled HuM195 and vorinostat; (ii) ^{225}Ac -labeled HuM195 and romidepsin; (iii) ^{225}Ac -labeled HuM195 and belinostat; and (iv) ^{225}Ac -labeled HuM195 and panobinostat.

[0107] This invention further provides a method for inducing the death of a myelodysplastic syndrome cell, including contacting the cell with (i) one or both of an HDAC inhibitor such as any of those disclosed herein and an LSD1 inhibitor such as any of those disclosed herein, in conjunction with (ii) a radiolabeled CD33 targeting agent such as ^{225}Ac -labeled HuM195, wherein the amounts of HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled CD33 targeting agent, when contacted with the cell in conjunction with one another, are effective to induce the cell's death. Specific exemplifications envisioned in this method are the following combinations: (i) ^{225}Ac -labeled HuM195 and vorinostat; (ii) ^{225}Ac -labeled HuM195 and romidepsin; (iii) ^{225}Ac -labeled HuM195 and belinostat; and (iv) ^{225}Ac -labeled HuM195 and panobinostat.

[0108] In addition to use in treating CD33-positive hematological proliferative disorders/cancers, radiolabeled CD33 targeting agents such as those disclosed herein may be used to treat a solid tumor cancer in a mammal by depleting CD33-positive myeloid-derived suppressor cells (MDSCs), in combination with treatment using one or both of an HDAC inhibitor such as any of those disclosed herein and an LSD1 inhibitor such as any of those disclosed herein. Accordingly, one embodiment of the invention provides a method for treating a solid tumor cancer such as any of those disclosed herein in a mammal, such as a human, that includes administering to the mammal: (i) one or both of an HDAC inhibitor such as any of those disclosed herein and an LSD1 inhibitor such as any of those disclosed herein, and (ii) a radiolabeled CD33 targeting agent such as ^{225}Ac -labeled HuM195, wherein the amounts of HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled CD33 targeting agent, when administered in conjunction with one another, are effective to treat the solid tumor cancer.

[0109] The HDAC inhibitor and/or LSD1 inhibitor may each, for example, be administered, such as daily or every other day, at a dose of 1.0 to 2,000 mg, or in any subrange bounded by integer values thereof, such as 1.0 to 1,000 mg, such as 25 to 1,000 mg, such as 10 to 500 mg. In one variation of the instant methods, each of the HDAC inhibitor and/or the LSD1 inhibitor is administered in doses and/or schedules provided with approved labeling or used in clinical trials of the agents. In one variation of the instant methods, the HDAC inhibitor, the LSD1 inhibitor, and/or the radiolabeled targeting agent, administered in doses that are less than, and/or in dosing regimens of shorter duration than, those presently prescribed on their respective labels or used in clinical trials of the agents. Embodiments of the invention in this regard are set forth in the examples section.

[0110] This invention also provides articles of manufacture. The first article includes (i) one or both of an HDAC inhibitor (e.g., vorinostat, romidepsin, belinostat, panobinostat and/or any of those disclosed herein) and an LSD1 inhibitor (e.g., seclidemstat, TCP (tranylcypromine), ORY-

1001 (iadademstat), GSK2879552 (GSK), INCB059872, IMG-7289 (bomedemstat), ORY-2001 (vafidemstat), CC-90011, and/or any of those disclosed herein), and (ii) a label instructing the user (e.g., a healthcare provider) to treat a subject (e.g., a human) afflicted with a hematologic malignancy (e.g., acute myeloid leukemia or myelodysplastic syndrome) by administering the HDAC inhibitor to the subject in conjunction with a radioisotope-labeled agent (e.g., ^{225}Ac -labeled HuM195) that targets cancer cells in the subject, wherein the amounts of the HDAC inhibitor and labeled agent, when administered in conjunction with one another, are therapeutically effective. Specific exemplifications envisioned in this article are the following combinations: (i) ^{225}Ac -labeled HuM195 and vorinostat; (ii) ^{225}Ac -labeled HuM195 and romidepsin; (iii) ^{225}Ac -labeled HuM195 and belinostat; and/or (iv) ^{225}Ac -labeled HuM195 and panobinostat.

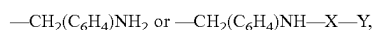
[0111] The second article of manufacture includes (i) a radioisotope-labeled agent (e.g., ^{225}Ac -labeled HuM195) that targets cancer cells and (ii) a label instructing the user (e.g., a healthcare provider) to treat a subject (e.g., a human) afflicted with a hematologic malignancy (e.g., acute myeloid leukemia, myelodysplastic syndrome, or multiple myeloma) by administering the labeled agent to the subject in conjunction with one or both of an HDAC inhibitor (e.g., vorinostat, romidepsin, belinostat, panobinostat and/or any of those disclosed herein) and an LSD1 inhibitor (e.g., seclidemstat, TCP (tranylcypromine), ORY-1001 (iadademstat), GSK2879552 (GSK), INCB059872, IMG-7289 (bomedemstat), ORY-2001 (vafidemstat), CC-90011, and/or any of those disclosed herein), wherein the amounts of the HDAC inhibitor and/or LSD1 inhibitor, and labeled agent, when administered in conjunction with one another, are therapeutically effective. Specific exemplifications envisioned in this article are the following combinations: (i) ^{225}Ac -labeled HuM195 and vorinostat; (ii) ^{225}Ac -labeled HuM195 and romidepsin; (iii) ^{225}Ac -labeled HuM195 and belinostat; and/or (iv) ^{225}Ac -labeled HuM195 and panobinostat.

[0112] Those skilled in the art will appreciate that methods for radiolabeling a targeting agent, directly or via a chelator group are well established.

[0113] Exemplary chelators that may be linked to targeting agents in the various aspects of the invention include: 1,4,7,10-tetraazacyclododecane-1,4,7-triacetic acid (DO3A) or a derivative thereof; 1,4,7-triazacyclononane-1,4-diacetic acid (NODA) or a derivative thereof; 1,4,7-triazacyclononane-1,4,7-triacetic acid (NOTA) or a derivative thereof; 1,4,7,10-tetraazacyclododecane-1,4,7,10-tetraacetic acid (DOTA) or a derivative thereof; 1,4,7-triazacyclononane-1-glutaric acid-4,7-diacetic acid (NODAGA) or a derivative thereof; 1,4,7,10-tetraazacyclododecane-1-glutaric acid-4,7,10-triacetic acid (DOTAGA) or a derivative thereof; 1,4,8,11-tetraazacyclotetradecane-1,4,8,11-tetraacetic acid (TETA) or a derivative thereof; 1,4,8,11-tetraazabicyclo[6.6.2]hexadecane-4,11-diacetic acid (CB-TE2A) or a derivative thereof; diethylene triamine pentaacetic acid (DTPA), its diester, or a derivative thereof; 2-cyclohexyl diethylene triamine pentaacetic acid (CHX-A"-DTPA) or a derivative thereof; deferoxamine (DFO) or a derivative thereof; 1,2-[[6-carboxypyridin-2-yl]methylamino]ethane (H_2dedpa) or a derivative thereof; DADA or a derivative thereof; 1,4,7,10-Tetraazacyclododecane-1,4,7,10-tetra(methylene phosphonic acid) (DOTP) or a derivative thereof; 4-amino-6-

[[16-[(6-carboxypyridin-2-yl)methyl]-1,4,10,13-tetraoxa-7,16-diazacyclooctadec-7-yl]methyl]pyridine-2-carboxylic acid (MACROPA-NH₂) or a derivative thereof; MACROPA or a derivative thereof; 1,4,7,10-tetrakis(carbamoylmethyl)-1,4,7,10-tetraazacyclododecane (TCMC) or a derivative thereof; {4-[2-(bis-carboxymethylamino)-ethyl]-7-carboxymethyl-[1,4,7]triazonan-1-yl]-acetic acid (NETA) or a derivative thereof; Diamsar or a derivative thereof; 1,4,7-triazacyclononane-1,4,7-tris[methyl(2-carboxyethyl)phosphinic acid (TRAP, PRP9, TRAP-Pr) or a derivative thereof; N,N'-bis(6-carboxy-2-pyridylmethyl)ethylenediamine-N,N'-diacetic acid (H4octapa) or a derivative thereof; N,N'-[1-benzyl-1,2,3-triazole-4-yl]methyl-N,N'-[6-(carboxy)pyridin-2-yl]-1,2-diaminoethane (H2azapa) or a derivative thereof; N,N'-[[6-(carboxy)pyridin-2-yl]methyl]diethylenetriamine-N,N'-triacetic acid (H5decapa) or a derivative thereof; N,N'-bis(2-hydroxy-5-sulfobenzyl)ethylenediamine-N,N'-diacetic acid (SHBED) or a derivative thereof; N,N'-bis(2-hydroxybenzyl)ethylenediamine-N,N'-diacetic acid (HBED) or a derivative thereof; 3,6,9,15-tetraazabicyclo[9.3.1]pentadeca-1(15),11,13-triene-3,6,9,-triacetic acid (PCTA) or a derivative thereof; desferrioxamine B (DFO) or a derivative thereof; N,N'-(methylenephosphonate)-N,N'-[6-(methoxycarbonyl)pyridin-2-yl]methyl-1,2-diaminoethane (H6phospa) or a derivative thereof; 1,4,7,10,13,16-hexaazacyclohexadecane-N,N',N'',N''',N''''-hexaacetic acid (HEHA) or a derivative thereof; 1,4,7,10,13-pentaazacyclopentadecane-N,N',N'',N''',N''''-pentaacetic acid (PEPA) or a derivative thereof; or 3,4,3-LI(1,2-HOPO) or a derivative thereof.

[0114] According to certain aspects, the targeting agent may be radiolabeled through chemical conjugation of suitable bifunctional chelators that can chelate one or more radionuclides. Exemplary chelator molecules that may be used include p-SCN-Bn-DOTA, NH₂-DOTA, NH₂—(CH₂)₁₋₂₀-DOTA, NH₂-(PEG)₁₋₂₀-DOTA, HS-DOTA, HS—(CH₂)₁₋₂₀-DOTA, HS-(PEG)₁₋₂₀-DOTA, dibromo-S—(CH₂)₁₋₂₀-DOTA, dibromo-S-(PEG)₁₋₂₀-DOTA, p-SCN-Bn-DOTP, NH₂-DOTP, NH₂—(CH₂)₁₋₂₀-DOTP, NH₂-(PEG)₁₋₂₀-DOTP, HS-DOTP, HS—(CH₂)₁₋₂₀-DOTP, HS-(PEG)₁₋₂₀-DOTP, dibromo-S—(CH₂)₁₋₂₀-DOTP, and dibromo-S-(PEG)₁₋₂₀-DOTP. The chelator molecules may, for example, be attached to a targeting agent through a linker molecule. Exemplary linker molecules include:



[0115] wherein X is

[0116] —R₂—CH₂CH₂O(CH₂CH₂O)_nCH₂CH₂—,

[0117] —R₂—CH₂CH₂NHC(O)CH₂CH₂O(CH₂CH₂O)_nCH₂CH₂—,

[0118] —R₂—(CH₂)_nCH₂—,

[0119] —R₂—CH₂CH₂NHC(O)(CH₂)_nCH₂—,

[0120] —R₂—CH(C(O)R₃)CH₂—, wherein R₃ is —OH or a short peptide (1-20 amino acids),

[0121] —R₂—CH₂CH₂O(CH₂CH₂O)_nCH₂C(O)O—, or

[0122] —R₂—CH₂CH₂NHC(O)CH₂CH₂O(CH₂CH₂O)_nCH₂CC(O)O—,

[0123] wherein n is 1-20, and

[0124] R₂ is —C(O)— or —C(S)NH—; and

[0125] Y is —NH₂ or —SR₄—, wherein R₄ is —H or —CH₂-3,5-bis(bromomethyl)benzene.

[0126] Targeting agents, such as protein targeting agents, for example antibodies and antigen-binding antibody frag-

ments, and peptide targeting agents may, for example, be conjugated with a chelator for radiolabeling the targeting agent via chelation of a radionuclide. Such protein or peptide targeting agents, for example, that include lysine(s), may conveniently be conjugated to a DOTA chelating moiety using the bifunctional agent S-2-(4-Isothiocyanatobenzyl)-1,4,7,10-tetraazacyclododecane tetraacetic acid a/k/a/“p-SCN-Bn-DOTA” (Catalog #B205; Macrocylics, Inc., Plano, TX, USA). p-SCN-Bn-DOTA may be synthesized by a multi-step organic synthesis fully described in U.S. Pat. No. 4,923,985. Chelation of a radionuclide by the DOTA moiety may be performed prior to chemical conjugation of the antibody with p-SCN-Bn-DOTA and/or after said conjugation.

[0127] Targeting agents containing one or more cysteine residues, such as peptides, proteins, antibodies and protein antibody mimetics that have one or more cysteines may, for example, be chemically conjugated to any of the chelator-bearing, such as DOTA-bearing, stable linkers disclosed in U.S. Pat. No. 11,000,604 titled “Reagent for site-selective bioconjugation of proteins or antibodies” for radionuclide labeling.

[0128] According to certain aspects of the present invention, administering a radiolabeled targeting agent to a subject in any of the embodiments may, for example, include administration of a composition that includes a radiolabeled fraction and a non-radiolabeled fraction of the targeting agent, such as an antibody, antibody fragment, etc. For example, the non-radiolabeled-labeled fraction may include the same antibody against the same epitope as the labeled fraction. In this way, the total radioactivity of the antibody may be varied or may be held constant while the overall antibody protein concentration may be held constant or may be varied, respectively. For example, the total protein concentration of non-radiolabeled labeled antibody fraction administered may be selected depending on the exact nature of the disease to be treated, age and weight of the patient, identity of the antibody, and the label (e.g., radionuclide) selected for labeling of the antibody. The composition may, for example, include the targeting agent in a ratio of radiolabeled: non-radiolabeled targeting agent of from about 0.01:10 to 1:1, such as 0.1:10 to 1:1 radiolabeled: non-radiolabeled. Various aspects of this invention will be better understood by reference to the following examples.

EXAMPLES

Example 1—Structure of ²²⁵Ac-Lintuzumab (²²⁵Ac-HuM195)

[0129] ²²⁵Ac-Lintuzumab includes three key components; humanized monoclonal antibody HuM195 (generic name, lintuzumab), the alpha-emitting radioisotope ²²⁵Ac, and the bi-functional chelate 2-(p-isothiocyanatobenzyl)-1,4,7,10-tetraazacyclododecane-1,4,7,10-tetraacetic acid (p-SCN-Bn-DOTA). As depicted in FIG. 4, HuM195 is radiolabeled using the bi-functional chelate p-SCN-Bn-DOTA that binds to ²²⁵Ac and that is covalently attached to the IgG via a lysine residue on the antibody.

Example 2—p-SCN-Bn-DOTA

[0130] DOTA, 2-(4-Isothiocyanatobenzyl)-1,4,7,10-tetraazacyclododecane tetraacetic acid (Macrocylics item

code B205-GMP) is synthesized by a multi-step organic synthesis that is fully described in U.S. Pat. No. 4,923,985.

Example 3—Preparation of ^{225}Ac -Lintuzumab (^{225}Ac -HuM195)

[0131] The procedure for preparing ^{225}Ac -Lintuzumab is based on the method described by Michael R. McDevitt, “Design and synthesis of ^{225}Ac radioimmuno-pharmaceuticals, Applied Radiation and Isotope”, 57 (2002), 841-847. The procedure involves radiolabeling the bi-functional chelate, p-SCN-Bn-DOTA, with the radioisotope ^{225}Ac , followed by binding of the radiolabeled p-SCN-Bn-DOTA to the antibody (HuM195). The construct, ^{225}Ac -p-SCN-Bn-DOTA-HuM195, is purified using 10 DG size exclusion chromatography and eluted with 1% human serum albumin (HSA). The resulting drug product, Ac^{225} -Lintuzumab, is then passed through a 0.2 μm sterilizing filter.

Example 4—Process Flow for Preparation of ^{225}Ac -Lintuzumab (^{225}Ac -HuM195)

[0132] The procedure, shown in FIG. 5, begins with confirming the identity of all components and the subsequent QC release of the components to production.

[0133] The ^{225}Ac is assayed to confirm the level of activity and is reconstituted to the desired activity concentration with hydrochloric acid. A vial of lyophilized p-SCN-Bn-DOTA is reconstituted with metal-free water to a concentration of 10 mg/mL. To the actinium reaction vial, 0.02 ml of ascorbic acid solution (150 mg/mL) and 0.05 ml of reconstituted p-SCN-Bn-DOTA are added and the pH adjusted to between 5 and 5.5 with 2M tetramethylammonium acetate (TMAA). The mixture is then heated at $55\pm 4^\circ\text{C}$. for 30 minutes.

[0134] To determine the labeling efficiency of the ^{225}Ac -p-SCN-Bn-DOTA, an aliquot of the reaction mixture is removed and applied to a 1 ml column of Sephadex C25 cation exchange resin. The product is eluted in 2-4 ml fractions with a 0.9% saline solution. The fraction of ^{225}Ac activity that elutes is ^{225}Ac -p-SCN-Bn-DOTA and the fraction that is retained on the column is un-chelated, unreactive ^{225}Ac . Typically, the labeling efficiency is greater than 95%.

[0135] To the reaction mixture, 0.22 ml of previously prepared HuM195 in DTPA (1 mg HuM195) and 0.02 ml of ascorbic acid are added. The DTPA is added to bind any trace amounts of metals that may compete with the labeling of the antibody. The ascorbic acid is added as a radio-protectant. The pH is adjusted with carbonate buffer to pH 8.5-9. The mixture is heated at $37\pm 3^\circ\text{C}$. for 30 minutes.

[0136] The final product is purified by size exclusion chromatography using 10DG resin and eluted with 2 ml of 1% HSA. Typical reaction yields are 10%.

Example 5—Vorinostat—Normal and Reduced Dosing Regimens

[0137] Vorinostat is sold by Merck & Co., Inc. under the brand name Zolinza®. Vorinostat is sold in capsule form at 100 mg. The dosage is 400 mg taken orally once daily. Dosing continues for five consecutive days each week, and may be continued as long as there is no evidence of progressive disease or toxicity. This dosing regimen is referred to herein as the “normal” human dosing regimen for vorinostat, regardless of the disorder treated. Any dosing regimen having a shorter duration (e.g., four days or fewer

per week) or involving the administration of less than 400 mg/day (e.g., 100 mg/day) is referred to herein as a “reduced” human dosing regimen. Examples of reduced human dosing regimens include the following: (i) 350 mg/day, 300 mg/day, 250 mg/day, 200 mg/day, 150 mg/day, 100 mg/day, 75 mg/day, 50 mg/day, 25 mg/day, and (ii) <400 mg/day; 300-350 mg/day, 250-300 mg/day, 200-250 mg/day, 150-200 mg/day, 100-150 mg/day, 75-100 mg/day, 50-75 mg/day, 25-50 mg/day, and <25 mg/day.

Example 6—Romidepsin—Normal and Reduced Dosing Regimens

[0138] Romidepsin is sold by Gloucester Pharmaceuticals, Inc., under the brand name Istodax®. Romidepsin is sold as an injectable formulation. The dosage is 14 mg/m² administered intravenously over a four-hour period on days 1, 8, and 15 of a 28-day cycle. Cycles are repeated every 28 days provided that the patient continues to benefit from and tolerate the therapy. This dosing regimen is referred to herein as the “normal” human dosing regimen for romidepsin, regardless of the disorder treated. Any dosing regimen having a shorter duration (e.g., intravenous administration over a four-hour period only on days 1 and 8 of a 28-day cycle, or a limited number of cycles (e.g., a total of two 28-day cycles)) or involving delivering less than 14 mg/m² per administration (e.g., 10 mg/m² per administration) is referred to herein as a “reduced” human dosing regimen. Examples of reduced human dosing regimens include the following: (i) 13 mg/m² per administration, 12 mg/m² per administration, 11 mg/m² per administration, 10 mg/m² per administration, 9 mg/m² per administration, 8 mg/m² per administration, 7 mg/m² per administration, 6 mg/m² per administration, 5 mg/m² per administration, 4 mg/m² per administration, 3 mg/m² per administration, 2 mg/m² per administration, 1 mg/m² per administration, and 0.5 mg/m² per administration; (ii) <14 mg/m² per administration, 12-13 mg/m² per administration, 11-12 mg/m² per administration, 10-11 mg/m² per administration, 9-10 mg/m² per administration, 8-9 mg/m² per administration, 7-8 mg/m² per administration, 6-7 mg/m² per administration, 5-6 mg/m² per administration, 4-5 mg/m² per administration, 3-4 mg/m² per administration, 2-3 mg/m² per administration, 1-2 mg/m² per administration, 0.5-1 mg/m² per administration, <0.5 mg/m² per administration; and (iii) intravenous administration only on days 1 and 8 of a 28-day cycle, intravenous administration only on days 1 and 15 of a 28-day cycle, intravenous administration only on day 1 of a 28-day cycle, intravenous administration only on day 1 of a 21-day cycle, intravenous administration only on day 1 of a 14-day cycle; and intravenous administration only on day 1 of a 7-day cycle.

Example 7—Belinostat—Normal and Reduced Dosing Regimens

[0139] Belinostat is sold by Spectrum Pharmaceuticals, Inc., under the brand name Beleodaq®. Belinostat is sold as an injectable formulation. The dosage is 1,000 mg/m² administered intravenously over a 30-minute period once daily on days 1-5 of a 21-day cycle. Cycles can be repeated until disease progression or unacceptable toxicity. This dosing regimen is referred to herein as the “normal” human dosing regimen for belinostat, regardless of the disorder treated. Any dosing regimen having a shorter duration (e.g.,

intravenous administration over a 30-minute period once daily only on days 1-3 of a 21-day cycle, or a limited number of cycles (e.g., a total of two 21-day cycles)) or involving delivering less than 1,000 mg/m² per administration (e.g., 500 mg/m² per administration) is referred to herein as a “reduced” human dosing regimen. Examples of reduced human dosing regimens include the following: (i) 900 mg/m² per administration, 800 mg/m² per administration, 700 mg/m² per administration, 600 mg/m² per administration, 500 mg/m² per administration, 400 mg/m² per administration, 300 mg/m² per administration, 250 mg/m² per administration, 200 mg/m² per administration, 150 mg/m² per administration, 100 mg/m² per administration, 75 mg/m² per administration, 50 mg/m² per administration, and 25 mg/m² per administration; (ii) <1,000 mg/m² per administration, 800-900 mg/m² per administration, 700-800 mg/m² per administration, 600-700 mg/m² per administration, 500-600 mg/m² per administration, 400-500 mg/m² per administration, 300-400 mg/m² per administration, 250-300 mg/m² per administration, 200-250 mg/m² per administration, 150-200 mg/m² per administration, 100-150 mg/m² per administration, 75-100 mg/m² per administration, 50-75 mg/m² per administration, 25-50 mg/m² per administration, and <25 mg/m² per administration; and (iii) intravenous administration only on days 1-4 of a 21-day cycle, intravenous administration only on days 1-3 of a 21-day cycle, intravenous administration only on days 1 and 2 of a 21-day cycle, intravenous administration only on day 1 of a 21-day cycle, intravenous administration only on day 1 of a 14-day cycle, and intravenous administration only on day 1 of a 7-day cycle.

Example 8—Panobinostat—Normal and Reduced Dosing Regimens

[0140] Panobinostat is sold by Novartis Pharmaceuticals Corporation under the brand name Farydak®. Panobinostat is sold in capsule form at 10 mg, 15 mg, and 20 mg. The dosage is 20 mg taken orally once every other day for three doses per week (on days 1, 3, 5, 8, 10, and 12) of weeks one and two of each 21-day cycle. Dosing continues for eight 21-day cycles. This dosing regimen is referred to herein as the “normal” human dosing regimen for panobinostat, regardless of the disorder treated. Any dosing regimen having a shorter duration (e.g., six 21-day cycles) or involving the administration of less than 20 mg/dose (e.g., 10 mg per dose) is referred to herein as a “reduced” human dosing regimen. Examples of reduced human dosing regimens include the following: (i) 15 mg/dose, 12.5 mg/dose, 10 mg/dose, 7.5 mg/dose, 5 mg/dose, 4 mg/dose, 3 mg/dose, 2 mg/dose, 1 mg/dose, and 0.5 mg/dose; (ii) <20 mg/dose, 12.5-15 mg/dose, 10-12.5 mg/dose, 7.5-10 mg/dose, 5-7.5 mg/dose, 4-5 mg/dose, 3-4 mg/dose, 2-3 mg/dose, 1-2 mg/dose, 0.5-1 mg/dose, and <0.5 mg/dose; and (iii) dosing on days 1, 3, 5, 8, and 10 of a 21-day cycle, dosing on days 1, 3, 5, 8, and 12 of a 21-day cycle, dosing on days 1, 3, 8, 10, and 12 of a 21-day cycle, dosing on days 1, 5, 8, 10, and 12 of a 21-day cycle, dosing on days 1, 5, 8, and 12 of a 21-day cycle, dosing on days 1 and 8 of a 21-day cycle, dosing on day 1 of a 21-day cycle, dosing on day 1 of a 14-day cycle, and dosing on day 1 of a 7-day cycle. In one embodiment, each of the normal and reduced human dosing regimens of panobinostat further includes the administration of bortezomib and dexamethasone as directed in the FDA label for panobinostat.

[0141] The terms “normal” human dosing regimen and “reduced” human dosing regimen also apply, *mutatis mutandis*, to any other HDAC inhibitor with respect to its approved or otherwise customary dosing regimen.

Example 9—²²⁵Ac-HuM195—Normal and Reduced Dosing Regimens

[0142] For an agent such as an antibody labeled with an alpha-emitting isotope, the majority of the drug administered to a subject typically consists of non-labeled antibody, with the minority being the labeled antibody. Doses of labeled agent used in connection with this invention include, for example, a single administration, and two or more administrations (i.e., fractions). The amount administered in each dose can be measured, for example, by labeled radiation activity (e.g., $\mu\text{Ci/kg}$) or antibody weight (e.g., $\mu\text{g/kg}$ or $\mu\text{g/m}^2$).

[0143] In the case of ²²⁵Ac-HuM195, the “normal” human dosing regimen (regardless of the disorder treated), as this term is used herein, includes either of the following: (i) 4.0 $\mu\text{Ci/kg}$ administered fractionally in multiple administrations over no less than 1 week apart between doses; or (ii) 4.0 $\mu\text{Ci/kg}$ when delivered in a single administration.

[0144] A dosing regimen involving the administration of less ²²⁵Ac-HuM195 (e.g., 2.0 $\mu\text{Ci/kg}$ when delivered in a single administration) is referred to herein as a “reduced” human dosing regimen. Additional reduced human dosing regimens include, for example: (i) $2 \times <0.25 \mu\text{Ci/kg}$, $2 \times 0.25 \mu\text{Ci/kg}$, $2 \times <0.5 \mu\text{Ci/kg}$, $2 \times 0.5 \mu\text{Ci/kg}$, $2 \times <0.75 \mu\text{Ci/kg}$, $2 \times 0.75 \mu\text{Ci/kg}$, $2 \times <1.0 \mu\text{Ci/kg}$, $2 \times 1.0 \mu\text{Ci/kg}$, $2 \times <1.25 \mu\text{Ci/kg}$, $2 \times 1.25 \mu\text{Ci/kg}$, $2 \times <1.5 \mu\text{Ci/kg}$, or $2 \times 1.5 \mu\text{Ci/kg}$, where the fractions are administered one week apart; or (ii) 0.25 $\mu\text{Ci/kg}$, 0.5 $\mu\text{Ci/kg}$, 0.75 $\mu\text{Ci/kg}$, 1.0 $\mu\text{Ci/kg}$, 1.25 $\mu\text{Ci/kg}$, 1.5 $\mu\text{Ci/kg}$, 1.75 $\mu\text{Ci/kg}$, 2.0 $\mu\text{Ci/kg}$, 2.5 $\mu\text{Ci/kg}$, 3.0 $\mu\text{Ci/kg}$ or 3.5 $\mu\text{Ci/kg}$ when delivered in a single administration. As a further example, reduced human dosing regimens of ²²⁵Ac-HuM195 include those corresponding to 25%, 50% or 75% of the normal dosing regimen.

[0145] The terms “normal” human dosing regimen and “reduced” human dosing regimen also apply, *mutatis mutandis*, to any other alpha-emitting isotope-labeled agent with respect to its approved or otherwise customary dosing regimen.

Example 10—Dosing Scenario I for ²²⁵Ac-HuM195 and One of Vorinostat, Romidepsin, Belinostat, or Panobinostat

[0146] A human AML patient is treated according to the following regimen. One of vorinostat, romidepsin, belinostat, or panobinostat (referred to in this Example as “HDACi”) is orally administered according to its normal dosing regimen, accompanied by intravenous administration of ²²⁵Ac-HuM195 according to its normal dosing regimen (either single or fractional administration). In this Example and the others where applicable, the dosing regimens include the following embodiments, by way of example: (a) the HDACi and antibody radioconjugate are administered concurrently, wherein (i) each is administered beginning on the same day, (ii) the antibody is administered in a single dose or fractionated doses not less than one week apart, and (iii) the HDACi is administered daily or less frequently (as appropriate), and for a duration equal to or exceeding that of the antibody administration; or (b) the HDACi and antibody

radioconjugate are administered concurrently, wherein (i) the HDACi administration precedes antibody radioconjugate administration by at least one week, (ii) the antibody is administered in a single dose or fractionated doses not less than one week apart, and (iii) the HDACi is administered daily or less frequently (as appropriate), and for a duration equal to or exceeding that of the antibody administration.

[0147] Also envisioned is the treatment of an experimental mouse model according to the treatment regimen in this scenario, whereby the appropriate dosing regimens are commensurate with mouse body weight and tumor xenograft size.

Example 11—Dosing Scenario II for
²²⁵Ac-HuM195 and One of Vorinostat, Romidepsin,
 Belinostat, or Panobinostat

[0148] A human AML patient is treated according to the following regimen. One of vorinostat, romidepsin, belinostat, or panobinostat (referred to in this Example as “HDACi”) is orally administered according to its normal dosing regimen, accompanied by intravenous administration of ²²⁵Ac-HuM195 according to a reduced dosing regimen (either single or fractional administration). In one embodiment, the reduced dosing regimen of ²²⁵Ac-HuM195 is (i) 2×0.5 μCi/kg, 2×1.0 μCi/kg, or 2×1.5 μCi/kg, where the fractions are administered one week apart; or (ii) 1×0.5 μCi/kg, 1×1.0 μCi/kg, 1×2.0 μCi/kg, or 1×3.0 μCi/kg, for a single administration. In another embodiment, the reduced human dosing regimen of HDACi includes that corresponding to 25%, 50% or 75% of its respective normal dosing regimen.

[0149] Also envisioned is the treatment of an experimental mouse model according to the treatment regimen in this scenario, whereby the appropriate dosing regimens are commensurate with mouse body weight and tumor xenograft size.

Example 12—Dosing Scenario III for
²²⁵Ac-HuM195 and One of Vorinostat, Romidepsin,
 Belinostat, or Panobinostat

[0150] A human AML patient is treated according to the following regimen. One of vorinostat, romidepsin, belinostat, or panobinostat (referred to in this Example as “HDACi”) is orally administered according to a reduced dosing regimen, accompanied by intravenous administration of the normal dosing regimen of ²²⁵Ac-HuM195 (either single or fractional administration). In one embodiment, the reduced dosing regimen of HDACi is one of the following: (i) where the HDACi is vorinostat, the reduced dosing regimen is 350 mg/day, 300 mg/day, 250 mg/day, 200 mg/day, 150 mg/day, 100 mg/day, 75 mg/day, 50 mg/day, or 25 mg/day; (ii) where the HDACi is romidepsin, the reduced dosing regimen is 13 mg/m² per administration, 12 mg/m² per administration, 11 mg/m² per administration, 10 mg/m² per administration, 9 mg/m² per administration, 8 mg/m² per administration, 7 mg/m² per administration, 6 mg/m² per administration, 5 mg/m² per administration, 4 mg/m² per administration, 3 mg/m² per administration, 2 mg/m² per administration, 1 mg/m² per administration, or 0.5 mg/m² per administration; (iii) where the HDACi is belinostat, the reduced dosing regimen is 900 mg/m² per administration, 800 mg/m² per administration, 700 mg/m² per administration, 600 mg/m² per administration, 500 mg/m² per admin-

istration, 400 mg/m² per administration, 300 mg/m² per administration, 250 mg/m² per administration, 200 mg/m² per administration, 150 mg/m² per administration, 100 mg/m² per administration, 75 mg/m² per administration, 50 mg/m² per administration, or 25 mg/m² per administration; and (iv) where the HDACi is panobinostat, the reduced dosing regimen is 15 mg/dose, 12.5 mg/dose, 10 mg/dose, 7.5 mg/dose, 5 mg/dose, 4 mg/dose, 3 mg/dose, 2 mg/dose, 1 mg/dose, or 0.5 mg/dose. In another embodiment, the reduced human dosing regimen of vorinostat, romidepsin, belinostat, and panobinostat includes that corresponding to 25%, 50% or 75% of its respective normal dosing regimen.

[0151] Also envisioned is the treatment of an experimental mouse model according to the treatment regimen in this scenario, whereby the appropriate dosing regimens are commensurate with mouse body weight and tumor xenograft size.

Example 13—Dosing Scenario IV for
²²⁵Ac-HuM195 and One of Vorinostat, Romidepsin,
 Belinostat, or Panobinostat

[0152] A human AML patient is treated according to the following regimen. One of vorinostat, romidepsin, belinostat, or panobinostat (referred to in this Example as “HDACi”) is orally administered according to a reduced dosing regimen, accompanied by intravenous administration of a reduced dosing regimen of ²²⁵Ac-HuM195 (either single or fractional administration). In one embodiment, (a) the reduced dosing regimen of ²²⁵Ac-HuM195 is one of (i) 2×0.5 μCi/kg, 2×1.0 μCi/kg, or 2×1.5 μCi/kg, where the fractions are administered one week apart; or (ii) 1×0.5 μCi/kg, 1×1.0 μCi/kg, 1×2.0 μCi/kg, or 1×3.0 μCi/kg, for a single administration, and (b) the reduced dosing regimen of HDACi is one of the following: (i) where the HDACi is vorinostat, the reduced dosing regimen is 350 mg/day, 300 mg/day, 250 mg/day, 200 mg/day, 150 mg/day, 100 mg/day, 75 mg/day, 50 mg/day, or 25 mg/day; (ii) where the HDACi is romidepsin, the reduced dosing regimen is 13 mg/m² per administration, 12 mg/m² per administration, 11 mg/m² per administration, 10 mg/m² per administration, 9 mg/m² per administration, 8 mg/m² per administration, 7 mg/m² per administration, 6 mg/m² per administration, 5 mg/m² per administration, 4 mg/m² per administration, 3 mg/m² per administration, 2 mg/m² per administration, 1 mg/m² per administration, or 0.5 mg/m² per administration; (iii) where the HDACi is belinostat, the reduced dosing regimen is 900 mg/m² per administration, 800 mg/m² per administration, 700 mg/m² per administration, 600 mg/m² per administration, 500 mg/m² per administration, 400 mg/m² per administration, 300 mg/m² per administration, 250 mg/m² per administration, 200 mg/m² per administration, 150 mg/m² per administration, 100 mg/m² per administration, 75 mg/m² per administration, 50 mg/m² per administration, or 25 mg/m² per administration; and (iv) where the HDACi is panobinostat, the reduced dosing regimen is 15 mg/dose, 12.5 mg/dose, 10 mg/dose, 7.5 mg/dose, 5 mg/dose, 4 mg/dose, 3 mg/dose, 2 mg/dose, 1 mg/dose, or 0.5 mg/dose. In another embodiment, the reduced human dosing regimen of vorinostat, romidepsin, belinostat, and panobinostat includes that corresponding to 25%, 50% or 75% of its respective normal dosing regimen.

[0153] Also envisioned is the treatment of an experimental mouse model according to the treatment regimen in this

scenario, whereby the appropriate dosing regimens are commensurate with mouse body weight and tumor xenograft size.

Example 14—Dosing Scenarios for ^{225}Ac -Labeled Anti-CD38 mAb and Vorinostat Or Romidepsin in Treating CTCL

[0154] Envisioned in this invention is an embodiment of the present therapeutic method wherein the hematologic malignancy is cutaneous T-cell lymphoma (CTCL), the radioisotope-labeled agent is an anti-CD38 antibody, such as Daratumumab and Isatuximab, and the HDAC inhibitor is vorinostat or romidepsin. A human CTCL patient may be treated according to one of the following two regimens.

[0155] In the first regimen, vorinostat is orally administered according to a reduced dosing regimen, accompanied by intravenous administration of a reduced dosing regimen of ^{225}Ac -Daratumumab or ^{225}Ac Isatuximab (either single or fractional administration). In one embodiment, (a) the reduced dosing regimen of ^{225}Ac -labeled antibody is one of (i) $2\times 0.5\ \mu\text{Ci/kg}$, $2\times 1.0\ \mu\text{Ci/kg}$, or $2\times 1.5\ \mu\text{Ci/kg}$, where the fractions are administered one week apart; or (ii) $1\times 0.5\ \mu\text{Ci/kg}$, $1\times 1.0\ \mu\text{Ci/kg}$, $1\times 2.0\ \mu\text{Ci/kg}$, or $1\times 3.0\ \mu\text{Ci/kg}$, for a single administration, and (b) the reduced dosing regimen of vorinostat is 350 mg/day, 300 mg/day, 250 mg/day, 200 mg/day, 150 mg/day, 100 mg/day, 75 mg/day, 50 mg/day, or 25 mg/day. In another embodiment, the reduced human dosing regimen of vorinostat includes that corresponding to 25%, 50% or 75% of its respective normal dosing regimen.

[0156] In the second regimen, romidepsin is intravenously administered according to a reduced dosing regimen, accompanied by intravenous administration of a reduced dosing regimen of the ^{225}Ac -labeled antibody (either single or fractional administration). In one embodiment, (a) the reduced dosing regimen of the ^{225}Ac -labeled antibody is one of (i) $2\times 0.5\ \mu\text{Ci/kg}$, $2\times 1.0\ \mu\text{Ci/kg}$, or $2\times 1.5\ \mu\text{Ci/kg}$, where the fractions are administered one week apart; or (ii) $1\times 0.5\ \mu\text{Ci/kg}$, $1\times 1.0\ \mu\text{Ci/kg}$, $1\times 2.0\ \mu\text{Ci/kg}$, or $1\times 3.0\ \mu\text{Ci/kg}$, for a single administration, and (b) the reduced dosing regimen of romidepsin is 13 mg/m² per administration, 12 mg/m² per administration, 11 mg/m² per administration, 10 mg/m² per administration, 9 mg/m² per administration, 8 mg/m² per administration, 7 mg/m² per administration, 6 mg/m² per administration, 5 mg/m² per administration, 4 mg/m² per administration, 3 mg/m² per administration, 2 mg/m² per administration, 1 mg/m² per administration, or 0.5 mg/m² per administration. In another embodiment, the reduced human dosing regimen of romidepsin includes that corresponding to 25%, 50% or 75% of its respective normal dosing regimen.

[0157] Also envisioned is the treatment of an experimental mouse model according to the treatment regimens in this scenario, whereby the appropriate dosing regimens are commensurate with mouse body weight and tumor xenograft size.

Example 15—Dosing Scenario for ^{225}Ac -Labeled Anti-CD38 mAb and Belinostat in Treating PTCL

[0158] Envisioned in this invention is an embodiment of the present therapeutic method wherein the hematologic malignancy is peripheral T-cell lymphoma (PTCL), the radioisotope-labeled agent is an anti-CD38 antibody, and the HDAC inhibitor is belinostat. A human PTCL patient may

be treated according to the following regimen. Belinostat is intravenously administered according to a reduced dosing regimen, accompanied by intravenous administration of a reduced dosing regimen of ^{225}Ac -Daratumumab or ^{225}Ac Isatuximab (either single or fractional administration). In one embodiment, (a) the reduced dosing regimen of ^{225}Ac -labeled antibody is one of (i) $2\times 0.5\ \mu\text{Ci/kg}$, $2\times 1.0\ \mu\text{Ci/kg}$, or $2\times 1.5\ \mu\text{Ci/kg}$, where the fractions are administered one week apart; or (ii) $1\times 0.5\ \mu\text{Ci/kg}$, $1\times 1.0\ \mu\text{Ci/kg}$, $1\times 2.0\ \mu\text{Ci/kg}$, or $1\times 3.0\ \mu\text{Ci/kg}$, for a single administration, and (b) the reduced dosing regimen of belinostat is 900 mg/m² per administration, 800 mg/m² per administration, 700 mg/m² per administration, 600 mg/m² per administration, 500 mg/m² per administration, 400 mg/m² per administration, 300 mg/m² per administration, 250 mg/m² per administration, 200 mg/m² per administration, 150 mg/m² per administration, 100 mg/m² per administration, 75 mg/m² per administration, 50 mg/m² per administration, or 25 mg/m² per administration. In another embodiment, the reduced human dosing regimen of belinostat includes that corresponding to 25%, 50% or 75% of its respective normal dosing regimen.

[0159] Also envisioned is the treatment of an experimental mouse model according to the treatment regimen in this scenario, whereby the appropriate dosing regimens are commensurate with mouse body weight and tumor xenograft size.

[0160] Any and all publications, patents, patent applications and other documents cited in this application are hereby incorporated by reference in their entireties for all purposes to the same extent as if each individual publication, patent, patent application or other document were individually indicated to be incorporated by reference for all purposes.

[0161] It should be understood that wherever in this disclosure an embodiment of the invention or an element or step thereof is described in terms of “including,” “include (s),” “comprising,” or “comprise(s),” corresponding embodiments, elements or steps thereof expressed, instead, in terms of “consisting essentially of” or “consisting of” are also intended to be disclosed and provided by this disclosure.

[0162] While various specific embodiments have been illustrated and described herein, it will be appreciated that various changes can be made without departing from the spirit and scope of the invention(s). Moreover, features described in connection with one embodiment of the invention may be used in conjunction with other embodiments, even if not explicitly exemplified in combination within.

REFERENCES

- [0163]** 1. Gansow, et al., U.S. Pat. No. 4,923,985.
- [0164]** 2. Scheinberg, et al., U.S. Pat. No. 6,683,162.
- [0165]** 3. Brandwein, et al., 2007. *Leukemia*. 21:821-4.
- [0166]** 4. Co, et al., *J. Immunol.* 148:1149-1154, 1992.
- [0167]** 5. Eckschlager, et al., *Int. J. Mol. Sci.* 2017, 18, 1414.
- [0168]** 6. McDevitt, *Applied Radiation and Isotope*, 57 (2002), 841-847.

SEQUENCE LISTING

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<210> SEQ ID NO 1

<211> LENGTH: 5

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<213> ORGANISM: Mus musculus

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<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

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Gly

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<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

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<210> SEQ ID NO 4

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

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1 5 10 15

<210> SEQ ID NO 5

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 5

Lys Val Ser Asn Arg Phe Ser
1 5

<210> SEQ ID NO 6

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

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<210> SEQ ID NO 7

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

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<400> SEQUENCE: 7

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Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Arg Pro Gly Thr
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Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser His
          20           25           30
Trp Leu His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
          35           40           45
Gly Val Leu Asp Pro Ser Asp Phe Tyr Ser Asn Tyr Asn Gln Asn Phe
          50           55           60
Lys Gly Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser Thr Ala Tyr
65           70           75           80
Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
          85           90           95
Ala Arg Gly Leu Leu Ser Gly Asp Tyr Ala Met Asp Tyr Trp Gly Ala
          100          105          110
Gly Thr Ser Val Thr Val Ser Ser
          115          120

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<210> SEQ ID NO 8

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 8

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Asp Val Leu Met Thr Gln Ile Pro Leu Ser Leu Pro Val Ser Leu Gly
1           5           10           15
Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile Val His Ser
          20           25           30
Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro Gly Gln Ser
          35           40           45
Pro Lys Ser Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser Gly Val Pro
          50           55           60
Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65           70           75           80
Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys Phe Gln Gly
          85           90           95
Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
          100          105          110

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<210> SEQ ID NO 9

<211> LENGTH: 463

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 9

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Met Gly Trp Ser Cys Ile Ile Val Leu Leu Val Ser Thr Ala Thr Gly
1           5           10           15
Val His Ser Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Arg
          20           25           30
Pro Gly Thr Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe
          35           40           45
Thr Ser His Trp Leu His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu
          50           55           60
Glu Trp Ile Gly Val Leu Asp Pro Ser Asp Phe Tyr Ser Asn Tyr Asn

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65		70		75		80
Gln Asn Phe Lys Gly Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser						
	85			90		95
Thr Ala Tyr Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val						
	100			105		110
Tyr Tyr Cys Ala Arg Gly Leu Leu Ser Gly Asp Tyr Ala Met Asp Tyr						
	115			120		125
Trp Gly Gln Gly Thr Ser Val Thr Val Ser Ser Ala Lys Thr Thr Pro						
	130			135		140
Pro Ser Val Tyr Pro Leu Ala Pro Gly Ser Ala Ala Gln Thr Asn Ser						
	145			150		155
Met Val Thr Leu Gly Cys Leu Val Lys Gly Tyr Phe Pro Glu Pro Val						
	165			170		175
Thr Val Thr Trp Asn Ser Gly Ser Leu Ser Ser Gly Val His Thr Phe						
	180			185		190
Pro Ala Val Leu Gln Ser Asp Leu Tyr Thr Leu Ser Ser Ser Val Thr						
	195			200		205
Val Pro Ser Ser Thr Trp Pro Ser Gln Thr Val Thr Cys Asn Val Ala						
	210			215		220
His Pro Ala Ser Ser Thr Lys Val Asp Lys Lys Ile Val Pro Arg Asp						
	225			230		235
Cys Gly Cys Lys Pro Cys Ile Cys Thr Val Pro Glu Val Ser Ser Val						
	245			250		255
Phe Ile Phe Pro Pro Lys Pro Lys Asp Val Leu Thr Ile Thr Leu Thr						
	260			265		270
Pro Lys Val Thr Cys Val Val Val Asp Ile Ser Lys Asp Asp Pro Glu						
	275			280		285
Val Gln Phe Ser Trp Phe Val Asp Asp Val Glu Val His Thr Ala Gln						
	290			295		300
Thr Gln Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Ser Val Ser						
	305			310		315
Glu Leu Pro Ile Met His Gln Asp Trp Leu Asn Gly Lys Glu Phe Lys						
	325			330		335
Cys Arg Val Asn Ser Ala Ala Phe Pro Ala Pro Ile Glu Lys Thr Ile						
	340			345		350
Ser Lys Thr Lys Gly Arg Pro Lys Ala Pro Gln Val Tyr Thr Ile Pro						
	355			360		365
Pro Pro Lys Glu Gln Met Ala Lys Asp Lys Val Ser Leu Thr Cys Met						
	370			375		380
Ile Thr Asp Phe Phe Pro Glu Asp Ile Thr Val Glu Trp Gln Trp Asn						
	385			390		395
Gly Gln Pro Ala Glu Asn Tyr Lys Asn Thr Gln Pro Ile Met Asp Thr						
	405			410		415
Asp Gly Ser Tyr Phe Val Tyr Ser Lys Leu Asn Val Gln Lys Ser Asn						
	420			425		430
Trp Glu Ala Gly Asn Thr Phe Thr Cys Ser Val Leu His Glu Gly Leu						
	435			440		445
His Asn His His Thr Glu Lys Ser Leu Ser His Ser Pro Gly Lys						
	450			455		460

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<211> LENGTH: 238
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<213> ORGANISM: Mus musculus

<400> SEQUENCE: 10
Met Lys Leu Pro Val Arg Leu Leu Val Leu Met Phe Trp Ile Pro Ala
1          5          10          15
Ser Ser Ser Asp Val Leu Met Thr Gln Ile Pro Leu Ser Leu Pro Val
20          25          30
Ser Leu Gly Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile
35          40          45
Val His Ser Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro
50          55          60
Gly Gln Ser Pro Lys Ser Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser
65          70          75          80
Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr
85          90          95
Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys
100         105         110
Phe Gln Gly Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu
115         120         125
Glu Ile Lys Arg Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro
130         135         140
Ser Ser Glu Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu
145         150         155         160
Asn Asn Phe Tyr Pro Arg Asp Ile Asn Val Lys Trp Lys Ile Asp Gly
165         170         175
Ser Glu Arg Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser
180         185         190
Lys Asp Ser Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp
195         200         205
Glu Tyr Glu Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr
210         215         220
Ser Thr Ser Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys
225         230         235

<210> SEQ ID NO 11
<211> LENGTH: 444
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 11
Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Arg Pro Gly Thr
1          5          10          15
Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser His
20          25          30
Trp Leu His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
35          40          45
Gly Val Leu Asp Pro Ser Asp Phe Tyr Ser Asn Tyr Asn Gln Asn Phe
50          55          60
Lys Gly Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser Thr Ala Tyr
65          70          75          80
Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
85          90          95

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Ala Arg Gly Leu Leu Ser Gly Asp Tyr Ala Met Asp Tyr Trp Gly Gln
 100 105 110
 Gly Thr Ser Val Thr Val Ser Ser Ala Lys Thr Thr Pro Pro Ser Val
 115 120 125
 Tyr Pro Leu Ala Pro Gly Ser Ala Ala Gln Thr Asn Ser Met Val Thr
 130 135 140
 Leu Gly Cys Leu Val Lys Gly Tyr Phe Pro Glu Pro Val Thr Val Thr
 145 150 155 160
 Trp Asn Ser Gly Ser Leu Ser Ser Gly Val His Thr Phe Pro Ala Val
 165 170 175
 Leu Gln Ser Asp Leu Tyr Thr Leu Ser Ser Ser Val Thr Val Pro Ser
 180 185 190
 Ser Thr Trp Pro Ser Gln Thr Val Thr Cys Asn Val Ala His Pro Ala
 195 200 205
 Ser Ser Thr Lys Val Asp Lys Lys Ile Val Pro Arg Asp Cys Gly Cys
 210 215 220
 Lys Pro Cys Ile Cys Thr Val Pro Glu Val Ser Ser Val Phe Ile Phe
 225 230 235 240
 Pro Pro Lys Pro Lys Asp Val Leu Thr Ile Thr Leu Thr Pro Lys Val
 245 250 255
 Thr Cys Val Val Val Asp Ile Ser Lys Asp Asp Pro Glu Val Gln Phe
 260 265 270
 Ser Trp Phe Val Asp Asp Val Glu Val His Thr Ala Gln Thr Gln Pro
 275 280 285
 Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Ser Val Ser Glu Leu Pro
 290 295 300
 Ile Met His Gln Asp Trp Leu Asn Gly Lys Glu Phe Lys Cys Arg Val
 305 310 315 320
 Asn Ser Ala Ala Phe Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr
 325 330 335
 Lys Gly Arg Pro Lys Ala Pro Gln Val Tyr Thr Ile Pro Pro Pro Lys
 340 345 350
 Glu Gln Met Ala Lys Asp Lys Val Ser Leu Thr Cys Met Ile Thr Asp
 355 360 365
 Phe Phe Pro Glu Asp Ile Thr Val Glu Trp Gln Trp Asn Gly Gln Pro
 370 375 380
 Ala Glu Asn Tyr Lys Asn Thr Gln Pro Ile Met Asp Thr Asp Gly Ser
 385 390 395 400
 Tyr Phe Val Tyr Ser Lys Leu Asn Val Gln Lys Ser Asn Trp Glu Ala
 405 410 415
 Gly Asn Thr Phe Thr Cys Ser Val Leu His Glu Gly Leu His Asn His
 420 425 430
 His Thr Glu Lys Ser Leu Ser His Ser Pro Gly Lys
 435 440

<210> SEQ ID NO 12

<211> LENGTH: 219

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 12

Asp Val Leu Met Thr Gln Ile Pro Leu Ser Leu Pro Val Ser Leu Gly


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1           5           10           15
Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile Val His Ser
      20           25           30

Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro Gly Gln Ser
      35           40           45

Pro Lys Ser Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser Gly Val Pro
      50           55           60

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65           70           75           80

Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys Phe Gln Gly
      85           90           95

Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
      100          105          110

Arg Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro Ser Ser Glu
      115          120          125

Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu Asn Asn Phe
130          135          140

Tyr Pro Arg Asp Ile Asn Val Lys Trp Lys Ile Asp Gly Ser Glu Arg
145          150          155          160

Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser Lys Asp Ser
      165          170          175

Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp Glu Tyr Glu
      180          185          190

Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr Ser Thr Ser
      195          200          205

Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys
      210          215

<210> SEQ ID NO 13
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 13

Gln Val Gln Leu Gln Gln Pro Gly Ala Glu
1           5           10

<210> SEQ ID NO 14
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 14

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<210> SEQ ID NO 15
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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<400> SEQUENCE: 15

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 16

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1 5 10 15

Gly

<210> SEQ ID NO 17
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 17

Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr
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<210> SEQ ID NO 18
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 18

Arg Ala Ser Gln Glu Ile Ser Gly Tyr Leu Ser
1 5 10

<210> SEQ ID NO 19
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 19

Ala Ala Ser Thr Leu Asp Ser
1 5

<210> SEQ ID NO 20
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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Leu Gln Tyr Asp Ser Tyr Pro Tyr Thr
1 5

<210> SEQ ID NO 21
<211> LENGTH: 119

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<212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 21

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 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
 20 25 30
 Ala Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly
 100 105 110
 Thr Leu Val Thr Val Ser Ser
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 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 22

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 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
 20 25 30
 Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Arg Leu Ile
 35 40 45
 Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 23
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 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 23

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	20	25	30
Leu Val Lys	Pro Gly Gly Ser	Leu Arg Leu Ser Cys Ala Ala Ser Gly	
	35	40	45
Phe Thr Phe	Ser Asp Tyr Ala Met Ser Trp Ile Arg Gln Ala Pro Gly		
	50	55	60
Lys Gly Leu Glu Trp Val Ser Thr Ile Ser Asp Gly Gly Thr Tyr Thr			
65	70	75	80
Tyr Tyr Pro Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn			
	85	90	95
Ala Lys Asn Ser Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp			
	100	105	110
Thr Ala Val Tyr Tyr Cys Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe			
	115	120	125
Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr			
	130	135	140
Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser			
145	150	155	160
Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu			
	165	170	175
Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His			
	180	185	190
Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser			
	195	200	205
Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys			
	210	215	220
Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu			
225	230	235	240
Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro			
	245	250	255
Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys			
	260	265	270
Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val			
	275	280	285
Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp			
	290	295	300
Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr			
305	310	315	320
Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp			
	325	330	335
Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu			
	340	345	350
Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg			
	355	360	365
Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys			
	370	375	380
Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp			
385	390	395	400
Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys			
	405	410	415

-continued

Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser
420 425 430

Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser
435 440 445

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
450 455 460

Leu Ser Leu Ser Pro Gly Lys
465 470

<210> SEQ ID NO 24

<211> LENGTH: 236

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 24

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
1 5 10 15

Leu Arg Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
20 25 30

Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
35 40 45

Gln Glu Ile Ser Gly Tyr Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys
50 55 60

Ala Pro Lys Arg Leu Ile Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val
65 70 75 80

Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr
85 90 95

Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln
100 105 110

Tyr Asp Ser Tyr Pro Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile
115 120 125

Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
130 135 140

Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
145 150 155 160

Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
165 170 175

Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
180 185 190

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
195 200 205

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
210 215 220

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
225 230 235

<210> SEQ ID NO 25

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

peptide

<400> SEQUENCE: 25

Ser His Trp Leu His
1 5

<210> SEQ ID NO 26

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 26

Val Leu Asp Pro Ser Asp Phe Tyr Ser Asn Tyr Asn Gln Asn Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 27

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 27

Gly Leu Leu Ser Gly Asp Tyr Ala Met Asp Tyr
1 5 10

<210> SEQ ID NO 28

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 28

Arg Ser Ser Gln Ser Ile Val His Ser Asn Gly Asn Thr Tyr Leu Glu
1 5 10 15

<210> SEQ ID NO 29

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 29

Lys Val Ser Asn Arg Phe Ser
1 5

<210> SEQ ID NO 30

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 30

-continued

Phe Gln Gly Ser Tyr Val Pro Trp Thr
1 5

<210> SEQ ID NO 31
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 31

Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Arg Pro Gly Thr
 1 5 10 15
 Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser His
 20 25 30
 Trp Leu His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Val Leu Asp Pro Ser Asp Phe Tyr Ser Asn Tyr Asn Gln Asn Phe
 50 55 60
 Lys Gly Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser Thr Ala Tyr
 65 70 75 80
 Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Gly Leu Leu Ser Gly Asp Tyr Ala Met Asp Tyr Trp Gly Gln
 100 105 110
 Gly Thr Ser Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 32
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 32

Asp Val Leu Met Thr Gln Ile Pro Leu Ser Leu Pro Val Ser Leu Gly
 1 5 10 15
 Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile Val His Ser
 20 25 30
 Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro Gly Gln Ser
 35 40 45
 Pro Lys Ser Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser Gly Val Pro
 50 55 60
 Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
 65 70 75 80
 Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys Phe Gln Gly
 85 90 95
 Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105 110

<210> SEQ ID NO 33
 <211> LENGTH: 463
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 33

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Met Gly Trp Ser Cys Ile Ile Val Leu Leu Val Ser Thr Ala Thr Gly
1      5      10      15
Val His Ser Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Arg
20     25     30
Pro Gly Thr Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe
35     40     45
Thr Ser His Trp Leu His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu
50     55     60
Glu Trp Ile Gly Val Leu Asp Pro Ser Asp Phe Tyr Ser Asn Tyr Asn
65     70     75     80
Gln Asn Phe Lys Gly Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser
85     90     95
Thr Ala Tyr Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val
100    105    110
Tyr Tyr Cys Ala Arg Gly Leu Leu Ser Gly Asp Tyr Ala Met Asp Tyr
115    120    125
Trp Gly Gln Gly Thr Ser Val Thr Val Ser Ser Ala Lys Thr Thr Pro
130    135    140
Pro Ser Val Tyr Pro Leu Ala Pro Gly Ser Ala Ala Gln Thr Asn Ser
145    150    155    160
Met Val Thr Leu Gly Cys Leu Val Lys Gly Tyr Phe Pro Glu Pro Val
165    170    175
Thr Val Thr Trp Asn Ser Gly Ser Leu Ser Ser Gly Val His Thr Phe
180    185    190
Pro Ala Val Leu Gln Ser Asp Leu Tyr Thr Leu Ser Ser Ser Val Thr
195    200    205
Val Pro Ser Ser Thr Trp Pro Ser Gln Thr Val Thr Cys Asn Val Ala
210    215    220
His Pro Ala Ser Ser Thr Lys Val Asp Lys Lys Ile Val Pro Arg Asp
225    230    235    240
Cys Gly Cys Lys Pro Cys Ile Cys Thr Val Pro Glu Val Ser Ser Val
245    250    255
Phe Ile Phe Pro Pro Lys Pro Lys Asp Val Leu Thr Ile Thr Leu Thr
260    265    270
Pro Lys Val Thr Cys Val Val Val Asp Ile Ser Lys Asp Asp Pro Glu
275    280    285
Val Gln Phe Ser Trp Phe Val Asp Asp Val Glu Val His Thr Ala Gln
290    295    300
Thr Gln Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Ser Val Ser
305    310    315    320
Glu Leu Pro Ile Met His Gln Asp Trp Leu Asn Gly Lys Glu Phe Lys
325    330    335
Cys Arg Val Asn Ser Ala Ala Phe Pro Ala Pro Ile Glu Lys Thr Ile
340    345    350
Ser Lys Thr Lys Gly Arg Pro Lys Ala Pro Gln Val Tyr Thr Ile Pro
355    360    365
Pro Pro Lys Glu Gln Met Ala Lys Asp Lys Val Ser Leu Thr Cys Met
370    375    380

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Ile Thr Asp Phe Phe Pro Glu Asp Ile Thr Val Glu Trp Gln Trp Asn
385                390                395                400

Gly Gln Pro Ala Glu Asn Tyr Lys Asn Thr Gln Pro Ile Met Asp Thr
                405                410                415

Asp Gly Ser Tyr Phe Val Tyr Ser Lys Leu Asn Val Gln Lys Ser Asn
                420                425                430

Trp Glu Ala Gly Asn Thr Phe Thr Cys Ser Val Leu His Glu Gly Leu
                435                440                445

His Asn His His Thr Glu Lys Ser Leu Ser His Ser Pro Gly Lys
450                455                460

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<210> SEQ ID NO 34
<211> LENGTH: 238
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

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<400> SEQUENCE: 34

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Met Lys Leu Pro Val Arg Leu Leu Val Leu Met Phe Trp Ile Pro Ala
1                5                10                15

Ser Ser Ser Asp Val Leu Met Thr Gln Ile Pro Leu Ser Leu Pro Val
20                25                30

Ser Leu Gly Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile
35                40                45

Val His Ser Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro
50                55                60

Gly Gln Ser Pro Lys Ser Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser
65                70                75                80

Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr
85                90                95

Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys
100               105               110

Phe Gln Gly Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu
115               120               125

Glu Ile Lys Arg Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro
130               135               140

Ser Ser Glu Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu
145               150               155               160

Asn Asn Phe Tyr Pro Arg Asp Ile Asn Val Lys Trp Lys Ile Asp Gly
165               170               175

Ser Glu Arg Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser
180               185               190

Lys Asp Ser Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp
195               200               205

Glu Tyr Glu Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr
210               215               220

Ser Thr Ser Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys
225               230               235

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<210> SEQ ID NO 35
<211> LENGTH: 7
<212> TYPE: PRT

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-continued

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 35

Thr Phe Gly Leu Ser Val Gly
1 5

<210> SEQ ID NO 36
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 36

His Ile Trp Trp Asp Asp Asp Lys Tyr Tyr Asn Pro Ala Leu Lys Ser
1 5 10 15

<210> SEQ ID NO 37
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 37

Ile Gly Ala Asp Ala Leu Pro Phe Asp Tyr
1 5 10

<210> SEQ ID NO 38
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 38

Arg Ser Ser Lys Ser Leu Leu His Ser Asn Gly Asn Thr Tyr Leu Tyr
1 5 10 15

<210> SEQ ID NO 39
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 39

Arg Met Ser Asn Leu Ala Ser
1 5

<210> SEQ ID NO 40
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 40

-continued

Met Gln His Leu Glu Tyr Pro Phe Thr
1 5

<210> SEQ ID NO 41
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 41

Gln Val Thr Leu Lys Glu Ser Gly Pro Gly Ile Leu Arg Pro Ser Gln
1 5 10 15
Thr Leu Ser Leu Thr Cys Ser Phe Ser Gly Phe Ser Leu Ser Thr Phe
20 25 30
Gly Leu Ser Val Gly Trp Ile Arg Gln Pro Ser Gly Lys Gly Leu Glu
35 40 45
Trp Leu Ala His Ile Trp Trp Asp Asp Asp Lys Tyr Tyr Asn Pro Ala
50 55 60
Leu Lys Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val
65 70 75 80
Phe Leu Lys Ile Ala Asn Val Asp Thr Ala Asp Thr Ala Thr Tyr Tyr
85 90 95
Cys Ala Arg Ile Gly Ala Asp Ala Leu Pro Phe Asp Tyr Trp Gly Gln
100 105 110
Gly Thr Thr Leu Thr Val Ser Ser
115 120

<210> SEQ ID NO 42
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 42

Asp Ile Val Leu Thr Gln Thr Ala Pro Ser Val Pro Val Thr Pro Gly
1 5 10 15
Glu Ser Val Ser Ile Ser Cys Arg Ser Ser Lys Ser Leu Leu His Ser
20 25 30
Asn Gly Asn Thr Tyr Leu Tyr Trp Phe Leu Gln Arg Pro Gly Gln Ser
35 40 45
Pro Gln Leu Leu Ile Tyr Arg Met Ser Asn Leu Ala Ser Gly Val Pro
50 55 60
Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Ala Phe Thr Leu Arg Ile
65 70 75 80
Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Met Gln His
85 90 95
Leu Glu Tyr Pro Phe Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys
100 105 110

<210> SEQ ID NO 43
<211> LENGTH: 475
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 43

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Met Gly Arg Leu Thr Ser Ser Phe Leu Leu Leu Ile Val Pro Ala Tyr
1           5           10          15

Val Leu Ser Gln Val Thr Leu Lys Glu Ser Gly Pro Gly Ile Leu Arg
20          25          30

Pro Ser Gln Thr Leu Ser Leu Thr Cys Ser Phe Ser Gly Phe Ser Leu
35          40          45

Ser Thr Phe Gly Leu Ser Val Gly Trp Ile Arg Gln Pro Ser Gly Lys
50          55          60

Gly Leu Glu Trp Leu Ala His Ile Trp Trp Asp Asp Asp Lys Tyr Tyr
65          70          75          80

Asn Pro Ala Leu Lys Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys
85          90          95

Asn Gln Val Phe Leu Lys Ile Ala Asn Val Asp Thr Ala Asp Thr Ala
100         105         110

Thr Tyr Tyr Cys Ala Arg Ile Gly Ala Asp Ala Leu Pro Phe Asp Tyr
115         120         125

Trp Gly Gln Gly Thr Thr Leu Thr Val Ser Ser Ala Lys Thr Thr Pro
130         135         140

Pro Ser Val Tyr Pro Leu Ala Pro Gly Cys Gly Asp Thr Thr Gly Ser
145         150         155         160

Ser Val Thr Ser Gly Cys Leu Val Lys Gly Tyr Phe Pro Glu Pro Val
165         170         175

Thr Val Thr Trp Asn Ser Gly Ser Leu Ser Ser Ser Val His Thr Phe
180         185         190

Pro Ala Leu Leu Gln Ser Gly Leu Tyr Thr Met Ser Ser Ser Val Thr
195         200         205

Val Pro Ser Ser Thr Trp Pro Ser Gln Thr Val Thr Cys Ser Val Ala
210         215         220

His Pro Ala Ser Ser Thr Thr Val Asp Lys Lys Leu Glu Pro Ser Gly
225         230         235         240

Pro Ile Ser Thr Ile Asn Pro Cys Pro Pro Cys Lys Glu Cys His Lys
245         250         255

Cys Pro Ala Pro Asn Leu Glu Gly Gly Pro Ser Val Phe Ile Phe Pro
260         265         270

Pro Asn Ile Lys Asp Val Leu Met Ile Ser Leu Thr Pro Lys Val Thr
275         280         285

Cys Val Val Val Asp Val Ser Glu Asp Asp Pro Asp Val Gln Ile Ser
290         295         300

Trp Phe Val Asn Asn Val Glu Val His Thr Ala Gln Thr Gln Thr His
305         310         315         320

Arg Glu Asp Tyr Asn Ser Thr Ile Arg Val Val Ser Thr Leu Pro Ile
325         330         335

Gln His Gln Asp Trp Met Ser Gly Lys Glu Phe Lys Cys Lys Val Asn
340         345         350

Asn Lys Asp Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ile Lys
355         360         365

Gly Leu Val Arg Ala Pro Gln Val Tyr Thr Leu Pro Pro Pro Ala Glu

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370	375	380
Gln Leu Ser Arg Lys Asp Val Ser Leu Thr Cys Leu Val Val Gly Phe		
385	390	395 400
Asn Pro Gly Asp Ile Ser Val Glu Trp Thr Ser Asn Gly His Thr Glu		
	405	410 415
Glu Asn Tyr Lys Asp Thr Ala Pro Val Leu Asp Ser Asp Gly Ser Tyr		
	420	425 430
Phe Ile Tyr Ser Lys Leu Asn Met Lys Thr Ser Lys Trp Glu Lys Thr		
	435	440 445
Asp Ser Phe Ser Cys Asn Val Arg His Glu Gly Leu Lys Asn Tyr Tyr		
	450	455 460
Leu Lys Lys Thr Ile Ser Arg Ser Pro Gly Lys		
465	470	475

<210> SEQ ID NO 44
 <211> LENGTH: 239
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 44

Met Arg Cys Leu Ala Glu Phe Leu Gly Leu Leu Val Leu Trp Ile Pro		
1	5	10 15
Gly Ala Ile Gly Asp Ile Val Leu Thr Gln Thr Ala Pro Ser Val Pro		
	20	25 30
Val Thr Pro Gly Glu Ser Val Ser Ile Ser Cys Arg Ser Ser Lys Ser		
	35	40 45
Leu Leu His Ser Asn Gly Asn Thr Tyr Leu Tyr Trp Phe Leu Gln Arg		
	50	55 60
Pro Gly Gln Ser Pro Gln Leu Leu Ile Tyr Arg Met Ser Asn Leu Ala		
65	70	75 80
Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Ala Phe		
	85	90 95
Thr Leu Arg Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr		
	100	105 110
Cys Met Gln His Leu Glu Tyr Pro Phe Thr Phe Gly Ser Gly Thr Lys		
	115	120 125
Leu Glu Ile Lys Arg Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro		
	130	135 140
Pro Ser Ser Glu Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe		
145	150	155 160
Leu Asn Asn Phe Tyr Pro Arg Asp Ile Asn Val Lys Trp Lys Ile Asp		
	165	170 175
Gly Ser Glu Arg Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp		
	180	185 190
Ser Lys Asp Ser Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys		
	195	200 205
Asp Glu Tyr Glu Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys		
	210	215 220
Thr Ser Thr Ser Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys		
225	230	235

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<210> SEQ ID NO 45
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 45

Asp His Ile Ile His
1 5

<210> SEQ ID NO 46
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 46

Tyr Ile Tyr Pro Arg Asp Gly Tyr Ile Lys Tyr Asn Glu Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 47
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 47

Gly Tyr Tyr Tyr Ala Met Asp Tyr
1 5

<210> SEQ ID NO 48
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 48

Arg Ser Ser Gln Ser Ile Val His Ser Ile Gly Asn Thr Tyr Leu Glu
1 5 10 15

<210> SEQ ID NO 49
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 49

Phe Gln Gly Ser His Val Pro Phe Thr
1 5

<210> SEQ ID NO 50
<211> LENGTH: 117
<212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 50

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Gln Val Gln Leu Gln Gln Ser Asp Ala Glu Leu Val Lys Pro Gly Ala
1           5           10          15
Ser Val Lys Ile Ser Cys Lys Val Ser Gly Tyr Thr Phe Thr Asp His
20          25          30
Ile Ile His Trp Met Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile
35          40          45
Gly Tyr Ile Tyr Pro Arg Asp Gly Tyr Ile Lys Tyr Asn Glu Lys Phe
50          55          60
Lys Gly Lys Ala Thr Leu Thr Ala Asp Lys Ser Ser Ser Thr Ala Tyr
65          70          75          80
Met Gln Val Asn Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Phe Cys
85          90          95
Ala Arg Gly Tyr Tyr Tyr Ala Met Asp Tyr Trp Gly Gln Gly Thr Ser
100         105         110
Val Thr Val Ser Ser
115

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<210> SEQ ID NO 51
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 51

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Asp Val Leu Met Thr Gln Thr Pro Leu Ser Leu Pro Val Ser Leu Gly
1           5           10          15
Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile Val His Ser
20          25          30
Ile Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35          40          45
Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser Gly Val Pro
50          55          60
Glu Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65          70          75          80
Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys Phe Gln Gly
85          90          95
Ser His Val Pro Phe Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys
100         105         110

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<210> SEQ ID NO 52
 <211> LENGTH: 460
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 52

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Met Glu Trp Ser Trp Val Ser Leu Phe Phe Leu Ser Val Thr Thr Gly
1           5           10          15

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Val	His	Ser	Gln	Val	Gln	Leu	Gln	Gln	Ser	Asp	Ala	Glu	Leu	Val	Lys	
			20					25					30			
Pro	Gly	Ala	Ser	Val	Lys	Ile	Ser	Cys	Lys	Val	Ser	Gly	Tyr	Thr	Phe	
		35						40				45				
Thr	Asp	His	Ile	Ile	His	Trp	Met	Lys	Gln	Arg	Pro	Glu	Gln	Gly	Leu	
	50					55					60					
Glu	Trp	Ile	Gly	Tyr	Ile	Tyr	Pro	Arg	Asp	Gly	Tyr	Ile	Lys	Tyr	Asn	
65					70					75					80	
Glu	Lys	Phe	Lys	Gly	Lys	Ala	Thr	Leu	Thr	Ala	Asp	Lys	Ser	Ser	Ser	
			85						90					95		
Thr	Ala	Tyr	Met	Gln	Val	Asn	Ser	Leu	Thr	Ser	Glu	Asp	Ser	Ala	Val	
			100					105						110		
Tyr	Phe	Cys	Ala	Arg	Gly	Tyr	Tyr	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	
		115					120					125				
Gly	Thr	Ser	Val	Thr	Val	Ser	Ser	Ala	Lys	Thr	Thr	Pro	Pro	Ser	Val	
	130					135						140				
Tyr	Pro	Leu	Ala	Pro	Gly	Ser	Ala	Ala	Gln	Thr	Asn	Ser	Met	Val	Thr	
145					150					155					160	
Leu	Gly	Cys	Leu	Val	Lys	Gly	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Thr	
			165						170					175		
Trp	Asn	Ser	Gly	Ser	Leu	Ser	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	
		180						185					190			
Leu	Gln	Ser	Asp	Leu	Tyr	Thr	Leu	Ser	Ser	Ser	Val	Thr	Val	Pro	Ser	
		195					200					205				
Ser	Thr	Trp	Pro	Ser	Gln	Thr	Val	Thr	Cys	Asn	Val	Ala	His	Pro	Ala	
	210					215					220					
Ser	Ser	Thr	Lys	Val	Asp	Lys	Lys	Ile	Val	Pro	Arg	Asp	Cys	Gly	Cys	
225				230						235					240	
Lys	Pro	Cys	Ile	Cys	Thr	Val	Pro	Glu	Val	Ser	Ser	Val	Phe	Ile	Phe	
			245					250						255		
Pro	Pro	Lys	Pro	Lys	Asp	Val	Leu	Thr	Ile	Thr	Leu	Thr	Pro	Lys	Val	
		260					265						270			
Thr	Cys	Val	Val	Val	Asp	Ile	Ser	Lys	Asp	Asp	Pro	Glu	Val	Gln	Phe	
	275					280					285					
Ser	Trp	Phe	Val	Asp	Asp	Val	Glu	Val	His	Thr	Ala	Gln	Thr	Gln	Pro	
	290					295					300					
Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Ser	Val	Ser	Glu	Leu	Pro	
305				310						315					320	
Ile	Met	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Phe	Lys	Cys	Arg	Val	
			325					330						335		
Asn	Ser	Ala	Ala	Phe	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	
		340						345					350			
Lys	Gly	Arg	Pro	Lys	Ala	Pro	Gln	Val	Tyr	Thr	Ile	Pro	Pro	Pro	Lys	
	355						360					365				
Glu	Gln	Met	Ala	Lys	Asp	Lys	Val	Ser	Leu	Thr	Cys	Met	Ile	Thr	Asp	
	370					375					380					
Phe	Phe	Pro	Glu	Asp	Ile	Thr	Val	Glu	Trp	Gln	Trp	Asn	Gly	Gln	Pro	
385					390					395					400	
Ala	Glu	Asn	Tyr	Lys	Asn	Thr	Gln	Pro	Ile	Met	Asp	Thr	Asp	Gly	Ser	
				405					410					415		

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Tyr Phe Val Tyr Ser Lys Leu Asn Val Gln Lys Ser Asn Trp Glu Ala
420 425 430

Gly Asn Thr Phe Thr Cys Ser Val Leu His Glu Gly Leu His Asn His
435 440 445

His Thr Glu Lys Ser Leu Ser His Ser Pro Gly Lys
450 455 460

<210> SEQ ID NO 53

<211> LENGTH: 238

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 53

Met Lys Leu Pro Val Arg Leu Leu Val Leu Met Phe Trp Ile Pro Ala
1 5 10 15

Ser Arg Ser Asp Val Leu Met Thr Gln Thr Pro Leu Ser Leu Pro Val
20 25 30

Ser Leu Gly Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile
35 40 45

Val His Ser Ile Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro
50 55 60

Gly Gln Ser Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser
65 70 75 80

Gly Val Pro Glu Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr
85 90 95

Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys
100 105 110

Phe Gln Gly Ser His Val Pro Phe Thr Phe Gly Ser Gly Thr Lys Leu
115 120 125

Glu Ile Lys Arg Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro
130 135 140

Ser Ser Glu Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu
145 150 155 160

Asn Asn Phe Tyr Pro Lys Asp Ile Asn Val Lys Trp Lys Ile Asp Gly
165 170 175

Ser Glu Arg Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser
180 185 190

Lys Asp Ser Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp
195 200 205

Glu Tyr Glu Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr
210 215 220

Ser Thr Ser Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys
225 230 235

<210> SEQ ID NO 54

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 54

-continued

Ser Tyr Trp Met His
1 5

<210> SEQ ID NO 55
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
 <400> SEQUENCE: 55

Met Ile Asp Pro Ser Asp Val Tyr Thr Asn Tyr Asn Pro Lys Phe Lys
 1 5 10 15

Gly

<210> SEQ ID NO 56
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
 <400> SEQUENCE: 56

Asn Tyr Ser Gly Asp Tyr
 1 5

<210> SEQ ID NO 57
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <400> SEQUENCE: 57

Gln Val Gln Leu Gln Pro Gly Ala Glu Leu Val Arg Pro Gly Thr
 1 5 10 15

Ser Val Lys Leu Ser Cys Lys Thr Ser Gly Tyr Thr Phe Ser Ser Tyr
 20 25 30

Trp Met His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Met Ile Asp Pro Ser Asp Val Tyr Thr Asn Tyr Asn Pro Lys Phe
 50 55 60

Lys Gly Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser Thr Ala Tyr
 65 70 75 80

Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Asn Tyr Ser Gly Asp Tyr Trp Gly Gln Gly Thr Thr Leu Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 58
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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<400> SEQUENCE: 58

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Asp Val Leu Met Thr Gln Ile Pro Leu Ser Leu Pro Val Ser Leu Gly
1           5           10           15
Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile Val His Ser
20           25           30
Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35           40           45
Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser Gly Val Pro
50           55           60
Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65           70           75           80
Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys Phe Gln Gly
85           90           95
Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100          105          110

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<210> SEQ ID NO 59

<211> LENGTH: 458

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 59

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Met Gly Trp Ser Cys Ile Ile Val Leu Leu Val Ser Thr Ala Thr Cys
1           5           10           15
Val His Ser Gln Val Gln Leu Leu Gln Pro Gly Ala Glu Leu Val Arg
20           25           30
Pro Gly Thr Ser Val Lys Leu Ser Cys Lys Thr Ser Gly Tyr Thr Phe
35           40           45
Ser Ser Tyr Trp Met His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu
50           55           60
Glu Trp Ile Gly Met Ile Asp Pro Ser Asp Val Tyr Thr Asn Tyr Asn
65           70           75           80
Pro Lys Phe Lys Gly Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser
85           90           95
Thr Ala Tyr Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val
100          105          110
Tyr Tyr Cys Ala Arg Asn Tyr Ser Gly Asp Tyr Trp Gly Gln Gly Thr
115          120          125
Thr Leu Thr Val Ser Ser Ala Lys Thr Thr Pro Pro Ser Val Tyr Pro
130          135          140
Leu Ala Pro Gly Ser Ala Ala Gln Thr Asn Ser Met Val Thr Leu Gly
145          150          155          160
Cys Leu Val Lys Gly Tyr Phe Pro Glu Pro Val Thr Val Thr Trp Asn
165          170          175
Ser Gly Ser Leu Ser Ser Gly Val His Thr Phe Pro Ala Val Leu Gln
180          185          190
Ser Asp Leu Tyr Thr Leu Ser Ser Ser Val Thr Val Pro Ser Ser Thr
195          200          205
Trp Pro Ser Gln Thr Val Thr Cys Asn Val Ala His Pro Ala Ser Ser
210          215          220

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Thr Lys Val Asp Lys Lys Ile Val Pro Arg Asp Cys Gly Cys Lys Pro
 225 230 235 240
 Cys Ile Cys Thr Val Pro Glu Val Ser Ser Val Phe Ile Phe Pro Pro
 245 250 255
 Lys Pro Lys Asp Val Leu Thr Ile Thr Leu Thr Pro Lys Val Thr Cys
 260 265 270
 Val Val Val Asp Ile Ser Lys Asp Asp Pro Glu Val Gln Phe Ser Trp
 275 280 285
 Phe Val Asp Asp Val Glu Val His Thr Ala Gln Thr Gln Pro Arg Glu
 290 295 300
 Glu Gln Phe Asn Ser Thr Phe Arg Ser Val Ser Glu Leu Pro Ile Met
 305 310 315 320
 His Gln Asp Trp Leu Asn Gly Lys Glu Phe Lys Cys Arg Val Asn Ser
 325 330 335
 Ala Ala Phe Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly
 340 345 350
 Arg Pro Lys Ala Pro Gln Val Tyr Thr Ile Pro Pro Pro Lys Glu Gln
 355 360 365
 Met Ala Lys Asp Lys Val Ser Leu Thr Cys Met Ile Thr Asp Phe Phe
 370 375 380
 Pro Glu Asp Ile Thr Val Glu Trp Gln Trp Asn Gly Gln Pro Ala Glu
 385 390 395 400
 Asn Tyr Lys Asn Thr Gln Pro Ile Met Asp Thr Asp Gly Ser Tyr Phe
 405 410 415
 Val Tyr Ser Lys Leu Asn Val Gln Lys Ser Asn Trp Glu Ala Gly Asn
 420 425 430
 Thr Phe Thr Cys Ser Val Leu His Glu Gly Leu His Asn His His Thr
 435 440 445
 Glu Lys Ser Leu Ser His Ser Pro Gly Lys
 450 455

<210> SEQ ID NO 60

<211> LENGTH: 238

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 60

Met Lys Leu Pro Val Arg Leu Leu Val Leu Met Phe Trp Ile Pro Ala
 1 5 10 15
 Ser Ser Ser Asp Val Leu Met Thr Gln Ile Pro Leu Ser Leu Pro Val
 20 25 30
 Ser Leu Gly Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile
 35 40 45
 Val His Ser Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro
 50 55 60
 Gly Gln Ser Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser
 65 70 75 80
 Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr
 85 90 95
 Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys

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100	105	110
Phe Gln Gly Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu		
115	120	125
Glu Ile Lys Arg Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro		
130	135	140
Ser Ser Glu Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu		
145	150	155
Asn Asn Phe Tyr Pro Arg Asp Ile Asn Val Lys Trp Lys Ile Asp Gly		
165	170	175
Ser Glu Arg Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser		
180	185	190
Lys Asp Ser Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp		
195	200	205
Glu Tyr Glu Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr		
210	215	220
Ser Thr Ser Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys		
225	230	235

<210> SEQ ID NO 61
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 61

Thr Tyr Gly Met Ser
1 5

<210> SEQ ID NO 62
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 62

Trp Ile Asn Thr Tyr Ser Gly Val Pro Thr Tyr Ala Asp Asp Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 63
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 63

Gly Arg Asp Gly Tyr Gln Val Ala Trp Phe Ala Tyr
1 5 10

<210> SEQ ID NO 64
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 64

Ile Thr Ser Thr Asp Ile Asp Asp Asp Met Asn
1 5 10

<210> SEQ ID NO 65

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 65

Glu Gly Asn Thr Leu Arg Pro
1 5

<210> SEQ ID NO 66

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 66

Leu Gln Ser Asp Asn Leu Pro Tyr Thr
1 5

<210> SEQ ID NO 67

<211> LENGTH: 121

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 67

Gln Ile Gln Leu Val Gln Ser Gly Pro Glu Leu Lys Lys Pro Gly Glu
1 5 10 15

Ala Val Lys Ile Ser Cys Lys Ser Ser Gly Tyr Thr Phe Thr Thr Tyr
20 25 30

Gly Met Ser Trp Val Lys Gln Ala Pro Gly Arg Ala Leu Lys Trp Met
35 40 45

Gly Trp Ile Asn Thr Tyr Ser Gly Val Pro Thr Tyr Ala Asp Asp Phe
50 55 60

Lys Gly Arg Phe Ala Phe Ser Leu Glu Ser Ser Ala Ser Thr Ala Tyr
65 70 75 80

Leu Gln Ile Asn Asn Leu Lys Asn Glu Asp Thr Ala Thr Tyr Phe Cys
85 90 95

Ala Arg Gly Arg Asp Gly Tyr Gln Val Ala Trp Phe Ala Tyr Trp Gly
100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ala
115 120

<210> SEQ ID NO 68

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 68

Glu Thr Thr Val Thr Gln Ser Pro Ala Ser Leu Ser Met Ala Ile Gly
 1 5 10 15
 Asp Lys Val Thr Ile Arg Cys Ile Thr Ser Thr Asp Ile Asp Asp Asp
 20 25 30
 Met Asn Trp Phe Gln Gln Lys Pro Gly Glu Pro Pro Lys Leu Leu Ile
 35 40 45
 Ser Glu Gly Asn Thr Leu Arg Pro Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Tyr Gly Thr Asp Phe Ile Phe Thr Ile Glu Asn Met Leu Ser
 65 70 75 80
 Glu Asp Val Ala Asp Tyr Tyr Cys Leu Gln Ser Asp Asn Leu Pro Tyr
 85 90 95
 Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 69

<211> LENGTH: 464

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 69

Met Gly Trp Leu Trp Asn Leu Leu Phe Leu Met Ala Ala Ala Gln Ser
 1 5 10 15
 Ala Gln Ala Gln Ile Gln Leu Val Gln Ser Gly Pro Glu Leu Lys Lys
 20 25 30
 Pro Gly Glu Ala Val Lys Ile Ser Cys Lys Ser Ser Gly Tyr Thr Phe
 35 40 45
 Thr Thr Tyr Gly Met Ser Trp Val Lys Gln Ala Pro Gly Arg Ala Leu
 50 55 60
 Lys Trp Met Gly Trp Ile Asn Thr Tyr Ser Gly Val Pro Thr Tyr Ala
 65 70 75 80
 Asp Asp Phe Lys Gly Arg Phe Ala Phe Ser Leu Glu Ser Ser Ala Ser
 85 90 95
 Thr Ala Tyr Leu Gln Ile Asn Asn Leu Lys Asn Glu Asp Thr Ala Thr
 100 105 110
 Tyr Phe Cys Ala Arg Gly Arg Asp Gly Tyr Gln Val Ala Trp Phe Ala
 115 120 125
 Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ala Ala Lys Thr Thr
 130 135 140
 Pro Pro Ser Val Tyr Pro Leu Ala Pro Gly Ser Ala Ala Gln Thr Asn
 145 150 155 160
 Ser Met Val Thr Leu Gly Cys Leu Val Lys Gly Tyr Phe Pro Glu Pro
 165 170 175
 Val Thr Val Thr Trp Asn Ser Gly Ser Leu Ser Ser Gly Val His Thr
 180 185 190
 Phe Pro Ala Val Leu Gln Ser Asp Leu Tyr Thr Leu Ser Ser Ser Val
 195 200 205

Thr	Val	Pro	Ser	Ser	Thr	Trp	Pro	Ser	Gln	Thr	Val	Thr	Cys	Asn	Val
210						215					220				
Ala	His	Pro	Ala	Ser	Ser	Thr	Lys	Val	Asp	Lys	Lys	Ile	Val	Pro	Arg
225					230					235				240	
Asp	Cys	Gly	Cys	Lys	Pro	Cys	Ile	Cys	Thr	Val	Pro	Glu	Val	Ser	Ser
				245					250					255	
Val	Phe	Ile	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Val	Leu	Thr	Ile	Thr	Leu
			260					265					270		
Thr	Pro	Lys	Val	Thr	Cys	Val	Val	Val	Asp	Ile	Ser	Lys	Asp	Asp	Pro
	275					280						285			
Glu	Val	Gln	Phe	Ser	Trp	Phe	Val	Asp	Asp	Val	Glu	Val	His	Thr	Ala
290						295					300				
Gln	Thr	Gln	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Ser	Val
305					310					315					320
Ser	Glu	Leu	Pro	Ile	Met	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Phe
				325					330					335	
Lys	Cys	Arg	Val	Asn	Ser	Ala	Ala	Phe	Pro	Ala	Pro	Ile	Glu	Lys	Thr
			340					345					350		
Ile	Ser	Lys	Thr	Lys	Gly	Arg	Pro	Lys	Ala	Pro	Gln	Val	Tyr	Thr	Ile
355						360					365				
Pro	Pro	Pro	Lys	Glu	Gln	Met	Ala	Lys	Asp	Lys	Val	Ser	Leu	Thr	Cys
370						375					380				
Met	Ile	Thr	Asp	Phe	Phe	Pro	Glu	Asp	Ile	Thr	Val	Glu	Trp	Gln	Trp
385					390					395					400
Asn	Gly	Gln	Pro	Ala	Glu	Asn	Tyr	Lys	Asn	Thr	Gln	Pro	Ile	Met	Asp
				405					410					415	
Thr	Asp	Gly	Ser	Tyr	Phe	Val	Tyr	Ser	Lys	Leu	Asn	Val	Gln	Lys	Ser
			420					425					430		
Asn	Trp	Glu	Ala	Gly	Asn	Thr	Phe	Thr	Cys	Ser	Val	Leu	His	Glu	Gly
	435					440					445				
Leu	His	Asn	His	His	Thr	Glu	Lys	Ser	Leu	Ser	His	Ser	Pro	Gly	Lys
450						455					460				

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<210> SEQ ID NO 70
<211> LENGTH: 234
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
```

<400> SEQUENCE: 70

Met	Phe	Ser	Leu	Ala	Leu	Leu	Leu	Ser	Leu	Leu	Leu	Leu	Cys	Val	Ser
1				5					10					15	
Asp	Ser	Arg	Ala	Glu	Thr	Thr	Val	Thr	Gln	Ser	Pro	Ala	Ser	Leu	Ser
			20					25					30		
Met	Ala	Ile	Gly	Asp	Lys	Val	Thr	Ile	Arg	Cys	Ile	Thr	Ser	Thr	Asp
		35					40					45			
Ile	Asp	Asp	Asp	Met	Asn	Trp	Phe	Gln	Gln	Lys	Pro	Gly	Glu	Pro	Pro
	50					55					60				
Lys	Leu	Leu	Ile	Ser	Glu	Gly	Asn	Thr	Leu	Arg	Pro	Gly	Val	Pro	Ser
65					70					75				80	
Arg	Phe	Ser	Gly	Ser	Gly	Tyr	Gly	Thr	Asp	Phe	Ile	Phe	Thr	Ile	Glu

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	85		90		95
Asn Met Leu Ser Glu Asp Val Ala Asp Tyr Tyr Cys Leu Gln Ser Asp					
	100		105		110
Asn Leu Pro Tyr Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg					
	115		120		125
Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro Ser Ser Glu Gln					
	130		135		140
Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu Asn Asn Phe Tyr					
145		150		155	160
Pro Arg Asp Ile Asn Val Lys Trp Lys Ile Asp Gly Ser Glu Arg Gln					
	165		170		175
Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser Lys Asp Ser Thr					
	180		185		190
Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp Glu Tyr Glu Arg					
	195		200		205
His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr Ser Thr Ser Pro					
	210		215		220
Ile Val Lys Ser Phe Asn Arg Asn Glu Cys					
225		230			

<210> SEQ ID NO 71
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

 <400> SEQUENCE: 71

 Asn Tyr Trp Met His
 1 5

<210> SEQ ID NO 72
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

 <400> SEQUENCE: 72

 Met Ile Asp Pro Ser Asp Ser Tyr Thr Asn Tyr Asn Pro Lys Phe Lys
 1 5 10 15

 Gly

<210> SEQ ID NO 73
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

 <400> SEQUENCE: 73

 Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Arg Pro Gly Thr
 1 5 10 15

 Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
 20 25 30

-continued

Trp Met His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Met Ile Asp Pro Ser Asp Ser Tyr Thr Asn Tyr Asn Pro Lys Phe
 50 55 60

Lys Gly Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser Thr Ala Tyr
 65 70 75 80

Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Asn Tyr Ser Gly Asp Tyr Trp Gly Gln Gly Thr Thr Leu Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 74
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 74

Asp Val Leu Met Thr Gln Thr Pro Leu Ser Leu Pro Val Ser Leu Gly
 1 5 10 15

Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile Val His Ser
 20 25 30

Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro Gly Gln Ser
 35 40 45

Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser Gly Val Pro
 50 55 60

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
 65 70 75 80

Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys Phe Gln Gly
 85 90 95

Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105 110

<210> SEQ ID NO 75
 <211> LENGTH: 458
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 75

Met Gly Trp Ser Cys Ile Ile Val Leu Leu Val Ser Thr Ala Thr Gly
 1 5 10 15

Val His Ser Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Arg
 20 25 30

Pro Gly Thr Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe
 35 40 45

Thr Asn Tyr Trp Met His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu
 50 55 60

Glu Trp Ile Gly Met Ile Asp Pro Ser Asp Ser Tyr Thr Asn Tyr Asn
 65 70 75 80

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Pro	Lys	Phe	Lys	Gly	Lys	Ala	Thr	Leu	Thr	Val	Asp	Thr	Ser	Ser	Ser	85	90	95
Thr	Ala	Tyr	Met	Gln	Leu	Ser	Ser	Leu	Thr	Ser	Glu	Asp	Ser	Ala	Val	100	105	110
Tyr	Tyr	Cys	Ala	Arg	Asn	Tyr	Ser	Gly	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	115	120	125
Thr	Leu	Thr	Val	Ser	Ser	Ala	Lys	Thr	Thr	Pro	Pro	Ser	Val	Tyr	Pro	130	135	140
Leu	Ala	Pro	Gly	Ser	Ala	Ala	Gln	Thr	Asn	Ser	Met	Val	Thr	Leu	Gly	145	150	155
Cys	Leu	Val	Lys	Gly	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Thr	Trp	Asn	165	170	175
Ser	Gly	Ser	Leu	Ser	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	180	185	190
Ser	Asp	Leu	Tyr	Thr	Leu	Ser	Ser	Ser	Val	Thr	Val	Pro	Ser	Ser	Thr	195	200	205
Trp	Pro	Ser	Gln	Thr	Val	Thr	Cys	Asn	Val	Ala	His	Pro	Ala	Ser	Ser	210	215	220
Thr	Lys	Val	Asp	Lys	Lys	Ile	Val	Pro	Arg	Asp	Cys	Gly	Cys	Lys	Pro	225	230	235
Cys	Ile	Cys	Thr	Val	Pro	Glu	Val	Ser	Ser	Val	Phe	Ile	Phe	Pro	Pro	245	250	255
Lys	Pro	Lys	Asp	Val	Leu	Thr	Ile	Thr	Leu	Thr	Pro	Lys	Val	Thr	Cys	260	265	270
Val	Val	Val	Asp	Ile	Ser	Lys	Asp	Asp	Pro	Glu	Val	Gln	Phe	Ser	Trp	275	280	285
Phe	Val	Asp	Asp	Val	Glu	Val	His	Thr	Ala	Gln	Thr	Gln	Pro	Arg	Glu	290	295	300
Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Ser	Val	Ser	Glu	Leu	Pro	Ile	Met	305	310	315
His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Phe	Lys	Cys	Arg	Val	Asn	Ser	325	330	335
Ala	Ala	Phe	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	340	345	350
Arg	Pro	Lys	Ala	Pro	Gln	Val	Tyr	Thr	Ile	Pro	Pro	Pro	Lys	Glu	Gln	355	360	365
Met	Ala	Lys	Asp	Lys	Val	Ser	Leu	Thr	Cys	Met	Ile	Thr	Asp	Phe	Phe	370	375	380
Pro	Glu	Asp	Ile	Thr	Val	Glu	Trp	Gln	Trp	Asn	Gly	Gln	Pro	Ala	Glu	385	390	395
Asn	Tyr	Lys	Asn	Thr	Gln	Pro	Ile	Met	Asp	Thr	Asp	Gly	Ser	Tyr	Phe	405	410	415
Val	Tyr	Ser	Lys	Leu	Asn	Val	Gln	Lys	Ser	Asn	Trp	Glu	Ala	Gly	Asn	420	425	430
Thr	Phe	Thr	Cys	Ser	Val	Leu	His	Glu	Gly	Leu	His	Asn	His	His	Thr	435	440	445
Glu	Lys	Ser	Leu	Ser	His	Ser	Pro	Gly	Lys							450	455	

<210> SEQ ID NO 76

<211> LENGTH: 238

<212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 76

Met Lys Leu Pro Val Arg Leu Leu Val Leu Met Phe Trp Ile Pro Ala
1 5 10 15
Ser Ser Ser Asp Val Leu Met Thr Gln Thr Pro Leu Ser Leu Pro Val
20 25 30
Ser Leu Gly Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Ile
35 40 45
Val His Ser Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu Gln Lys Pro
50 55 60
Gly Gln Ser Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn Arg Phe Ser
65 70 75 80
Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr
85 90 95
Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys
100 105 110
Phe Gln Gly Ser Tyr Val Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu
115 120 125
Glu Ile Lys Arg Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro
130 135 140
Ser Ser Glu Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu
145 150 155 160
Asn Asn Phe Tyr Pro Arg Asp Ile Asn Val Lys Trp Lys Ile Asp Gly
165 170 175
Ser Glu Arg Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser
180 185 190
Lys Asp Ser Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp
195 200 205
Glu Tyr Glu Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr
210 215 220
Ser Thr Ser Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys
225 230 235

<210> SEQ ID NO 77
<211> LENGTH: 449
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 77

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20 25 30
Ala Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr

-continued

65	70	75	80
Leu Gln Met Asn Ser	Leu Arg Ala Glu Asp	Thr Ala Val Tyr Tyr Cys	
	85	90	95
Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly			
	100	105	110
Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe			
	115	120	125
Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu			
	130	135	140
Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp			
	145	150	155
Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu			
	165	170	175
Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser			
	180	185	190
Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro			
	195	200	205
Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys			
	210	215	220
Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro			
	225	230	235
Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser			
	245	250	255
Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp			
	260	265	270
Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn			
	275	280	285
Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val			
	290	295	300
Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu			
	305	310	315
Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys			
	325	330	335
Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr			
	340	345	350
Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr			
	355	360	365
Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu			
	370	375	380
Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu			
	385	390	395
Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys			
	405	410	415
Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu			
	420	425	430
Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly			
	435	440	445

Lys

<210> SEQ ID NO 78

<211> LENGTH: 214

-continued

<212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 78

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1           5           10           15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
20          25          30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Arg Leu Ile
35          40          45
Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
50          55          60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65          70          75          80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
85          90          95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala
100         105         110
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
115         120         125
Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
130         135         140
Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
145         150         155         160
Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
165         170         175
Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
180         185         190
Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
195         200         205
Phe Asn Arg Gly Glu Cys
210

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<210> SEQ ID NO 79
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 79

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ala Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Asn Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
65          70          75          80

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-continued

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly
100 105 110

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 80
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 80

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20 25 30

Ala Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Asn Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly
100 105 110

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 81
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 81

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20 25 30

Ala Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly
100 105 110

-continued

 Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 82
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 82

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
 20 25 30
 Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Asn Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly
 100 105 110
 Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 83
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 83

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
 20 25 30
 Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Asn Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly
 100 105 110
 Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 84
 <211> LENGTH: 119

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 84

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ala Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Asn Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly
100 105 110
Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 85
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 85

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ala Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Asn Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly
100 105 110
Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 86
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 86

Asp Ile Gln Leu Thr Gln Ser Pro Ser Phe Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
20 25 30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45
Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 87

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 87

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
20 25 30
Leu Ser Trp Phe Gln Gln Lys Pro Gly Lys Ala Pro Lys Ser Leu Ile
35 40 45
Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 88

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 88

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
20 25 30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Arg Leu Ile
35 40 45
Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly

-continued

50	55	60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro		
65	70	75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr		
	85	90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys		
	100	105

<210> SEQ ID NO 89
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 89

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
20 25 30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45
Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Ile Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 90
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 90

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
20 25 30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45
Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

-continued

<210> SEQ ID NO 91
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 91

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
20 25 30
Leu Ser Trp Leu Gln Gln Lys Pro Gly Gly Ala Ile Lys Arg Leu Ile
35 40 45
Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Ser Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 92
<211> LENGTH: 468
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 92

Met Asn Phe Gly Leu Ser Leu Met Phe Leu Val Leu Val Leu Lys Gly
1 5 10 15
Val Gln Cys Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys
20 25 30
Pro Gly Gly Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
35 40 45
Ser Asp Tyr Ala Met Ser Trp Val Arg Gln Thr Pro Glu Lys Arg Leu
50 55 60
Glu Trp Val Ala Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro
65 70 75 80
Asp Asn Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn
85 90 95
Asn Leu Tyr Leu Gln Met Ser His Leu Lys Ser Glu Asp Thr Ala Met
100 105 110
Tyr Tyr Cys Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp
115 120 125
Gly Gln Gly Thr Thr Leu Thr Val Ser Ser Ala Ser Thr Lys Gly Pro
130 135 140
Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr
145 150 155 160
Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr
165 170 175
Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro

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<210> SEQ ID NO 93
<211> LENGTH: 236
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 93
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Met	Asp	Met	Arg	Val	Pro	Ala	His	Val	Phe	Gly	Phe	Leu	Leu	Leu	Trp
1				5					10					15	
Phe	Pro	Gly	Thr	Arg	Cys	Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser
			20					25					30		
Leu	Ser	Ala	Ser	Leu	Gly	Glu	Arg	Val	Ser	Leu	Thr	Cys	Arg	Ala	Ser
		35					40					45			

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Gln	Glu	Ile	Ser	Gly	Tyr	Leu	Ser	Trp	Leu	Gln	Gln	Lys	Pro	Asp	Gly
	50					55					60				
Thr	Ile	Lys	Arg	Leu	Ile	Tyr	Ala	Ala	Ser	Thr	Leu	Asp	Ser	Gly	Val
65					70					75					80
Pro	Lys	Arg	Phe	Ser	Gly	Ser	Arg	Ser	Gly	Ser	Asp	Tyr	Ser	Leu	Thr
			85						90					95	
Ile	Gly	Ser	Leu	Glu	Ser	Glu	Asp	Leu	Ala	Asp	Tyr	Tyr	Cys	Leu	Gln
			100					105					110		
Tyr	Asp	Ser	Tyr	Pro	Tyr	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile
		115					120					125			
Lys	Arg	Thr	Val	Ala	Ala	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp
	130					135					140				
Glu	Gln	Leu	Lys	Ser	Gly	Thr	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Asn
145					150					155					160
Phe	Tyr	Pro	Arg	Glu	Ala	Lys	Val	Gln	Trp	Lys	Val	Asp	Asn	Ala	Leu
			165					170						175	
Gln	Ser	Gly	Asn	Ser	Gln	Glu	Ser	Val	Thr	Glu	Gln	Asp	Ser	Lys	Asp
			180					185					190		
Ser	Thr	Tyr	Ser	Leu	Ser	Ser	Thr	Leu	Thr	Leu	Ser	Lys	Ala	Asp	Tyr
		195					200					205			
Glu	Lys	His	Lys	Val	Tyr	Ala	Cys	Glu	Val	Thr	His	Gln	Gly	Leu	Ser
	210					215					220				
Ser	Pro	Val	Thr	Lys	Ser	Phe	Asn	Arg	Gly	Glu	Cys				
225					230					235					

<210> SEQ ID NO 94

<211> LENGTH: 471

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 94

Met	Asp	Met	Arg	Val	Pro	Ala	Gln	Leu	Leu	Gly	Leu	Leu	Leu	Leu	Trp
1				5					10					15	
Leu	Arg	Gly	Ala	Arg	Cys	Gln	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly
	20					25							30		
Leu	Val	Lys	Pro	Gly	Gly	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly
	35					40					45				
Phe	Thr	Phe	Ser	Asp	Tyr	Ala	Met	Ser	Trp	Ile	Arg	Gln	Ala	Pro	Gly
	50					55				60					
Lys	Gly	Leu	Glu	Trp	Val	Ser	Thr	Ile	Ser	Asp	Gly	Gly	Thr	Tyr	Thr
65				70					75					80	
Tyr	Tyr	Pro	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn
			85					90						95	
Ala	Lys	Asn	Ser	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp
		100						105					110		
Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Glu	Trp	Gly	Asp	Tyr	Asp	Gly	Phe
	115					120						125			
Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr
	130					135					140				
Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser
145				150						155					160

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<210> SEQ ID NO 95
<211> LENGTH: 467
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 95
```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
1 5 10 15

Leu Arg Gly Ala Arg Cys Gln Val Gln Leu Val Glu Ser Gly Gly Gly

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20							25					30				
Leu	Val	Lys	Pro	Gly	Gly	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	
		35					40					45				
Phe	Thr	Phe	Ser	Asp	Tyr	Ala	Met	Ser	Trp	Ile	Arg	Gln	Ala	Pro	Gly	
	50					55					60					
Lys	Gly	Leu	Glu	Trp	Val	Ser	Thr	Ile	Ser	Asp	Gly	Gly	Thr	Tyr	Thr	
65					70					75					80	
Tyr	Tyr	Pro	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	
				85					90					95		
Ala	Lys	Asn	Ser	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	
			100					105					110			
Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Glu	Trp	Gly	Asp	Tyr	Asp	Gly	Phe	
		115					120					125				
Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	
	130					135					140					
Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	
145					150					155					160	
Glu	Ser	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	
				165					170					175		
Pro	Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	
			180						185				190			
Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	
		195					200					205				
Val	Val	Thr	Val	Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	
	210						215				220					
Asn	Val	Asp	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	
225					230					235					240	
Arg	Lys	Cys	Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	
				245					250					255		
Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	
			260					265					270			
Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	
		275					280					285				
Glu	Asp	Pro	Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	
	290					295					300					
His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	
305					310					315					320	
Arg	Val	Val	Ser	Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	
				325					330					335		
Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	
			340					345					350			
Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	
		355					360					365				
Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	
		370				375						380				
Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	
385					390					395					400	
Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	
			405						410					415		
Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	
			420					425					430			

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Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
 435 440 445

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
 450 455 460

Pro Gly Lys
 465

<210> SEQ ID NO 96
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 96

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
 1 5 10 15

Leu Arg Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
 20 25 30

Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
 35 40 45

Gln Glu Ile Ser Gly Tyr Leu Ser Trp Phe Gln Gln Lys Pro Gly Lys
 50 55 60

Ala Pro Lys Ser Leu Ile Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val
 65 70 75 80

Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 85 90 95

Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln
 100 105 110

Tyr Asp Ser Tyr Pro Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile
 115 120 125

Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
 130 135 140

Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
 145 150 155 160

Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
 165 170 175

Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
 180 185 190

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
 195 200 205

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
 210 215 220

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230 235

<210> SEQ ID NO 97
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 97

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Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
1          5          10          15
Leu Arg Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
          20          25          30
Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
          35          40          45
Gln Glu Ile Ser Gly Tyr Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys
          50          55          60
Ala Pro Lys Arg Leu Ile Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val
65          70          75          80
Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr
          85          90          95
Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln
          100          105          110
Tyr Asp Ser Tyr Pro Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile
          115          120          125
Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
          130          135          140
Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
145          150          155          160
Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
          165          170          175
Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
          180          185          190
Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
          195          200          205
Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
          210          215          220
Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
225          230          235

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<210> SEQ ID NO 98

<211> LENGTH: 449

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 98

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Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
          20          25          30
Ala Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45
Ser Thr Ile Ser Asp Gly Gly Thr Tyr Thr Tyr Tyr Pro Asp Ser Val
          50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95
Ala Arg Glu Trp Gly Asp Tyr Asp Gly Phe Asp Tyr Trp Gly Gln Gly

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100					105					110					
Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe
	115						120					125			
Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu
	130					135					140				
Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp
	145					150					155				160
Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu
				165					170					175	
Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser
			180						185					190	
Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro
		195					200					205			
Ser	Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys
	210					215					220				
Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro
	225					230					235				240
Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser
				245					250					255	
Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp
			260						265					270	
Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn
	275						280					285			
Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val
	290					295					300				
Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu
	305					310					315				320
Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys
				325					330					335	
Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr
			340						345					350	
Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr
	355						360					365			
Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu
	370					375					380				
Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu
	385					390					395				400
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys
				405					410					415	
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu
			420					425						430	
Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly
	435						440					445			

Lys

<210> SEQ ID NO 99

<211> LENGTH: 445

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 99

Gln	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Lys	Pro	Gly	Gly	1	5	10	15
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Asp	Tyr	20	25	30	
Ala	Met	Ser	Trp	Ile	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val	35	40	45	
Ser	Thr	Ile	Ser	Asp	Gly	Gly	Thr	Tyr	Thr	Tyr	Tyr	Pro	Asp	Ser	Val	50	55	60	
Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Asn	Ser	Leu	Tyr	65	70	75	80
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	85	90	95	
Ala	Arg	Glu	Trp	Gly	Asp	Tyr	Asp	Gly	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	100	105	110	
Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	115	120	125	
Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala	Leu	130	135	140	
Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	145	150	155	160
Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	165	170	175	
Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	180	185	190	
Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys	Pro	195	200	205	
Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys	Cys	Val	Glu	210	215	220	
Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	225	230	235	240
Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	245	250	255	
Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	260	265	270	
Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	275	280	285	
Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	290	295	300	
Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	305	310	315	320
Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	325	330	335	
Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	340	345	350	
Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	355	360	365	
Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	370	375	380	
Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	385	390	395	400

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Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln
405 410 415

Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn
420 425 430

His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
435 440 445

<210> SEQ ID NO 100

<211> LENGTH: 214

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 100

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
20 25 30

Leu Ser Trp Phe Gln Gln Lys Pro Gly Lys Ala Pro Lys Ser Leu Ile
35 40 45

Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
85 90 95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala
100 105 110

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
115 120 125

Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
130 135 140

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
145 150 155 160

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
165 170 175

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
180 185 190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
195 200 205

Phe Asn Arg Gly Glu Cys
210

<210> SEQ ID NO 101

<211> LENGTH: 214

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 101

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

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Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Glu Ile Ser Gly Tyr
 20 25 30
 Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Arg Leu Ile
 35 40 45
 Tyr Ala Ala Ser Thr Leu Asp Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr Asp Ser Tyr Pro Tyr
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala
 100 105 110
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
 115 120 125
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
 130 135 140
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
 145 150 155 160
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
 165 170 175
 Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
 180 185 190
 Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
 195 200 205
 Phe Asn Arg Gly Glu Cys
 210

<210> SEQ ID NO 102

<211> LENGTH: 450

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 102

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
 20 25 30
 Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
 100 105 110
 Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
 115 120 125
 Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala

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130					135					140					
Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser
145					150					155					160
Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val
			165						170					175	
Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro
			180					185					190		
Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys
			195				200					205			
Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp
			210			215					220				
Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly
225						230					235				240
Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile
			245						250					255	
Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu
			260					265					270		
Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His
			275				280					285			
Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg
			290			295					300				
Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys
305						310					315				320
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu
			325						330					335	
Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr
			340				345						350		
Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu
		355					360					365			
Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp
		370				375					380				
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val
385						390					395				400
Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp
			405						410					415	
Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His
			420				425						430		
Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro
		435					440					445			
Gly	Lys														
	450														

<210> SEQ ID NO 103

<211> LENGTH: 214

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 103

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1				5						10				15	

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Asp	Arg	Val	Thr	Ile	Thr	Cys	Arg	Ala	Ser	Gln	Asp	Val	Asn	Thr	Ala
			20					25					30		
Val	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile
		35					40					45			
Tyr	Ser	Ala	Ser	Phe	Leu	Tyr	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50					55					60				
Ser	Arg	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro
65				70						75				80	
Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	His	Tyr	Thr	Thr	Pro	Pro
				85					90					95	
Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Arg	Thr	Val	Ala	Ala
			100					105					110		
Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser	Gly
		115					120					125			
Thr	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu	Ala
	130					135					140				
Lys	Val	Gln	Trp	Lys	Val	Asp	Asn	Ala	Leu	Gln	Ser	Gly	Asn	Ser	Gln
145					150					155					160
Glu	Ser	Val	Thr	Glu	Gln	Asp	Ser	Lys	Asp	Ser	Thr	Tyr	Ser	Leu	Ser
			165						170					175	
Ser	Thr	Leu	Thr	Leu	Ser	Lys	Ala	Asp	Tyr	Glu	Lys	His	Lys	Val	Tyr
		180						185					190		
Ala	Cys	Glu	Val	Thr	His	Gln	Gly	Leu	Ser	Ser	Pro	Val	Thr	Lys	Ser
		195					200					205			
Phe	Asn	Arg	Gly	Glu	Cys										
		210													

<210> SEQ ID NO 104

<211> LENGTH: 448

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 104

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly
1			5					10					15		
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Thr	Asp	Tyr
		20					25					30			
Thr	Met	Asp	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
	35					40						45			
Ala	Asp	Val	Asn	Pro	Asn	Ser	Gly	Gly	Ser	Ile	Tyr	Asn	Gln	Arg	Phe
	50				55					60					
Lys	Gly	Arg	Phe	Thr	Leu	Ser	Val	Asp	Arg	Ser	Lys	Asn	Thr	Leu	Tyr
65				70					75					80	
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
		85						90					95		
Ala	Arg	Asn	Leu	Gly	Pro	Ser	Phe	Tyr	Phe	Asp	Tyr	Trp	Gly	Gln	Gly
		100					105						110		
Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe
	115						120					125			
Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu
	130					135					140				

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Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp
 145 150 155 160
 Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu
 165 170 175
 Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser
 180 185 190
 Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro
 195 200 205
 Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys
 210 215 220
 Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro
 225 230 235 240
 Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
 245 250 255
 Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp
 260 265 270
 Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
 275 280 285
 Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val
 290 295 300
 Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu
 305 310 315 320
 Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys
 325 330 335
 Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr
 340 345 350
 Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr
 355 360 365
 Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu
 370 375 380
 Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu
 385 390 395 400
 Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys
 405 410 415
 Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu
 420 425 430
 Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly
 435 440 445

<210> SEQ ID NO 105

<211> LENGTH: 214

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 105

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asp Val Ser Ile Gly
 20 25 30
 Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile

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35	40	45
Tyr Ser Ala Ser Tyr Arg	Tyr Thr Gly Val Pro	Ser Arg Phe Ser Gly
50	55	60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile	Ser Ser Leu Gln Pro	
65	70	75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Tyr Ile Tyr Pro Tyr		
	85	90 95
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala		
	100	105 110
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly		
	115	120 125
Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala		
	130	135 140
Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln		
	145	150 155 160
Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser		
	165	170 175
Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr		
	180	185 190
Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser		
	195	200 205
Phe Asn Arg Gly Glu Cys		
	210	

<210> SEQ ID NO 106

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 106

Gln Val Gln Leu Gln Gln Trp Gly Ala Gly Leu Leu Lys Pro Ser Glu
1 5 10 15
Thr Leu Ser Leu Thr Cys Ala Val Tyr Gly Gly Ser Phe Ser Gly Tyr
20 25 30
Tyr Trp Ser Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Ile
35 40 45
Gly Glu Ile Asn His Ser Gly Ser Thr Asn Tyr Asn Pro Ser Leu Lys
50 55 60
Ser Arg Val Thr Ile Ser Val Glu Thr Ser Lys Asn Gln Phe Ser Leu
65 70 75 80
Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Asp Lys Trp Thr Trp Tyr Phe Asp Leu Trp Gly Arg Gly Thr Leu
100 105 110
Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu
115 120 125
Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys
130 135 140
Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser
145 150 155 160

Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	
				165					170					175		
Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser	
				180					185					190		
Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn	
				195					200					205		
Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr	His	
				210					215					220		
Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	
				225					230					235		
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	
				245					250					255		
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	
				260					265					270		
Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	
				275					280					285		
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	
				290					295					300		
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	
				305					310					315		
Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	
				325					330					335		
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	
				340					345					350		
Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	
				355					360					365		
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	
				370					375					380		
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	
				385					390					395		
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	
				405					410					415		
Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	
				420					425					430		
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys		
				435					440					445		

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<210> SEQ ID NO 107
<211> LENGTH: 220
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide
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<400> SEQUENCE: 107

Asp	Ile	Glu	Met	Thr	Gln	Ser	Pro	Asp	Ser	Leu	Ala	Val	Ser	Leu	Gly
1				5					10					15	
Glu	Arg	Ala	Thr	Ile	Asn	Cys	Arg	Ser	Ser	Gln	Ser	Val	Leu	Tyr	Ser
			20					25					30		
Ser	Ser	Asn	Arg	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Asn	Pro	Gly	Gln
		35					40					45			
Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Trp	Ala	Ser	Thr	Arg	Glu	Ser	Gly	Val
	50					55					60				

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Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65          70          75          80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Gln Gln
          85          90          95

Tyr Tyr Ser Thr Pro Arg Thr Phe Gly Gln Gly Thr Lys Val Glu Ile
          100          105          110

Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
          115          120          125

Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
          130          135          140

Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
145          150          155          160

Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
          165          170          175

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
          180          185          190

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
          195          200          205

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
          210          215          220

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<210> SEQ ID NO 108

<211> LENGTH: 445

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 108

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr
          20          25          30

Val Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45

Ser Ser Ile Ser Ser Ser Gly Gly Trp Thr Leu Tyr Ala Asp Ser Val
          50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95

Thr Arg Gly Leu Lys Met Ala Thr Ile Phe Asp Tyr Trp Gly Gln Gly
          100          105          110

Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe
          115          120          125

Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu
          130          135          140

Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp
145          150          155          160

Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu
          165          170          175

Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser

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180					185					190					
Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys	Pro
	195						200					205			
Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys	Cys	Val	Glu
	210					215					220				
Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu
	225					230					235				240
Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu
			245						250					255	
Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln
			260					265					270		
Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys
	275					280						285			
Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu
	290					295					300				
Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys
	305					310					315				320
Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys
			325						330					335	
Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser
			340					345					350		
Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys
	355					360						365			
Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln
	370					375					380				
Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly
	385					390					395				400
Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln
			405					410						415	
Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn
			420					425					430		
His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys			
	435					440						445			

<210> SEQ ID NO 109

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 109

Gln	Ser	Ala	Leu	Thr	Gln	Pro	Ala	Ser	Val	Ser	Gly	Ser	Pro	Gly	Gln
1				5					10					15	
Ser	Ile	Thr	Ile	Ser	Cys	Thr	Gly	Thr	Ser	Ser	Asp	Val	Gly	Ser	Tyr
		20						25					30		
Asn	Val	Val	Ser	Trp	Tyr	Gln	Gln	His	Pro	Gly	Lys	Ala	Pro	Lys	Leu
		35						40				45			
Ile	Ile	Tyr	Glu	Val	Ser	Gln	Arg	Pro	Ser	Gly	Val	Ser	Asn	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Lys	Ser	Gly	Asn	Thr	Ala	Ser	Leu	Thr	Ile	Ser	Gly	Leu
	65				70					75					80

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Gln	Thr	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Cys	Ser	Tyr	Ala	Gly	Ser
				85					90					95	
Ser	Ile	Phe	Val	Ile	Phe	Gly	Gly	Gly	Thr	Lys	Val	Thr	Val	Leu	Gly
			100					105					110		
Gln	Pro	Lys	Ala	Ala	Pro	Ser	Val	Thr	Leu	Phe	Pro	Pro	Ser	Ser	Glu
		115					120					125			
Glu	Leu	Gln	Ala	Asn	Lys	Ala	Thr	Leu	Val	Cys	Leu	Val	Ser	Asp	Phe
	130					135					140				
Tyr	Pro	Gly	Ala	Val	Thr	Val	Ala	Trp	Lys	Ala	Asp	Gly	Ser	Pro	Val
145					150					155					160
Lys	Val	Gly	Val	Glu	Thr	Thr	Lys	Pro	Ser	Lys	Gln	Ser	Asn	Asn	Lys
			165					170						175	
Tyr	Ala	Ala	Ser	Ser	Tyr	Leu	Ser	Leu	Thr	Pro	Glu	Gln	Trp	Lys	Ser
			180					185					190		
His	Arg	Ser	Tyr	Ser	Cys	Arg	Val	Thr	His	Glu	Gly	Ser	Thr	Val	Glu
	195						200					205			
Lys	Thr	Val	Ala	Pro	Ala	Glu	Cys	Ser							
	210					215									

<210> SEQ ID NO 110

<211> LENGTH: 449

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 110

Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala
1			5						10					15	
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Arg	Ser	Ser
		20				25							30		
Tyr	Ile	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Met
	35					40						45			
Gly	Trp	Ile	Tyr	Ala	Gly	Thr	Gly	Ser	Pro	Ser	Tyr	Asn	Gln	Lys	Leu
	50				55						60				
Gln	Gly	Arg	Val	Thr	Met	Thr	Thr	Asp	Thr	Ser	Thr	Ser	Thr	Ala	Tyr
65				70					75						80
Met	Glu	Leu	Arg	Ser	Leu	Arg	Ser	Asp	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85					90						95	
Ala	Arg	His	Arg	Asp	Tyr	Tyr	Ser	Asn	Ser	Leu	Thr	Tyr	Trp	Gly	Gln
		100					105						110		
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val
	115					120						125			
Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala
	130				135						140				
Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser
145					150					155					160
Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val
			165					170						175	
Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro
		180						185					190		
Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys
		195					200					205			

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Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp
 210                215                220

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
 225                230                235                240

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
                245                250                255

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
                260                265                270

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 275                280                285

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
 290                295                300

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 305                310                315                320

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
                325                330                335

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
                340                345                350

Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu
 355                360                365

Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 370                375                380

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 385                390                395                400

Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
                405                410                415

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
                420                425                430

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 435                440                445

Gly

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<210> SEQ ID NO 111
<211> LENGTH: 220
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

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<400> SEQUENCE: 111

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Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
 1                5                10                15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Val Leu Asn Ser
                20                25                30

Gly Asn Gln Lys Asn Tyr Leu Thr Trp Tyr Gln Gln Lys Pro Gly Gln
 35                40                45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
 50                55                60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 65                70                75                80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Gln Ser
 85                90                95

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Asp Tyr Ser Tyr Pro Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile
100 105 110

Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
115 120 125

Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
130 135 140

Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
145 150 155 160

Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
165 170 175

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
180 185 190

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
195 200 205

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
210 215 220

<210> SEQ ID NO 112

<211> LENGTH: 446

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 112

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Ala Ile Asn Ser Gln Gly Lys Ser Thr Tyr Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Trp Gly Asp Glu Gly Phe Asp Ile Trp Gly Gln Gly Thr Leu
100 105 110

Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu
115 120 125

Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys
130 135 140

Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser
145 150 155 160

Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser
165 170 175

Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser
180 185 190

Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn
195 200 205

Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys Asp Lys Thr His

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210					215					220						
Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	
225					230					235					240	
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	
					245					250					255	
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	
					260					265				270		
Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	
					275					280				285		
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	
					290					295				300		
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	
					305					310				315		320
Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	
					325					330					335	
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	
					340					345				350		
Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	
					355					360				365		
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	
					370					375				380		
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	
					385					390				395		400
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	
					405					410					415	
Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	
					420					425				430		
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly			
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<210> SEQ ID NO 113

<211> LENGTH: 214

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 113

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				20				25						30	
Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile
				35			40					45			
Tyr	Gly	Ala	Ser	Ser	Leu	Gln	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
				50			55					60			
Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro
				65			70					75		80	
Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Tyr	Ser	Ser	Phe	Pro	Thr
				85				90						95	
Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Arg	Thr	Val	Ala	Ala
				100				105						110	

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Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser	Gly
		115					120					125			
Thr	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu	Ala
	130					135					140				
Lys	Val	Gln	Trp	Lys	Val	Asp	Asn	Ala	Leu	Gln	Ser	Gly	Asn	Ser	Gln
145					150					155					160
Glu	Ser	Val	Thr	Glu	Gln	Asp	Ser	Lys	Asp	Ser	Thr	Tyr	Ser	Leu	Ser
			165						170					175	
Ser	Thr	Leu	Thr	Leu	Ser	Lys	Ala	Asp	Tyr	Glu	Lys	His	Lys	Val	Tyr
		180						185					190		
Ala	Cys	Glu	Val	Thr	His	Gln	Gly	Leu	Ser	Ser	Pro	Val	Thr	Lys	Ser
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Phe	Asn	Arg	Gly	Glu	Cys										
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<210> SEQ ID NO 114

<211> LENGTH: 1952

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 114

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agtctagaaa ttctaaactc tgaggggggtc ggatgacgtg gccattcttt gcctaaagca	480
ttgagtttac tgcaagggtc gaaaagcatg caaagccctc agaatggctg caaagagctc	540
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aactttgtta cttaaacacc atcctgtttg cttcttttct caggaactgt ggctgcacca	780
tctgtcttca tcttcccgc atctgatgag cagttgaaat ctggaactgc ctctgttgtg	840
tgctctgtga ataacttcta tcccagagag gccaaagtac agtgggaagg ggataacgcc	900
ctccaatcgg gtaactccca ggagagtgtc acagagcagg acagcaagga cagcacctac	960
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tgttggttta ccaactactc aatttctctt ataagggact aaatatgtag tcatcctaag 1380
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<210> SEQ ID NO 115

<211> LENGTH: 238

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 115

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20          25          30

Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Ser
35          40          45

Val Asp Asn Tyr Gly Ile Ser Phe Met Asn Trp Phe Gln Gln Lys Pro
50          55          60

Gly Lys Ala Pro Lys Leu Leu Ile Tyr Ala Ala Ser Asn Gln Gly Ser
65          70          75          80

Gly Val Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr
85          90          95

Leu Thr Ile Ser Ser Leu Gln Pro Asp Asp Phe Ala Thr Tyr Tyr Cys
100         105         110

Gln Gln Ser Lys Glu Val Pro Trp Thr Phe Gly Gln Gly Thr Lys Val
115         120         125

Glu Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro
130         135         140

Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu
145         150         155         160

Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn
165         170         175

Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser
180         185         190

Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala
195         200         205

Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly
210         215         220

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<211> LENGTH: 3246			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
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<210> SEQ ID NO 117

<211> LENGTH: 465

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 117

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20          25          30
Pro Gly Ser Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe
35          40          45
Thr Asp Tyr Asn Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu
50          55          60
Glu Trp Ile Gly Tyr Ile Tyr Pro Tyr Asn Gly Gly Thr Gly Tyr Asn
65          70          75          80
Gln Lys Phe Lys Ser Lys Ala Thr Ile Thr Ala Asp Glu Ser Thr Asn
85          90          95

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Thr	Ala	Tyr	Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	100	105	110
Tyr	Tyr	Cys	Ala	Arg	Gly	Arg	Pro	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	115	120	125
Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	130	135	140
Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	145	150	155
Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	165	170	175
Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	180	185	190
Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	195	200	205
Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	210	215	220
Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	225	230	235
Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	245	250	255
Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	260	265	270
Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	275	280	285
Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	290	295	300
Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	305	310	315
Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	325	330	335
Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	340	345	350
Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	355	360	365
Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	370	375	380
Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	385	390	395
Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	405	410	415
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	420	425	430
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	435	440	445
Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	450	455	460
Lys																465		

What is claimed is:

1. A method for treating a human subject afflicted with acute myeloid leukemia, comprising administering to the subject (i) one or both of a histone deacetylase (HDAC) inhibitor and an LSD1 inhibitor, and (ii) ^{225}Ac -labeled HuM195, wherein the amounts of the HDAC inhibitor and/or LSD1 inhibitor, and ^{225}Ac -labeled HuM195, when administered in conjunction with one another, are therapeutically effective.

2. The method of claim 1, wherein the administering step comprises:

administering the HDAC inhibitor to the subject.

3. The method of claim 1 or 2, wherein the administering step comprises:

administering the LSD1 inhibitor to the subject.

4. A method for treating a mammalian subject afflicted with cancer, comprising administering to the subject (i) one or both of a histone deacetylase (HDAC) inhibitor and an LSD1 inhibitor, and (ii) a radioisotope-labeled agent that targets cancer cells in the subject, wherein the amounts of the HDAC inhibitor and/or the LSD1 inhibitor, and radioisotope-labeled agent, when administered in conjunction with one another, are therapeutically effective.

5. The method of claim 4, wherein the subject is human.

6. The method of claim 4 or 5, wherein the cancer is a solid tumor.

7. The method of any one of claims 4-6, wherein the cancer is selected from the group consisting of breast cancer, ovarian cancer, prostate cancer, lung cancer, squamous cell carcinoma of the head and neck, gastric cancer, pancreatic cancer, brain cancer, liver cancer, sarcoma and melanoma.

8. The method of any one of claims 4-7, wherein the cancer is selected from the group consisting of breast cancer and ovarian cancer.

9. The method of claim 4 or 5, wherein the cancer is a hematologic malignancy.

10. The method of any one of claims 4, 5 and 9, wherein the cancer is selected from the group consisting of acute myeloid leukemia, myelodysplastic syndrome, and multiple myeloma.

11. The method of claim 10, wherein the hematologic malignancy is acute myeloid leukemia.

12. The method of any of claims 4-11, wherein the HDAC inhibitor is selected from the group consisting of vorinostat, romidepsin, belinostat, and panobinostat.

13. The method of any of claims 4-12, wherein the radioisotope-labeled agent is an antibody that specifically binds to a moiety selected from the group consisting of CD33, CD45, CD38, Her3, DR5 (TRAIL-R2), 5T4, Lewis-Y, CAIX, Her2, Claudin 18.2, CEACAM5, MUC1, and mesothelin.

14. The method of claim 13, wherein the antibody is selected from the group consisting of alemtuzumab, ibritumomab, brentuximab, trastuzumab, trastuzumab emtansine, gemtuzumab, lintuzumab, B1836858, BC8/apamistamab, daratumumab, felzartamab/MOR202, isatuximab/SAR650984, TAK-169, AV-203, CDX-3379, HMBD 001, patritumab, seribantumab, elgemtumab/LJM716, lumretuzumab/RG7116/RO5479599, GSK2849330, mapatumumab, tigatuzumab/CS-1008, drozitumab, lexatumumab, conatumumab, MEDI10641, PF-06263507/A1 mcMMAF, Anti-5T4 SYD1875, ASN004, AVA-020, Tb535, Hu3S193, B3, IGN311, BR96/BMS-182248-01, girentuximab, BAY 794620, BAY 2701439, MEDI-4276, XMT-1522, zolbetux-

imab, TST001, A315, labetuzumab, SAR566658, DMOT4039, anetumab, BAY 2287411, and amatuximab.

15. The method of any one of claims 4, 5, and 9-12, wherein the radioisotope-labeled agent is an anti-CD33 antibody labeled with an isotope selected from the group consisting of ^{90}Y , ^{89}Sr , ^{153}Sm , ^{32}P , ^{225}Ac , ^{213}Bi , ^{213}Po , ^{211}At , ^{212}Bi , ^{213}Bi , ^{223}Ra , ^{227}Th , ^{149}Tb , ^{131}I , ^{137}Cs , ^{212}Pb , ^{103}Pd , ^{166}Ho , ^{186}Re , ^{188}Re , ^{67}Cu , ^{199}Au , ^{105}Rh , ^{211}As , and ^{177}Lu .

16. The method of claim 15, wherein the radioisotope-labeled agent is ^{225}Ac -labeled HuM195 or ^{177}Lu -labeled HuM195.

17. A method for inducing the death of a mammalian cancer cell, comprising contacting the cell with (i) one or both of a histone deacetylase (HDAC) inhibitor and an LSD1 inhibitor, in conjunction with (ii) a radioisotope-labeled agent that targets the cancer cell, wherein the amounts of HDAC inhibitor and/or LSD1 inhibitor, and radiolabeled agent, when contacted with the cell in conjunction with one another, are effective to induce the cell's death.

18. The method of claim 17, wherein the cancer cell is a human cancer cell.

19. The method of claim 17 or 18, wherein the cancer cell is selected from the group consisting of a breast cancer cell, an ovarian cancer cell, a prostate cancer cell, a lung cancer cell, a squamous cell carcinoma of the head and neck cell, a gastric cancer cell, a pancreatic cancer cell, a brain cancer cell, a liver cancer cell, a sarcoma cell, and a melanoma cell.

20. The method of any of claims 17-19, wherein the cancer cell is selected from the group consisting of a breast cancer cell and an ovarian cancer cell.

21. The method of claim 17 or 19, wherein the cancer cell is a hematologic cancer cell.

22. The method of any of claims 17, 18, and 21, wherein the hematologic cancer cell is an acute myeloid leukemia cell, a myelodysplastic syndrome cell, or a multiple myeloma cell.

23. The method of any of claims 17, 18, 21, and 22, wherein the hematologic cancer cell is an acute myeloid leukemia cell.

24. The method of any of claims 17-23, wherein the HDAC inhibitor is selected from the group consisting of vorinostat, romidepsin, belinostat, and panobinostat.

25. The method of any of claims 17-23, wherein the radioisotope-labeled agent is an antibody that specifically binds to an antigen selected from the group consisting of CD33, CD45, CD38, Her3, DR5 (TRAIL-R2), 5T4, Lewis-Y, CAIX, Her2, Claudin 18.2, CEACAM5, MUC1, and mesothelin.

26. The method of claim 25, wherein the antibody is selected from the group consisting of alemtuzumab, ibritumomab, brentuximab, trastuzumab, gemtuzumab, lintuzumab, B1836858, BC8/apamistamab, daratumumab, felzartamab/MOR202, isatuximab/SAR650984, TAK-169, AV-203, CDX-3379, HMBD 001, patritumab, seribantumab, elgemtumab/LJM716, lumretuzumab/RG7116/RO5479599, GSK2849330, mapatumumab, tigatuzumab/CS-1008, drozitumab, lexatumumab, conatumumab, MEDI10641, PF-06263507/A1 mcMMAF, Anti-5T4 SYD1875, ASN004, AVA-020, Tb535, Hu3S193, B3, IGN311, BR96/BMS-182248-01, girentuximab, BAY 794620, BAY 2701439, MEDI-4276, XMT-1522, zolbetuximab, TST001, A315, labetuzumab, SAR566658, DMOT4039, anetumab, BAY 2287411, and amatuximab.

27. The method of any of claims **17-19** and **21-24**, wherein the radioisotope-labeled agent is an anti-CD33 antibody labeled with a radioisotope selected from the group consisting of ^{90}Y , ^{89}Sr , ^{153}Sm , ^{32}P , ^{225}Ac , ^{213}Bi , ^{213}Po , ^{211}At , ^{212}Bi , ^{213}Bi , ^{223}Ra , ^{227}Th , ^{149}Tb , ^{131}I , ^{137}Cs , ^{212}Pb , ^{103}Pd , ^{166}Ho , ^{186}Re , ^{188}Re , ^{67}Cu , ^{199}Au , ^{105}Rh , ^{211}As , and ^{177}Lu .

28. The method of any of claims **17-19** and **21-24**, wherein the radioisotope-labeled agent is ^{225}Ac -labeled HuM195.

29. A method for inducing the death of a mammalian acute myeloid leukemic cell, comprising contacting the cell with (i) one or both of a histone deacetylase (HDAC) inhibitor and an LSD1 inhibitor, and (ii) ^{225}Ac -labeled HuM195, wherein the amounts of HDAC inhibitor and/or LSD1 inhibitor, and ^{225}Ac -labeled HuM195, when con-

tacted with the cell in conjunction with one another, are effective to induce the cell's death.

30. The method of claim **29**, wherein the contacting step comprises:

contacting the cell with the HDAC inhibitor.

31. The method of claim **29** or **30**, wherein the contacting step comprises:

contacting the cell with the LSD1 inhibitor.

32. Combination use of a cancer targeting agent labeled with an alpha-particle emitting radionuclide and one or both of an HDAC inhibitor and an LSD1 inhibitor for the treatment of cancer in a mammalian subject.

33. Combination use of a cancer targeting agent labeled with a beta-particle emitting radionuclide and one or both of an HDAC inhibitor and an LSD1 inhibitor for the treatment of cancer in a mammalian subject.

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